



Probing the Electroweak Phase Transition in a Light-Singlet Extension of the SM

– Master's Thesis –

Diego García Tejada 

– Supervisors –

Dr. Thomas Biekötter 

Dr. Jose Miguel No Redondo 

Para las *super estrellas* de la selección masculina de fútbol
española por ganar la copa del mundo del 2026

Acknowledgments

There are a lot of people that I would like to thank for their help, either direct or indirect, in writing this thesis. By the presence of any grammatical errors that may still remain, any reader may come to the conclusion that I am by no means a great writer, still I will try to acknowledge all of the persons that have made this work possible.

To my supervisors Thomas and Josemi: To say that I am fortunate to have had the opportunity to have worked with you both would be the understatement of the century, just below saying that Betis is the best football club of Seville. Through this past year I have learned so much that I honestly had to stop using footnotes so as to not go over the page limit. But not only have I learned from you both but I have also felt super supported, through all the hard situations and doubts, you were always a message away. The time spent together alongside you is one of the main things I take away from this experience and I hope that we continue to work together for the years to come.

To my sister Loreto: To be honest the only person who I think wants to be named here. Being able to live together this past year has been an experience to be sure but I am always happy to know that I have someone to talk to at home. I do not know what I am going to do without you next year, watching ANHQP was a break from all the work and you driving me to and from the IFT was truly a blessing.

To my *colegones*: Julia, Isaac, Nico, Gus, Lucia, Juanjo and Nahia. Solo os conozco desde hace un año y aun así parece que os llevo conociendo toda mi vida. Desde los museos en el BBVA, las horas perdidas en el ping pong, las sevillanas en la feria, y el incidente con la taza, todo eso son anécdotas que superan las locuras que hice en 4 años de carrera y las hemos conseguido en menos de un año. Habéis conseguido que Madrid sea la mejor ciudad del mundo solo por vuestra presencia, gracias por este año increíble.

To my friends that have known me since before I started my journey into physics: Ana, Fátima and Simon. Simon, first time you can understand something I write. You have no idea how grateful I am to still consider you a friend after all these years without seeing each other in person, I am so looking forward to spending time together in vice city, but not too much I do need to get a PhD. Fátima, we see each other once in a blue moon but damn I am always looking forward to those occasions, it is thanks to you that I consider Tres Cantos my home and I am going to be looking forward to coming back. Ana, you know why you are here, as they say, it is always a blessing to have reasons to make saying goodbye hard.

*“All that is gold does not glitter,
Not all those who wander are lost;
The old that is strong does not wither,
Deep roots are not reached by the frost.”*

J.R.R. Tolkien

List of Abbreviations

1PI one-particle irreducible. 4–6	RSB Radiative Symmetry Breaking. 3, 9, 13, 17
BAU Baryon Asymmetry of the Universe. 1, 31	RxSM Real Singlet Extension of the Standard Model. 33, 37, 41, 42, 48, 52, 54–56
BSM Beyond the Standard Model. 1, 16, 32, 33, 51	
CW Coleman-Weinberg. 6–9, 12, 37, 43, 51	SFOEWPT Strong First-Order Electroweak Phase Transition. 1, 44, 45, 49–52, 54, 55
EW Electroweak. 15, 18, 33–35, 38, 41, 43–47, 56–58	SFOPT Strong First-Order Phase Transition. 28, 29, 31
EWPT Electroweak Phase Transition. 1, 19, 31, 32, 54	SM Standard Model. 1, 2, 15, 16, 18, 26, 31–33, 35, 41, 44, 45, 51, 52, 54, 56, 57
EWSB Electroweak Symmetry Breaking. 18	SSB Spontaneous Symmetry Breaking. 3, 5, 15, 26, 35, 57
FOEWPT First-Order Electroweak Phase Transition. 32, 33, 41	TFT Thermal Field Theory. 19, 20, 30, 31
FOPT First-Order Phase Transition. 26–28, 31, 44, 46	VEV Vacuum Expectation Value. 4, 5, 10, 14–16, 22, 26, 28, 33, 35, 41, 56
QFT Quantum Field Theory. 3, 19, 20, 31, 36	

Contents

1	Introduction	1
2	Zero-Temperature Effective Potential	3
2.1	Generating Functionals, Effective action and Effective Potential	3
2.2	Effective Potential of a Real Scalar Field Theory	5
2.2.1	Coleman-Weinberg Effective Potential	6
2.2.2	Counter Term Effective Potential	8
2.2.3	Radiative Symmetry Breaking	9
2.3	Effective Potential of an Abelian Higgs Model	10
2.3.1	Review of the Abelian Higgs Mechanism	11
2.3.2	One-Loop Effective Potential	12
2.3.3	Radiative Symmetry Breaking	13
2.4	Effective Potential of the Standard Model	15
2.4.1	Coleman-Weinberg Effective Potential	16
2.4.2	Observables and Renormalization	16
2.4.3	The Impossibility of Radiative Symmetry Breaking in the SM	17
3	Field Theory and Effective Potentials at Finite Temperatures	19
3.1	Review of Statistical Mechanics	19
3.2	Partition Functions and Path Integrals	19
3.3	Matsubara Formalism	20
3.3.1	Thermal Propagator	21
3.4	Evaluating Matsubara Sums	21
3.4.1	Example of a Matsubara Sum: Free Boson/Fermion Propagator	22
3.5	Green's Functions at Finite Temperature	22
3.6	Finite Temperature Effective Potential	23
3.7	High Temperature Expansion	25
3.8	Phase Transitions	26
3.8.1	Second-Order Phase Transitions	27
3.8.2	First-Order Phase Transitions	27
3.9	Thermal Tunneling and Bubble Nucleation	29
3.10	On the Validity of Thermal Perturbation Theory	30
3.11	Electroweak Phase Transitions and Baryogenesis	31
3.11.1	Electroweak Baryogenesis	31
4	Real Singlet Extension of the Standard Model	33
4.1	Scalar and Tree-Level Potential	33
4.2	Tadpole Equations and Field Dependent Masses	33
4.3	Parameter Prescription	35
4.4	Feynman Rules and Higgs-Singlet Coupling	35
4.5	Tree-Level vacua	36
4.6	Coleman-Weinberg and Counter Term Potentials	37
4.6.1	Coleman-Weinberg Potential	37
4.6.2	Counter Term Potential	38
4.7	Thermal Potential and Thermal Masses	39
4.8	On the Z_2 Symmetric Models	41

5	First Order Phase Transitions in a Z_2 preserving RxSM	42
5.1	Model and Parameters	42
5.2	Tree-Level Vacua and Stability Constraints	42
5.3	High Temperature Limit	43
5.4	Types of Phase Transitions	44
5.4.1	Two-Step Phase Transitions	45
5.4.2	Three-Step Phase Transitions	46
5.5	Analytical Estimate for Vacuum Trapping	48
5.6	Parameter Scans	48
5.6.1	High Temperature Limit	49
5.6.2	Numeric Scan	49
5.7	Experimental Constraints on Exotic Higgs Decays at the LHC	51
5.7.1	Generalization to Mixed Scenarios	52
6	Conclusions and Outlook	54
A	RxSM with a Spontaneously Broken Z_2 Symmetry	56
A.1	Scalar and Tree-Level Potential	56
A.2	Tadpole Equations and Field Dependent Masses	56
A.3	Parameter Prescription	57
A.4	Higgs-Singlet Coupling	57
A.5	Tree-Level Vacuums	58
B	Auxiliary Figures	59
	Bibliography	60

1 Introduction

One of the biggest open problems in modern physics is that it seems that our Universe is made almost entirely of matter, with essentially no trace of primordial antimatter. Big Bang nucleosynthesis [1] and measurements of the cosmic microwave background [2] both point to a baryon-to-photon ratio around $\eta \equiv n_B/n_\gamma \sim 6 \cdot 10^{-10}$ [3], a number that is tiny, non-zero, and within the Standard Model (SM) of particle physics, extended by standard cosmology, impossible to explain from a matter-antimatter symmetric initial state. Due to this limitation of the SM, explaining the origin of this Baryon Asymmetry of the Universe (BAU) dynamically, instead of imposing it as an initial condition, is one of the enduring motivations for physics Beyond the Standard Model (BSM).

Sakharov formulated the three necessary ingredients for any dynamical mechanisms of baryogenesis: 1) Baryon number violation, 2) C and CP violation, and 3) a departure from thermal equilibrium [4]. The Standard Model, remarkably, contains all the necessary ingredients to in principle satisfy those conditions: baryon number is violated non-perturbatively by the electroweak sphaleron process in the early universe [5, 6]; the complex phase of the CKM matrix provides a source of CP violation [3]; and the Electroweak Phase Transition (EWPT), during which the Higgs field spontaneously breaks the $SU(2)_L \times U(1)_Y$ symmetry, offers a natural out-of-equilibrium environment if it proceeds sufficiently violently. This is the scenario of electroweak baryogenesis [7].

For electroweak baryogenesis to be able to satisfy the third Sakharov condition, the EWPT must be a Strong First-Order Electroweak Phase Transition (SFOEWPT): the Higgs field must tunnel through a potential barrier between a symmetric and a broken phase, nucleating bubbles whose expanding walls provide the departure from equilibrium needed for sphaleron processes to imprint a net baryon number before it is washed out again in the broken phase. The main issue with this is that in the SM, non-perturbative lattice studies have shown that the EWPT, for the measured mass of 125 GeV for the Higgs boson [8, 9], is not a phase transition in the thermodynamical sense, but a smooth crossover [10–12].

The Standard Model is therefore not sufficient for electroweak baryogenesis, and we must look to BSM theories to achieve that SFOEWPT. Most of the BSM models considered for this consist on extensions of the standard model’s Higgs-sector, modifying the shape of the effective potential and strengthening the phase transition through tree-level barriers or through strengthened thermal radiative effects.

Among the many proposed extensions, one of the simplest and most extensively studied is the addition of a real scalar field singlet under the SM gauge group, coupled to the Higgs doublet, commonly denoted xSM or RxSM. Despite its minimality, this model can generate a strong first-order EWPT either through tree-level barriers already present at zero temperature, or through thermal loop effects analogous to, but stronger than, those responsible for the (insufficient) SM transition. The RxSM therefore provides an economical and well-motivated physical model in which to study, in a controlled setting, how to transform the phase transition of SM into a SFOEWPT.

First-order electroweak phase transitions are also interesting from a phenomenological standpoint due to the possibility of producing new BSM observables. The collision of bubble walls and the associated bulk fluid motion become a source of a stochastic background of gravitational waves that could be probed by future space-based interferometers such as the Laser Interferometer Space Antenna (LISA) [13, 14]. Another avenue for probing this new physics is at the collider level, LHC and beyond [15]. Since the strength of the phase transition will be controlled by the parameters of our model, BSM searches of the Higgs sector offer a genuine, model-dependent but calculable window onto the thermal history of the EWPT [16].

In order to carry out the necessary computations to be able to relate the phase transition observables to the collider observables, however, one first requires a solid command of the quantum field theory machinery needed to compute the effective potential at both zero and finite temperature. We will need to compute the Coleman-Weinberg one-loop effective potential and its renormalization [17, 18], the extension to finite temperature via the imaginary-time (Matsubara) formalism [19], the resummation of infrared-sensitive daisy diagrams, and the final classification of the resulting phase transitions [6].

Because these techniques are subtle and filled with theoretical dependencies [20], a substantial part of this thesis is devoted to developing them systematically and pedagogically. First, we apply them to simple toy models (a real scalar theory, the Abelian Higgs model) where the physics can be checked against well-known limits, before moving on to the full Standard Model, and finally to the real singlet extension. This is done so that, rather than having to treat the effective-potential formalism as a black box imported from the literature, we can re-derive it in enough detail so that the approximations entering the phenomenological analysis are transparent.

The thesis is organized as follows. Chapter 2 introduces the zero-temperature effective potential from the path-integral definition of the effective action, derives the Coleman-Weinberg potential for a real scalar theory and for the Abelian Higgs model, and discusses renormalization and radiative symmetry breaking, before applying this formalism to the full SM. Chapter 3 extends this formalism to finite temperature, reviewing the Matsubara formalism, the high temperature expansion, and the classification of second- and first-order phase transitions, culminating in a discussion of vacuum trapping, and the physics of electroweak baryogenesis. Chapter 4 applies these tools to the real singlet extension of the Standard Model, deriving the tree-level potential, the tadpole conditions, the field-dependent masses, and the one-loop thermal effective potential, finally concluding with a discussion of the Z_2 symmetric sub-models. Chapter 5 presents the phase transition analysis proper for the Z_2 preserving RxSM: the stability constraints on the parameter space, the high-temperature limit, the classification of one-, two-, and three-step transitions, and a parameter scan comparing analytic and **CosmoTransitions**-based numerical results. Finally, chapter 6 summarizes the results and outlines different future studies needed for a full LHC-oriented phenomenological study of the RxSM model.

2 Zero-Temperature Effective Potential

The potential in a classical field theory allows us to study the vacuum structure of the theory using geometric arguments. The effective potential allows us to extend this geometric argument to a Quantum Field Theory (QFT) [21]. At tree-level the effective potential recovers the classical potential, but at one-loop we can start to include the quantum corrections and therefore extend our analysis to include those quantum effects.

The effective potential was originally developed by Euler, Heisenberg [22] and Schwinger [23], and applied to studies of Spontaneous Symmetry Breaking (SSB) by Goldstone, Salam and S. Weinberg [24]. Calculations of the effective potential at one-loop were performed by Coleman and E. Weinberg [17], where they also introduced the possibility of Radiative Symmetry Breaking (RSB) [18].

In this chapter we will introduce the effective potential for quantum field theories and compute it, up to one-loop level, for multiple theories, allowing us to discuss the concepts of renormalization scheme choices, and RSB.

2.1 Generating Functionals, Effective action and Effective Potential

Let us start by considering a theory that describes a scalar field ϕ with Lagrangian density $\mathcal{L}$, and action

$$S[\phi] = \int d^4x \mathcal{L}(\phi(x)). \quad (2.1)$$

Using the path integral formalism [6, 21, 25, 26], we can write the partition function or the vacuum to vacuum amplitude as

$$Z[J] = \int D\phi e^{iS(\phi) + J\phi}, \quad (2.2)$$

where $J(x)$ is a source function of our field¹, written using the notation

$$J\phi = \int d^4x \phi(x) J(x). \quad (2.3)$$

The partition function (2.2), is also called the generating functional since by taking functional derivatives, w.r.t. $J(x)$, of it we can obtain the n-point Green's functions

$$G^{(n)} = (-i)^n \frac{\delta^n Z[J]}{\delta J(x_1) \dots \delta J(x_n)} \Big|_{J=0}. \quad (2.4)$$

The connected generating functional $W[J]$ is defined in terms of (2.2) as

$$Z[J] = e^{iW[J]} \Rightarrow iW[J] = \ln Z[J]. \quad (2.5)$$

Following the same result of (2.4) [21, 26], the functional derivatives of $W[J]$ give the connected n-point Green's functions

$$G_C^{(n)} = (-i)^n \frac{\delta^n iW[J]}{\delta J(x_1) \dots \delta J(x_n)} \Big|_{J=0}. \quad (2.6)$$

The effective action $\Gamma[\bar{\phi}]$ is defined as the Legendre transform of $W[J]$

$$\Gamma[\bar{\phi}] = W[J] - \int d^4x J\bar{\phi} \quad (2.7)$$

¹ $J(x)$ is an external, in general unphysical, perturbation that we introduce as a bookkeeping device. Turning it on allows us to deform the theory so that its vacuum can support field configurations away from the true, source-free ground state. We will switch it back off at the end, once it has served its purpose.

where $\bar{\phi}$ is the classical [18] or the background [27] field, and it is defined as

$$\bar{\phi}(x) = \frac{\delta W[J]}{\delta J(x)} = \frac{-i}{Z[J]} \frac{\delta Z[J]}{\delta J(x)} = \langle 0 | \phi(x) | 0 \rangle_J, \quad (2.8)$$

which allows us to identify the background field $\bar{\phi}$ as the expectation value of the field ϕ in the presence of an external source $J(x)$ [25, 26].

The effective action is also a generating functional. Its variations in terms of the classical field give the one-particle irreducible (1PI) Green's functions $\Gamma^{(n)}$ [6, 21, 26]. This allows us to write

$$\Gamma[\bar{\phi}] = \sum_{n=0}^{\infty} \frac{1}{n!} \int d^4x_1 \dots d^4x_n \Gamma^{(n)}(x_1, \dots, x_n) \bar{\phi}(x_1) \dots \bar{\phi}(x_n). \quad (2.9)$$

Taking the functional derivative w.r.t. $\bar{\phi}(x)$ we obtain the following relation

$$\begin{aligned} \frac{\delta \Gamma[\bar{\phi}]}{\delta \bar{\phi}(x)} &= \frac{\delta W[J]}{\delta \bar{\phi}(x)} - \int d^4y \frac{\delta J(y)}{\delta \bar{\phi}(x)} \bar{\phi}(y) - \int d^4y J(y) \frac{\delta \bar{\phi}(y)}{\delta \bar{\phi}(x)} \\ &= \int d^4y \frac{\delta J(y)}{\delta \bar{\phi}(x)} \frac{\delta W[J]}{\delta J(y)} - \int d^4y \frac{\delta J(y)}{\delta \bar{\phi}(x)} \bar{\phi}(y) - \int d^4y J(y) \delta^{(4)}(y-x) \\ &= \int d^4y \frac{\delta J(y)}{\delta \bar{\phi}(x)} \bar{\phi}(y) - \int d^4y \frac{\delta J(y)}{\delta \bar{\phi}(x)} \bar{\phi}(y) - J(x) = -J(x), \end{aligned} \quad (2.10)$$

where in the second step we used the functional derivative property $\delta f(y)/\delta f(x) = \delta^{(4)}(y-x)$ [21], and in the last step we used definition (2.8).

Equation (2.10) tells us that setting $J = 0$, i.e. switching off the external sources, we obtain a stationary condition of the effective action,

$$\left. \frac{\delta \Gamma[\bar{\phi}]}{\delta \bar{\phi}(x)} \right|_{J=0} = 0, \quad (2.11)$$

which defines the vacuum of our theory [6]. We are interested in theories where the vacuum is translationally invariant, so as to preserve Poincaré invariance. This forces the background field, i.e. the Vacuum Expectation Value (VEV), to be a constant value

$$\bar{\phi}(x) = \varphi_c. \quad (2.12)$$

This restriction becomes extremely powerful once we expand the effective action as [17, 18]

$$\Gamma[\bar{\phi}] = \int d^4x \left[-V_{\text{eff}}(\bar{\phi}) + \frac{1}{2} Z(\bar{\phi}) (\partial_\mu \bar{\phi})^2 + \dots \right], \quad (2.13)$$

where $V_{\text{eff}}(\bar{\phi})$ is the effective potential, and setting $\bar{\phi}(x) = \varphi_c$ results in²

$$\Gamma[\varphi_c] = - \int d^4x V_{\text{eff}}(\varphi_c). \quad (2.14)$$

This expression combined with (2.11), provides the more familiar way to find the vacuum of our theory

$$\left. \frac{\delta \Gamma[\bar{\phi}]}{\delta \bar{\phi}(x)} \right|_{J=0} = 0 \Rightarrow \left. \frac{\partial V_{\text{eff}}(\varphi_c)}{\partial \varphi_c} \right|_{\varphi_c = \langle \phi \rangle} = 0, \quad (2.15)$$

where $\langle \phi \rangle$ is the VEV of our field ϕ [6, 18, 25].

²where we also have removed an overall factor of space-time volume [6].

Before going into the diagrammatic computation of V_{eff} , it is important to understand the significance of (2.15). The effective potential $V_{\text{eff}}(\varphi_c)$, as we said in the preamble for this chapter, is a quantum corrected version of the potential $V(\phi)$ in our classical field theory. When we want to find the vacuum of a classical field theory we just look for the stationary points of $V(\phi)$, now thanks to (2.15), we know that the procedure is identical after replacing the fields with their background configurations³ and the classical (tree-level) potential with the quantum corrected effective one.

In order to compute the V_{eff} , we introduce (2.14) into (2.9)

$$-\int d^4x V_{\text{eff}}(\varphi_c) = \sum_{n=0}^{\infty} \frac{\varphi_c^n}{n!} \int d^4x_1 \dots d^4x_n \Gamma^{(n)}(x_1, \dots, x_n), \quad (2.16)$$

which allows us to rewrite the effective potential in terms of the 1PI Green's functions.

Expanding the effective action in momentum space

$$\int d^4x_1 \dots d^4x_n e^{i(p_1x_1 \dots p_nx_n)} \Gamma^{(n)}(x_1, \dots, x_n) = \tilde{\Gamma}^{(n)}(p_1, \dots, p_n) (2\pi)^4 \delta^{(4)}(p_1 + \dots + p_n), \quad (2.17)$$

where if we set $p_i = 0 \quad \forall p_i$, we obtain the same integral of (2.16), thus allowing us to write

$$\begin{aligned} V_{\text{eff}}(\varphi_c) \int d^4x &= - \sum_{n=0}^{\infty} \frac{\varphi_c^n}{n!} \tilde{\Gamma}^{(n)}(p_i = 0) (2\pi)^4 \delta^{(4)}(0) \\ V_{\text{eff}}(\varphi_c) &= - \sum_{n=0}^{\infty} \frac{\varphi_c^n}{n!} \tilde{\Gamma}^{(n)}(p_i = 0). \end{aligned} \quad (2.18)$$

This equation reads as follows: one can compute the effective potential by summing the n-point 1PI Green's functions. If we take a loop expansion of said green's functions we can perturbatively calculate V_{eff} by computing the 1PI diagrams with 0 external momentum [6, 17, 18, 25].

2.2 Effective Potential of a Real Scalar Field Theory

Let us now compute the one-loop effective potential for a real scalar field theory with a Lagrangian density

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{4!} \phi^4. \quad (2.19)$$

We can expand our dynamical field $\phi(x)$ in its background φ and perturbation $\tilde{\phi}(x)$ components. By doing that, we find that the tree-level effective potential $V_0(\varphi)$ is

$$V_0(\varphi) = \frac{m^2}{2} \varphi^2 + \frac{\lambda}{4!} \varphi^4. \quad (2.20)$$

Imposing the model to have a SSB of its Z_2 ($\phi \rightarrow -\phi$) symmetry, the VEV of the field will be located at

$$\langle \phi \rangle \equiv u = \pm \sqrt{-\frac{6m^2}{\lambda}}, \quad \Longleftrightarrow \quad -m^2 > 0. \quad (2.21)$$

The field-dependent mass is found by

$$m^2(\varphi) \equiv \frac{\partial^2 V_0}{\partial \phi^2} = \frac{\lambda \varphi^2}{2} + m^2, \quad (2.22)$$

while the physical mass of the ϕ field is obtained by evaluating the field-dependent mass at the VEV (2.21)

$$m^2(u) \equiv m_\phi^2 = \frac{\lambda u^2}{3}. \quad (2.23)$$

³This gives rise to two equivalent notations: an explicit notation $\phi(x) = \varphi + \tilde{\phi}(x)$, where $\langle \tilde{\phi}(x) \rangle = 0$, or an implicit notation $\phi(x) = \tilde{\phi}(x)$, where $\langle \tilde{\phi}(x) \rangle = \varphi$. Throughout this thesis we will use the first.

2.2.1 Coleman-Weinberg Effective Potential

In order to compute the effective potential of this theory, we take a loop expansion of (2.18). By truncating this expansion to one-loop, the effective potential becomes

$$V_{\text{eff}}(\varphi) = V_0(\varphi) + V_{\text{CW}}(\varphi), \quad (2.24)$$

where V_0 is the tree-level potential (2.20) [6], and V_{CW} , the Coleman-Weinberg (CW) potential, is computed as the sum of all 1PI diagrams with a single loop and zero external momenta.

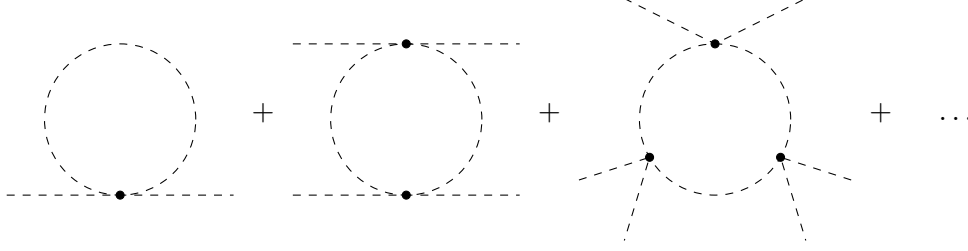


Figure 1: 1PI diagrams contributing to the one-loop effective potential the model (2.19).

In figure 1 we have drawn the 1PI with lowest number of external legs. From this we see that the general expression for a diagram with $2n$ external legs (and n propagators and vertices) is

$$\tilde{\Gamma}_1^{(2n)}(p_i = 0) = -i(-i\lambda)^n \int \frac{d^4k}{(2\pi)^4} \left(\frac{i}{k^2 - m^2 + i\epsilon} \right)^n \frac{(2n)!}{2^n 2n}, \quad (2.25)$$

where $\frac{(2n)!}{2^n 2n}$ is the symmetry factor of this diagrams [18, 25].

Therefore, using (2.18) we find that the one-loop, or CW, effective potential can be written as

$$\begin{aligned} V_{\text{CW}}(\varphi) &= \sum_{n=1}^{\infty} \frac{\varphi^{2n}}{(2n)!} i(-i)^n \int \frac{d^4k}{(2\pi)^4} \left(\frac{i\lambda}{k^2 - m^2 + i\epsilon} \right)^n \frac{(2n)!}{2^n 2n} \\ &= \frac{i}{2} \sum_{n=1}^{\infty} \frac{1}{n} \int \frac{d^4k}{(2\pi)^4} \left(\frac{\frac{\lambda\varphi^2}{2}}{k^2 - m^2 + i\epsilon} \right)^n = -\frac{i}{2} \int \frac{d^4k}{(2\pi)^4} \ln \left(1 - \frac{\frac{\lambda\varphi^2}{2}}{k^2 - m^2 + i\epsilon} \right), \end{aligned} \quad (2.26)$$

where, in the second line, we used the identity $\sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln(1-x)$. After a Wick rotation ($k^0 \rightarrow ik_E^0$), (2.26) becomes

$$V_{\text{CW}}(\varphi) = \frac{1}{2} \int \frac{d^4k_E}{(2\pi)^4} \ln \left(1 + \frac{\frac{\lambda\varphi^2}{2}}{k_E^2 + m^2} \right). \quad (2.27)$$

This integral has an ultraviolet divergence that we will need to tame by using dimensional regularization.

First, we use the logarithm properties to express (2.27) in a nicer form

$$V_{\text{CW}}(\varphi) = \frac{1}{2} \int \frac{d^4k_E}{(2\pi)^4} \ln \left(k_E^2 + m^2 + \frac{\lambda\varphi^2}{2} \right) - \frac{1}{2} \int \frac{d^4k_E}{(2\pi)^4} \ln (k_E^2 + m^2). \quad (2.28)$$

Both of the integral diverge, but the second one does not depend on φ . This is a constant infinity in a potential and hence, can be dropped. By doing this, the CW effective potential becomes

$$V_{\text{CW}} = \frac{1}{2} \int \frac{d^4k_E}{(2\pi)^4} \ln \left(k_E^2 + m^2 + \frac{\lambda\varphi^2}{2} \right) = \frac{1}{2} \int \frac{d^4k_E}{(2\pi)^4} \ln (k_E^2 + m^2(\varphi)), \quad (2.29)$$

where we have used the field dependent mass (2.22).

In order to renormalize the UV divergence of (2.29), we introduce the following counter terms for the mass and the coupling constants

$$V_{\text{eff}}(\varphi) = \frac{m^2 + \delta\tilde{m}^2}{2}\varphi^2 + \frac{\lambda + \delta\tilde{\lambda}}{4!}\varphi^4 + \frac{1}{2} \int \frac{d^4 k_E}{(2\pi)^4} \ln(k_E^2 + m^2(\varphi)). \quad (2.30)$$

We will now use dimensional regularization for the integral term by changing dimensions from $4 \rightarrow d$

$$V_{\text{CW}}(\varphi) \rightarrow V_1^R(\varphi) = \frac{1}{2} (\mu^2)^{2-\frac{d}{2}} \int \frac{d^d k_E}{(2\pi)^d} \ln(k_E^2 + m^2(\varphi)), \quad (2.31)$$

where we introduced the renormalization scale μ^2 .

This integral can be solved by differentiating under the integral:

$$\begin{aligned} \frac{dV_1^R(\varphi)}{dm^2(\varphi)} &= \frac{1}{2} (\mu^2)^{2-\frac{d}{2}} \int \frac{d^d k_E}{(2\pi)^d} \frac{\partial \ln(k_E^2 + m^2(\varphi))}{\partial m^2(\varphi)} = \frac{1}{2} (\mu^2)^{2-\frac{d}{2}} \int \frac{d^d k_E}{(2\pi)^d} \frac{1}{k_E^2 + m^2(\varphi)} \\ &= \frac{1}{2} (\mu^2)^{2-\frac{d}{2}} \frac{(m^2(\varphi))^{\frac{d}{2}-1}}{(4\pi)^{d/2}} \Gamma\left(1 - \frac{d}{2}\right), \end{aligned} \quad (2.32)$$

integrating back

$$\begin{aligned} V_1^R(\varphi) &= \int dm^2(\varphi) \frac{dV_1^R(\varphi)}{dm^2(\varphi)} = \frac{1}{2} \frac{(\mu^2)^{2-\frac{d}{2}}}{(4\pi)^{d/2}} \Gamma\left(1 - \frac{d}{2}\right) \int dm^2(\varphi) (m^2(\varphi))^{\frac{d}{2}-1} \\ &= -\frac{1}{32\pi^2} \frac{\Gamma\left(2 - \frac{d}{2}\right)}{\frac{d}{2} \left(\frac{d}{2} - 1\right)} \left(\frac{m^2(\varphi)}{4\pi\mu^2}\right)^{\frac{d}{2}-2} m^4(\varphi). \end{aligned} \quad (2.33)$$

Expanding (2.33) in powers of $2 - d/2$ [6]

$$\begin{aligned} V_1^R(\varphi) &= \frac{m^4(\varphi)}{64\pi^2} \left[-\left(\frac{1}{2 - \frac{d}{2}} - \gamma_E + \ln(4\pi)\right) \right. \\ &\quad \left. + \ln\left(\frac{m^2(\varphi)}{\mu^2}\right) - \frac{3}{2} + \mathcal{O}\left(\frac{d}{2} - 2\right) \right]. \end{aligned} \quad (2.34)$$

In the $\overline{MS}$ renormalization scheme we subtract the divergent term plus $\gamma_E - \ln(4\pi)$, i.e. the first line of (2.34), using the counter terms introduced in (2.30)

$$\frac{\delta\tilde{m}^2}{2}\varphi^2 + \frac{\delta\tilde{\lambda}}{4!}\varphi^4 = \frac{m^4(\varphi)}{64\pi^2} \left(\frac{1}{2 - \frac{d}{2}} - \gamma_E + \ln(4\pi) \right). \quad (2.35)$$

Finally, the resulting term is the CW effective potential, free from UV divergences,

$$V_{\text{CW}}(\varphi) = \frac{m^4(\varphi)}{64\pi^2} \left[\ln\left(\frac{m^2(\varphi)}{\mu^2}\right) - \frac{3}{2} \right]. \quad (2.36)$$

This expression can be generalized to include more complicated models such as those with multiple particles, multiple types of particles such as scalars, Weyl and Dirac fermions, and vector-bosons [6]. The general expression of the CW in the $\overline{MS}$ scheme is

$$V_{\text{CW}_i}(\varphi) = (-1)^{2s_i} \frac{N_i}{64\pi^2} \text{Tr} M_i^4(\varphi) \left[\ln\left(\frac{M_i^2(\varphi)}{\mu^2}\right) - C_i \right], \quad (2.37)$$

where i goes over the 3 types of particles (scalars, fermions, and gauge bosons), s_i is the spin of the particle and N_i its number of degrees of freedom, $M_i(\varphi)$ is the field-dependent mass matrix⁴, and C_i is a renormalization factor that its $3/2$ for scalars and fermions, and $5/6$ for gauge bosons.

If we substitute back the expression for the field-dependent mass (2.22) into (2.36), we find that the effective potential for our theory at one-loop is

$$V_{\text{eff}}(\varphi, \mu) = \frac{1}{2}m^2\varphi^2 + \frac{1}{4!}\lambda\varphi^4 + \frac{1}{64\pi^2} \left(\frac{\lambda\varphi^2}{2} + m^2 \right)^2 \left[\ln \left(\frac{\frac{\lambda\varphi^2}{2} + m^2}{\mu^2} \right) - \frac{3}{2} \right]. \quad (2.38)$$

The dependence of (2.38) on the renormalization scale is not a peculiarity of the $\overline{MS}$ scheme itself, but a generic feature of any fixed-order perturbative calculation: physical observables are only exactly μ -independent once resummed to all orders, and at finite order a residual μ -dependence remains as an artifact of the truncation. One can mitigate its numerical impact by choosing μ close to the characteristic energy scale of the theory (in our scalar case, m_ϕ or u). A more systematic treatment would track this dependence explicitly via the renormalization-group-improved effective potential [6]. Here, instead, we opt for an on-shell (OS) renormalization scheme [28].

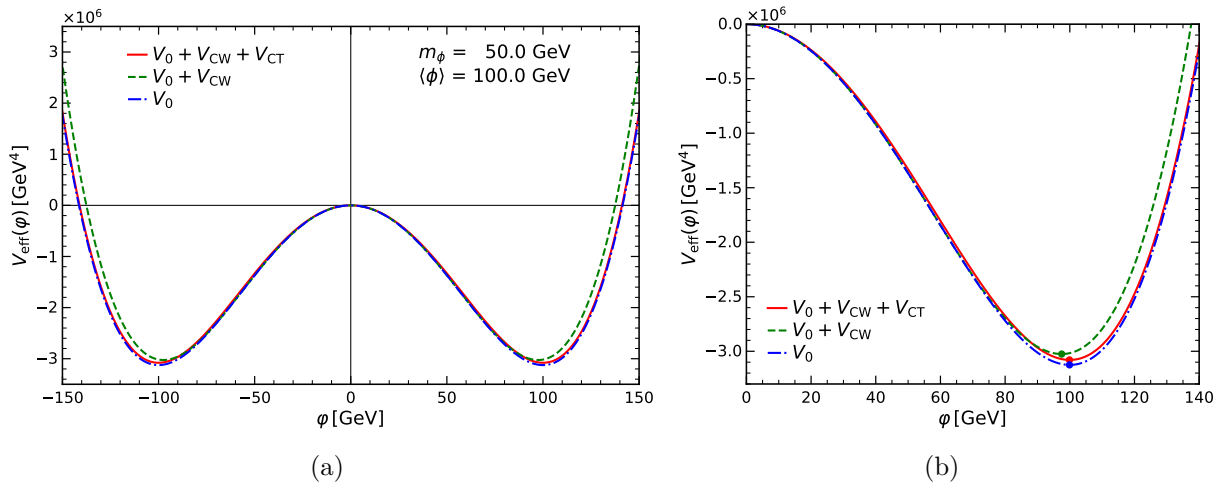


Figure 2: (a) Effective Potential of the real scalar field theory (2.19). The blue dash-dotted line represents the tree-level effective potential (2.20). The green dashed line shows the one-loop effective potential (2.38). And the red line is the same but with V_{CT} to preserve the tree-level values for $\langle\phi\rangle$, and m_ϕ (2.40). (b) Same as (a) but zoomed in on the VEVs.

2.2.2 Counter Term Effective Potential

As we can see from figure 2b, radiative corrections included by the quantum effects of the CW potential shift the vacuum and masses of our theory away from their tree-level values. To perform an efficient phenomenological scan through the parameter space we would like to use those physical quantities instead of the Lagrangian parameters. In order to do so, we will shift from the $\overline{MS}$ scheme to an OS scheme via the addition of a finite counter term effective potential [28–30]. This means that our full effective potential now becomes

$$V_{\text{eff}} = V_0 + V_{CW} + V_{CT}. \quad (2.39)$$

⁴Note that the expression is the trace of the matrix M_i , this ensures that the result is independent of the choice of basis in the case of mixing.

For our real scalar field theory, the counter term potential is

$$V_{\text{CT}}(\varphi) = \frac{1}{2}\delta m^2\varphi^2 + \frac{1}{4!}\delta\lambda\varphi^4 + \delta V_0, \quad (2.40)$$

where the parameters δm^2 , $\delta\lambda$, and δV_0 are determined by our renormalization conditions [31]. We want them to preserve the tree-level relations found previously, therefore they will be:

1. The position of the one-loop minima must coincide with the tree-level value u

$$\left. \frac{\partial (V_{\text{CW}} + V_{\text{CT}})}{\partial \varphi} \right|_u = 0 \Rightarrow V'_{\text{CW}}(u) + \delta m^2 u + \frac{\delta\lambda u^3}{6} = 0, \quad (2.41)$$

where $V'_{\text{CW}}(u)$ is a shorthand notation for the first derivative of the CW potential evaluated at u .

2. The second derivative of the field at u (m_ϕ) must coincide with the tree-level value

$$\left. \frac{\partial^2 (V_{\text{CW}} + V_{\text{CT}})}{\partial \varphi^2} \right|_u = 0 \Rightarrow V''_{\text{CW}}(u) + \delta m^2 + \frac{\delta\lambda u^2}{2} = 0. \quad (2.42)$$

3. The one-loop value of the potential at u is the same as the tree-level value

$$(V_{\text{CW}} + V_{\text{CT}})|_u = 0 \Rightarrow V_{\text{CW}} + \delta V_0 + \frac{\delta\lambda u^4}{24} + \frac{\delta m^2 u^2}{2} = 0. \quad (2.43)$$

Solving the system of equations formed by these conditions, we arrive at the following values for the counter terms

$$\begin{aligned} \delta m^2 &= \frac{3V'_{\text{CW}}(u) - uV''_{\text{CW}}(u)}{2u}, \quad \delta\lambda = \frac{3(V'_{\text{CW}}(u) - uV''_{\text{CW}}(u))}{u^3}, \\ \delta V_0 &= \frac{u(5V'_{\text{CW}}(u) - uV''_{\text{CW}}(u))}{8} - V_{\text{CW}}(0). \end{aligned} \quad (2.44)$$

2.2.3 Radiative Symmetry Breaking

Through this whole section on the real scalar field theory we have assumed that the Z_2 symmetry is spontaneously broken by a tachyonic mass term ($m^2 < 0$), but one of the main insights of Coleman's and Weinberg's work was to show that the tachyonic mass term is not necessary since one can design a model to have RSB, i.e. breaking of the symmetry due to purely quantum effects, by setting $m^2 = 0$.

Setting $m^2 = 0$ we get that our one-loop effective potential⁵ becomes

$$V_{\text{eff}}(\varphi) = \frac{\lambda\varphi^4}{24} + \frac{\lambda^2\varphi^4}{256\pi^2} \left(\ln \left(\frac{\lambda\varphi^2}{2\mu^2} \right) - \frac{3}{2} \right). \quad (2.45)$$

Differentiating w.r.t. the field φ , we find that the non-zero solutions for the minima are

$$\left. \frac{\lambda\varphi^3}{6} + \frac{\lambda^2\varphi^3}{128\pi^2} + \frac{\lambda^2\varphi^3 \left(\ln \left(\frac{\lambda\varphi^2}{2\mu^2} \right) - \frac{3}{2} \right)}{64\pi^2} \right|_{\langle\varphi\rangle} = 0 \Rightarrow \frac{\langle\varphi\rangle^2}{\mu^2} = \frac{2}{\lambda} e^{1 - \frac{32\pi^2}{3\lambda}}. \quad (2.46)$$

Setting the coupling λ to its perturbative ceiling (i.e. $\lambda = 4\pi$) we get that $\langle\varphi\rangle/\mu = \pm 9.97 \cdot 10^{-3}$. This is a small value in a best-case scenario, as we decrease λ the VEVs decrease rapidly as

⁵We don't need to include V_{CT} since there are no tree-level observables to preserve.

shown in figure 3b. The reason for this apparent need for breaking perturbation theory comes from the fact that we are imposing that the quantum corrections of the effective potential cancel out the tree-level effects [21].

The main reason for this model to fail is its simplicity since the only free parameter when we consider the scale invariant theory (no mass term) is the quartic coupling λ . As we can see from (2.46), if we perform a naive power counting we see that in order for that equation to be true we expect $\lambda \sim \lambda^2/16\pi^2$, something that already warns us about the need for non-perturbative values of λ in this scenario.

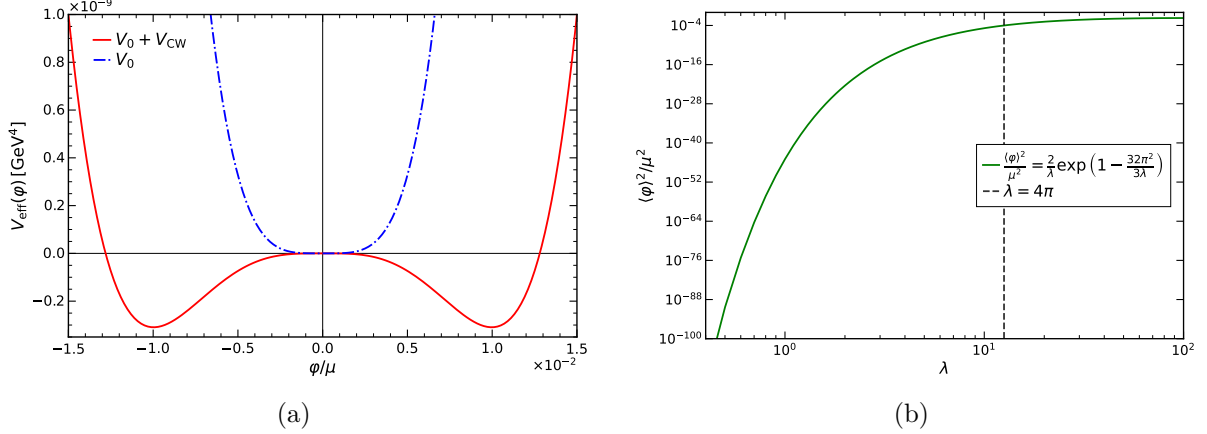


Figure 3: (a) Effective potential for a scale invariant real scalar theory. The red(blue) lines shows the effective potential (2.45) up to one-loop(tree-level). (b) Values of the $\langle\phi\rangle^2/\mu^2$ that satisfy the minima condition (2.46), as a function of the coupling λ .

2.3 Effective Potential of an Abelian Higgs Model

We will now consider a more complete abelian Higgs model, i.e. a spontaneously broken $U(1)$ symmetric gauge theory, with gauge boson A , coupled to a complex scalar field Φ . This model is described by the Lagrangian density

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\Phi)^*(D^\mu\Phi) - V(\Phi), \quad (2.47)$$

with scalar potential

$$V(\Phi) = -\mu^2|\Phi|^2 + \lambda|\Phi|^4, \quad (2.48)$$

where $F_{\mu\nu}$ is the field strength tensor, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, $D_\mu\Phi$ is the covariant derivative given by $D_\mu\Phi = \partial_\mu\Phi + igA_\mu\Phi$.

We can expand the field as follows

$$\Phi(x) = \frac{1}{\sqrt{2}}(\varphi + \phi_1(x) + i\phi_2(x)), \quad (2.49)$$

where $\phi_{1(2)}$ are the real(imaginary) components of the excitations, and φ is the background field.

Inserting this decomposition into the scalar potential (2.48) we get that the tree-level effective potential is

$$V_0(\varphi) = -\frac{1}{2}\mu^2\varphi^2 + \frac{1}{4}\lambda\varphi^4. \quad (2.50)$$

Imposing that the $U(1)$ symmetry is spontaneously broken, we find that the VEV of the field is located at

$$\left.\frac{\partial V_0}{\partial \varphi}\right|_{\langle\Phi\rangle} = 0 \Rightarrow \langle\Phi\rangle \equiv v = \frac{\mu}{\sqrt{\lambda}}. \quad (2.51)$$

The field-dependent masses of the scalar fields are found to be

$$m_{\phi_1}^2(\varphi) = 3\lambda\varphi^2 - \mu^2, \quad m_{\phi_2}^2(\varphi) = \lambda\varphi^2 - \mu^2, \quad (2.52)$$

while their physical masses are $m_{\phi_1}^2(v) \equiv m_{\phi_1}^2 = 2v^2\lambda$, and $m_{\phi_2}^2(v) \equiv m_{\phi_2}^2 = 0$, respectively.

In order to find the field-dependent masses of the gauge bosons we must look at the kinetic part, mainly the term proportional to $-g^2 A^2 \varphi^2$, this results in

$$m_A^2(\varphi) = g^2 \varphi^2, \quad \Rightarrow m_A^2 = g^2 v^2. \quad (2.53)$$

This massive gauge boson is a well-known consequence of having a spontaneously broken gauge symmetry [21, 27], and it is commonly referred to as the Higgs mechanism [32–34].

2.3.1 Review of the Abelian Higgs Mechanism

We again decompose the complex scalar field, now around its vacuum expectation value v , but using a non-linear (radial) parametrization

$$\Phi(x) = \frac{1}{\sqrt{2}}(v + \rho(x))e^{i\chi(x)/v}, \quad (2.54)$$

where $\rho(x)$ is the physical excitation of the Higgs field around the vev, and $\chi(x)$ is the goldstone boson, whose normalization by v ensures it is canonically normalized.

Under the local $U(1)$ transformation $\Phi(x) \rightarrow e^{i\alpha(x)}\Phi(x)$, the decomposition (2.54) shows that

$$\rho(x) \rightarrow \rho(x), \quad \chi(x) \rightarrow \chi(x) + v\alpha(x), \quad (2.55)$$

i.e. ρ is gauge invariant while χ shifts by an amount proportional to the gauge parameter, the defining feature of a goldstone boson [27].

Inserting (2.54) into the covariant derivative $D_\mu\Phi = \partial_\mu\Phi + igA_\mu\Phi$ we find

$$D_\mu\Phi = \frac{1}{\sqrt{2}} \left[\partial_\mu\rho + i \frac{v+\rho}{v} (\partial_\mu\chi + gvA_\mu) \right] e^{i\chi/v}, \quad (2.56)$$

so that the kinetic term becomes

$$|D_\mu\Phi|^2 = \frac{1}{2}\partial_\mu\rho\partial^\mu\rho + \frac{1}{2}\frac{(v+\rho)^2}{v^2}(\partial_\mu\chi + gvA_\mu)^2. \quad (2.57)$$

Expanding around $\rho = 0$ and keeping only the terms bilinear in the fields, we obtain

$$|D_\mu\Phi|_{\text{bilinear}}^2 = \frac{1}{2}\partial_\mu\rho\partial^\mu\rho + \frac{1}{2}\partial_\mu\chi\partial^\mu\chi + \frac{1}{2}g^2v^2A_\mu A^\mu + gvA^\mu\partial_\mu\chi, \quad (2.58)$$

where we recover the gauge boson mass $m_A^2 = g^2v^2$ found above, together with a bilinear *mixing* term $gvA^\mu\partial_\mu\chi$ between the gauge field and the goldstone boson. This term signals that A_μ and χ are not mass eigenstates, and that we cannot read off a diagonal propagator for the gauge field without first specifying how this mixing is treated.

Unitary gauge. Since χ shifts as $\chi \rightarrow \chi + v\alpha(x)$, (2.55), we may choose the gauge parameter $\alpha(x) = -\chi(x)/v$ to remove χ from the theory altogether,

$$\Phi(x) \rightarrow \frac{1}{\sqrt{2}}(v + \rho(x)), \quad A_\mu(x) \rightarrow A_\mu(x) + \frac{1}{gv}\partial_\mu\chi(x). \quad (2.59)$$

In this, the unitary gauge, the goldstone field disappears entirely from (2.58), leaving

$$\mathcal{L}_{\text{unitary}} \supset -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}g^2v^2A_\mu A^\mu + \frac{1}{2}\partial_\mu\rho\partial^\mu\rho - V(\rho), \quad (2.60)$$

which manifestly describes a massive gauge boson A_μ and a real scalar ρ , with the mixing term of (2.58) completely absent. While physically transparent, calculations in unitary gauge are complicated in practice by the structure of the massive vector propagator, $\propto (g_{\mu\nu} - k_\mu k_\nu/m_A^2)/(k^2 - m_A^2)$, whose bad high-energy behavior obscures the renormalizability of the theory order by order in perturbation theory [21, 27].

R_ξ gauge. A more practical way of dealing with the mixing term in (2.58) is to add a gauge-fixing term that cancels it directly, rather than eliminating χ from the theory. We choose

$$\mathcal{L}_{gf} = -\frac{1}{2\xi}(\partial^\mu A_\mu - \xi g v \chi)^2, \quad (2.61)$$

with ξ a free gauge parameter [27]. Expanding (2.61),

$$\mathcal{L}_{gf} = -\frac{1}{2\xi}(\partial^\mu A_\mu)^2 + g v \chi \partial^\mu A_\mu - \frac{1}{2}\xi g^2 v^2 \chi^2, \quad (2.62)$$

and integrating the second term by parts, $g v \chi \partial^\mu A_\mu = -g v A^\mu \partial_\mu \chi$ up to a total derivative, so this term exactly cancels the mixing term in (2.58). Adding (2.61) to (2.58), the bilinear Lagrangian becomes

$$\begin{aligned} \mathcal{L}_{\text{bilinear}} = & -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial^\mu A_\mu)^2 + \frac{1}{2}g^2v^2A_\mu A^\mu \\ & + \frac{1}{2}\partial_\mu\rho\partial^\mu\rho + \frac{1}{2}\partial_\mu\chi\partial^\mu\chi - \frac{1}{2}\xi g^2v^2\chi^2, \end{aligned} \quad (2.63)$$

which is now fully diagonal: the mixing has disappeared, at the price of introducing a gauge-dependent mass for the goldstone boson, $m_\chi^2 = \xi m_A^2 = \xi g^2 v^2$. These are the so-called R_ξ gauges, and they interpolate between two familiar limits: as $\xi \rightarrow \infty$ the goldstone mass diverges and χ decouples, recovering the unitary-gauge result (2.60); at $\xi = 1$ (the Feynman gauge) both A_μ and χ acquire simple, Feynman-like propagators, which greatly simplifies loop computations; while $\xi = 0$ gives the Landau gauge. Since χ is manifestly gauge-dependent, as already seen from its shift under gauge transformations in (2.55), it cannot correspond to a physical degree of freedom for any choice of ξ ; it is, however, an indispensable bookkeeping device that keeps the theory manifestly renormalizable order by order in perturbation theory [21, 27].

2.3.2 One-Loop Effective Potential

Using the general expression⁶ for the CW (2.37), and substituting in the field-dependent masses we found in equations (2.52) and (2.53), we get that the full one-loop effective potential becomes

$$\begin{aligned} V_{\text{eff}}(\varphi) = & -\frac{1}{2}\mu^2\varphi^2 + \frac{1}{4}\lambda\varphi^4 - \frac{1}{2}\delta\mu^2\varphi^2 + \frac{1}{4}\delta\lambda\varphi^4 + \delta V_0 \\ & + \frac{3g^4\varphi^4\left(\ln\left(\frac{g^2\varphi^2}{Q^2}\right) - \frac{5}{6}\right)}{64\pi^2} + \frac{(\lambda\varphi^2 - \mu^2)^2\left(\ln\left(\frac{\lambda\varphi^2 - \mu^2}{Q^2}\right) - \frac{3}{2}\right)}{64\pi^2} \\ & + \frac{(3\lambda\varphi^2 - \mu^2)^2\left(\ln\left(\frac{3\lambda\varphi^2 - \mu^2}{Q^2}\right) - \frac{3}{2}\right)}{64\pi^2}, \end{aligned} \quad (2.64)$$

Where Q is the renormalization scale, and the finite counter-terms $\delta\mu^2, \delta\lambda, \delta V_0$, are determined by the following $\overline{\text{OS}}$ renormalization conditions [31]:

⁶We will always work in the Landau Gauge so as to avoid contribution from the ghost fields [6]. Although in this model we do not have that issue, this becomes relevant when working on theories like the SM.

1. The position of the one-loop minima must coincide with the tree-level value v

$$\left. \frac{\partial (V_{\text{CW}} + V_{\text{CT}})}{\partial \varphi} \right|_v = 0 \Rightarrow V'_{\text{CW}}(v) + \delta\lambda v^3 - \delta\mu^2 v = 0. \quad (2.65)$$

2. The second derivative of the field at v must coincide with the tree-level value

$$\left. \frac{\partial^2 (V_{\text{CW}} + V_{\text{CT}})}{\partial \varphi^2} \right|_v = 0 \Rightarrow V''_{\text{CW}}(v) + 3\delta\lambda v^2 - \delta\mu^2 = 0. \quad (2.66)$$

3. The one-loop value of the potential at v is the same as the tree-level value

$$(V_{\text{CW}} + V_{\text{CT}})|_v = 0 \Rightarrow V_{\text{CW}}(v) + \delta V_0 + \frac{\delta\lambda v^4}{4} - \frac{\delta\mu^2 v^2}{2} = 0. \quad (2.67)$$

The counter terms that satisfy these conditions are

$$\begin{aligned} \delta\mu^2 &= \frac{3V'_{\text{CW}}(v) - vV''_{\text{CW}}(v)}{2v}, \quad \delta\lambda = \frac{V'_{\text{CW}}(v) - vV''_{\text{CW}}(v)}{2v^3}, \\ \delta V_0 &= \frac{5vV'_{\text{CW}}(v) - v^2V''_{\text{CW}}(v)}{8} - V_{\text{CW}}(v). \end{aligned} \quad (2.68)$$

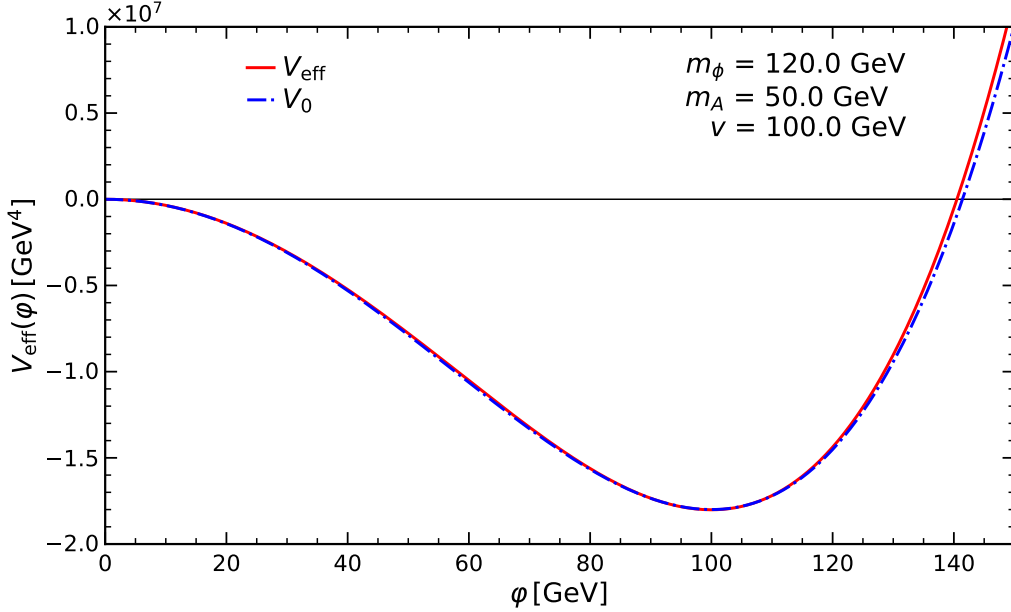


Figure 4: Effective potential for an example set of parameters, for the abelian Higgs model (2.47). The red solid line shows the full one-loop effective potential as described by (2.64), while the blue line shows only the tree-level potential (2.50).

2.3.3 Radiative Symmetry Breaking

Let us now focus on a version of the abelian Higgs model but with the symmetry being broken by radiative effects. Since we are interested in the perturbative limit, in order to allow for RSB we must set the mass term ($\propto \mu^2$) in (2.50) to 0

$$V_0(\varphi) = \frac{1}{4}\lambda\varphi^4. \quad (2.69)$$

The, tree-level, field dependent masses, following the procedure as in the general model, are

$$m_{\phi_1}^2(\varphi) = 3\lambda\varphi^2, \quad m_{\phi_2}^2(\varphi) = \lambda\varphi^2, \quad m_A^2(\varphi) = g^2\varphi^2, \quad (2.70)$$

and using (2.37) we find that the one-loop effective potential becomes

$$V_{\text{eff}}(\varphi) = \frac{\lambda\varphi^4}{4} + \frac{3g^4\varphi^4}{64\pi^2} \left(\ln \left(\frac{g^2\varphi^2}{Q^2} \right) - \frac{5}{6} \right) \quad (2.71)$$

$$+ \frac{\lambda^2\varphi^4}{64\pi^2} \left(\ln \left(\frac{\lambda\varphi^2}{Q^2} \right) - \frac{3}{2} \right) + \frac{9\lambda^2\varphi^4}{64\pi^2} \left(\ln \left(\frac{3\lambda\varphi^2}{Q^2} \right) - \frac{3}{2} \right). \quad (2.72)$$

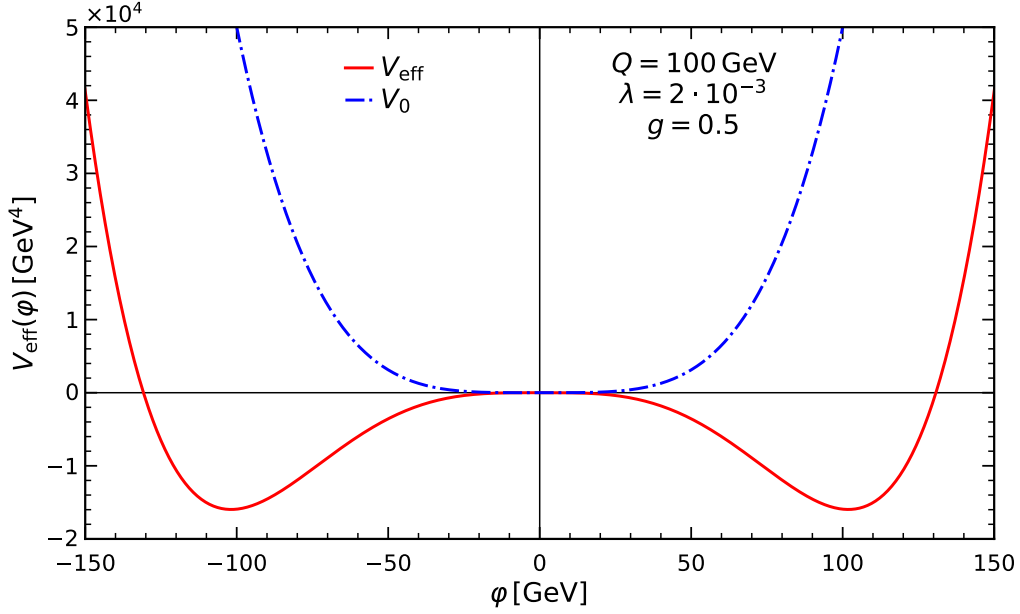


Figure 5: Effective Potential of an abelian Higgs model with radiative symmetry breaking. The blue line shows the tree-level effective potential (2.69), while the red one shows the one-loop effective potential (2.71).

Differentiating (2.71) w.r.t φ the loop-corrected effective potential we arrive at the following equation for finding the VEV

$$\begin{aligned} & -g^4 + 3g^4 \ln \left(\frac{g^2\varphi^2}{Q^2} \right) - 10\lambda^2 + 9\lambda^2 \ln(3) + 16\pi^2\lambda + 10\lambda^2 \ln \left(\frac{\lambda\varphi^2}{Q^2} \right) \Big|_{\varphi=v} = 0, \\ \Rightarrow \frac{v^2}{Q^2} &= \exp \left[\frac{g^4 + 10\lambda^2 - 16\pi^2\lambda - 3g^4 \ln(g^2) - 9\lambda^2 \ln(3) - 10\lambda^2 \ln(\lambda)}{3g^4 + 10\lambda^2} \right]. \end{aligned} \quad (2.73)$$

As we can see in figure 6, thanks to the fact that our model now contains two different parameters we can have acceptable values of v^2/Q^2 while keeping the perturbativity of our theory. As we can see in 6a, choosing $v^2/Q^2 = 1$, we can find a whole subspace of values for our parameters while keeping them both in their perturbative limit.

This perturbative validity is apparent when we look at (2.73), performing a naive power counting, we now get that the estimates for our parameters is more or less $\lambda \sim g^4/16\pi^2$. Comparing this to the result of the real scalar theory, it is apparent that introducing more parameters to our model has loosened the constraints on λ .

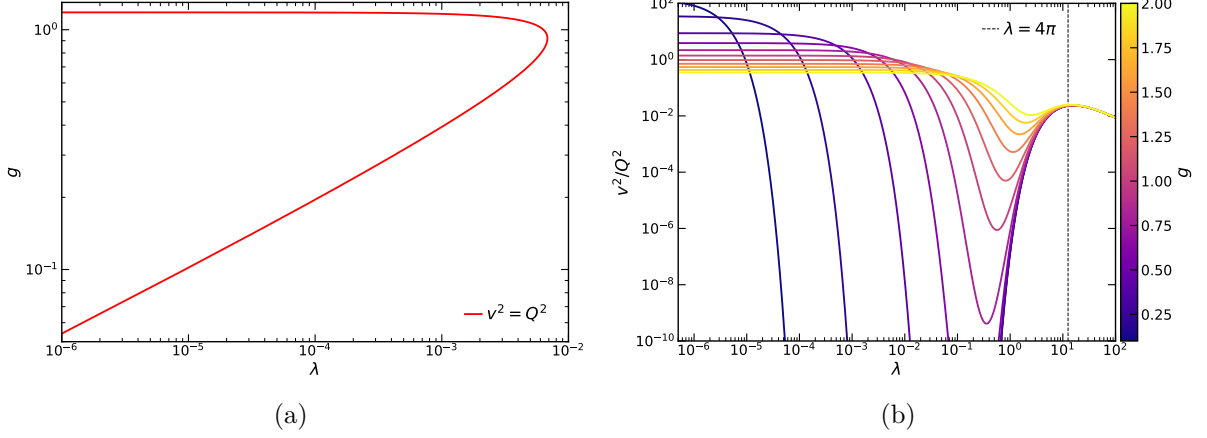


Figure 6: (a) Vales of λ and g that result in $v^2/Q^2 = 1$, following (2.73). (b) Values of v^2/Q^2 as a function of the quartic coupling λ , as predicted by (2.73). The lines are color coded according to their values of the gauge coupling g .

2.4 Effective Potential of the Standard Model

We will now study the effective potential of the SM. The standard model is a gauge theory with symmetry groups $SU(3)_C \times SU(2)_L \times U(1)_Y$. The Higgs field, H , is a doublet of $SU(2)_L$, with hypercharge $Y = 1$ [21, 27]. The field can be decomposed into its degrees of freedom as

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} \chi_1(x) + i\chi_2(x) \\ \varphi + h(x) + i\chi_3(x) \end{pmatrix} = \begin{pmatrix} G^+(x) \\ \frac{\varphi + h(x) + iG^0(x)}{\sqrt{2}} \end{pmatrix}, \quad (2.74)$$

where $G^{\pm(0)}$ are the three would-be goldstone bosons, $h(x)$ is the massive Higgs boson [35], and φ is the background field.

In order to induce SSB of the Electroweak (EW) sector, the SM is built with the following scalar potential

$$V(H) = -\mu^2 H^\dagger H + \lambda (H^\dagger H)^2. \quad (2.75)$$

From this we find that the tree-level effective potential is

$$V_0(\varphi) = -\frac{1}{2}\mu^2\varphi^2 + \frac{1}{4}\lambda\varphi^4. \quad (2.76)$$

Differentiating this we find that the VEV of the Higgs field will be located at

$$\lambda\varphi^3 - \mu^2\varphi \Big|_{\varphi=v} = 0 \Rightarrow v = \frac{\mu}{\sqrt{\lambda}}. \quad (2.77)$$

The field dependent masses of the SM are collected alongside their degrees of freedom in table 1. The masses of the scalars are found by differentiating the scalar potential, while the masses of the gauge bosons come from the kinetic term of the field H , and finally the fermion masses come from the Yukawa terms [27, 36].

Particle(s)	$m^2(\varphi)$	$\bar{N}$
χ	$-\mu^2 + \lambda\varphi^2$	3
h	$-\mu^2 + 3\lambda\varphi^2$	1
$W^\pm$	$\frac{1}{4}g^2\varphi^2$	6
Z	$\frac{1}{4}(g^2 + g'^2)\varphi^2$	3
f	$\frac{1}{2}g_f^2\varphi^2$	-12

Table 1: Particle content of the SM. The table collects the field dependent masses $m^2(\varphi)$, and their degrees of freedom. We use the shorthand notation $\bar{N}_i = (-1)^{2s_i} N_i$, with s_i referring to the particle's spin, for the internal degrees of freedom, f being all massive fermions in the standard model, and χ to represent the three goldstones.

It is customary, when computing the effective potential, to include only the top-quark contribution, since it dominates over the rest due to its large mass [6]. But, as a prelude of our work in BSM theories, we will also include the bottom quark since we will be dealing with BSM particles of similar mass range.

2.4.1 Coleman-Weinberg Effective Potential

Applying the same procedure we have used so far, the Coleman-Weinberg potential, in the Landau gauge and $\overline{MS}$ renormalization scheme, takes the form

$$V_{\text{CW}}(\varphi) = \sum_{i=h,\chi,W^\pm,Z,t,b} (-1)^{2s_i} N_i \frac{m_i^4(\varphi)}{64\pi^2} \left[\ln \left(\frac{m_i^2(\varphi)}{Q^2} \right) - C_i \right], \quad (2.78)$$

where $C_i = 5/6$ for gauge bosons, and $C_i = 3/2$ for the rest of the particles.

2.4.2 Observables and Renormalization

Before we start writing V_{CT} like we did for the previous two examples, it is important to note what exactly are the properties we have measured of the Higgs boson, and how they affect the renormalization conditions.

Among the plethora of observables predicted by the SM, we would like to focus on those that help us determine the shape of the SM's scalar potential. The potential is perfectly determined by the values of the Higgs quartic coupling λ , and the Higgs VEV v . The most precise way of measuring the VEV is through the Fermi constant G_F , extracted from muon decay⁷ $v = (\sqrt{2}G_F)^{-1/2} \simeq 246.22 \text{ GeV}$ [3, 37, 38]. The quartic coupling is in turn fixed by the mass of the Higgs boson $m_h \simeq 125 \text{ GeV}$ [8, 9, 39] through the tree-level relation $\lambda = m_h^2/2v^2$.

Therefore, in order to preserve these tree-level relations we use the same renormalization conditions we used for the abelian Higgs model, but now there will be no freedom in our parameter choice. The relations are: 1) preserve the tree-level VEV, 2) preserve the tree-level curvature at v , 3) preserve the tree-level value of the effective potential at v .

The counter term potential for the SM can be written as

$$V_{\text{CT}}(\varphi) = -\frac{1}{2}\delta\mu^2\varphi^2 + \frac{1}{4}\delta\lambda\varphi^4 + \delta V_0, \quad (2.79)$$

⁷Although one may also take the mass of the W boson as an input [3].

where the counter terms are given by

$$\begin{aligned}\delta\mu^2 &= \frac{3V'_{\text{CW}}(v) - vV''_{\text{CW}}(v)}{2v}, \quad \delta\lambda = \frac{V'_{\text{CW}}(v) - vV''_{\text{CW}}(v)}{2v^3}, \\ \delta V_0 &= \frac{5vV'_{\text{CW}}(v) - v^2V''_{\text{CW}}(v)}{8} - V_{\text{CW}}(v).\end{aligned}\tag{2.80}$$

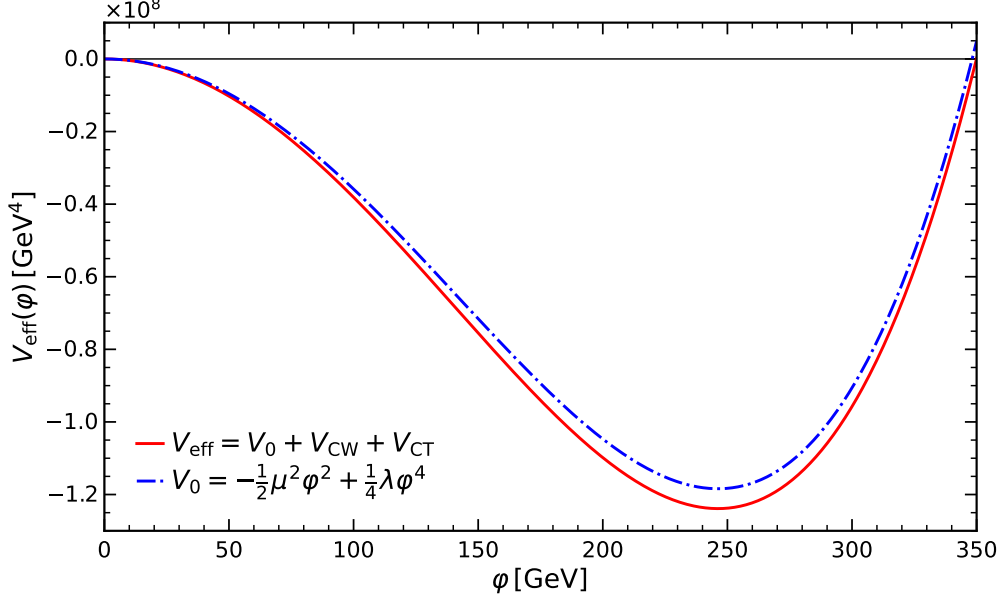


Figure 7: Effective Potential of the Standard Model. The blue, dash-dotted, line shows only the tree-level effective potential (2.76). The red line shows the one-loop effective potential, V_{CW} is the Coleman-Weinberg effective potential (2.78), and V_{CT} is the counter term potential (2.79).

2.4.3 The Impossibility of Radiative Symmetry Breaking in the SM

Let us now look at the possibility of RSB of the electroweak symmetry in the standard model, as first proposed by Coleman and Weinberg in [17], but now with the hindsight of the discoveries of the top quark and the Higgs boson. We will consider a scale invariant theory ($\mu^2 = 0$), and for simplicity only consider the standard model's heavy degrees of freedom. Therefore our effective potential becomes

$$V_{\text{eff}}(\varphi) = \frac{\lambda}{4}\varphi^4 + \sum_{i=h,W^\pm,Z,t} \bar{N}_i \frac{m_i^4(\varphi)}{64\pi^2} \left[\ln \left(\frac{m_i^2(\varphi)}{Q^2} \right) - C_i \right], \tag{2.81}$$

where the field-dependent masses are the same as in table 1 setting $\mu^2 = 0$. Since this means that all of the masses follow the form $m_i^2(\varphi) = c_i\varphi^2$ we can rewrite (2.81) as

$$\begin{aligned}V_{\text{eff}}(\varphi) &= \frac{\lambda}{4}\varphi^4 + \sum_i \bar{N}_i \frac{c_i^2\varphi^4}{64\pi^2} \left[\ln \left(\frac{c_i\varphi^2}{Q^2} \right) - C_i \right] \\ &= \frac{\lambda}{4}\varphi^4 + \sum_i \bar{N}_i \frac{c_i^2\varphi^4}{64\pi^2} \left[\ln \left(\frac{\varphi^2}{Q^2} \right) + \ln(c_i) - C_i \right] \\ &= \varphi^4 \left[\frac{\lambda}{4} + A \ln \left(\frac{\varphi^2}{Q^2} \right) + B \right],\end{aligned}\tag{2.82}$$

where

$$A \equiv \frac{1}{64\pi^2} \sum_i \bar{N}_i c_i^2, \quad B \equiv \frac{1}{64\pi^2} \sum_i \bar{N}_i c_i^2 (\ln(c_i) - C_i). \tag{2.83}$$

Setting $Q = v$, and imposing that the EW vacuum is a minimum results in the following constraints

$$\left. \frac{\partial V_{\text{eff}}}{\partial \varphi} \right|_{\varphi=v} = 0 \Rightarrow 2Av^3 + 4v^3 \left(B + \frac{\lambda}{4} \right) = 0 \Rightarrow A = -\frac{4B + \lambda}{2}, \quad (2.84)$$

thus leaving

$$V_{\text{eff}}(\varphi) = A\varphi^4 \left[\ln \left(\frac{\varphi^2}{v^2} \right) - \frac{1}{2} \right]. \quad (2.85)$$

As we saw in the abelian example, the scale invariant theories have the advantages of reducing the number of parameters. In this particular model, this translates into being allowed to predict a mass for the Higgs boson

$$m_h^2 = V''_{\text{eff}}(v) = 8Av^2 = \frac{v^2}{8\pi^2} \sum_i \bar{N}_i \frac{m_i^4(v)}{v^4} \Rightarrow m_h^2 = \frac{1}{8\pi^2 v^2} \sum_i \bar{N}_i m_i^4. \quad (2.86)$$

Writing out the sum

$$m_h^2 = \frac{6m_W^4 + 3m_Z^4 + m_h^4 - 12m_t^4}{8\pi^2 v^2} \Rightarrow m_h^2 \left(1 - \frac{m_h^2}{8\pi^2 v^2} \right) = \frac{6m_W^4 + 3m_Z^4 - 12m_t^4}{8\pi^2 v^2}, \quad (2.87)$$

Neglecting the $m_h^2/8\pi^2 v^2$ term we arrive at the following final prediction

$$m_h^2 = \frac{6m_W^4 + 3m_Z^4 - 12m_t^4}{8\pi^2 v^2} \approx -2 \cdot 10^3 \text{ GeV}^2, \quad (2.88)$$

where we have used the following values from the Particle Data Group [3]: $v = 246.22 \text{ GeV}$, $m_W = 80.36 \text{ GeV}$, $m_Z = 91.18 \text{ GeV}$, $m_t = 172 \text{ GeV}$.

Equation (2.88) confirms that Electroweak Symmetry Breaking (EWSB) can not be entirely due to radiative effects. The reason for this failure can be seen directly from the negative sign, of course a negative mass squared is unphysical but also it means that the EW vacuum is not even a minimum but a maximum ($V''_{\text{eff}}(v) < 0$). Therefore, we can conclude, that imposing scale invariance in the SM is impossible as the heavy mass of the top quark would turn our EW into an unstable vacuum.

3 Field Theory and Effective Potentials at Finite Temperatures

All the work we have performed so far using QFT has been carried out under the assumption that temperature can be neglected. This means, for example that the vacuum $|0\rangle$ of the theory is a state with 0 particles [21], but this assumption fails completely once we start studying particle physics of the early universe.

Thermal Field Theory (TFT), also called finite-temperature QFT, provides a formalism to unite QFT with statistical mechanics. This allows us to study thermodynamical effects in the early universe, like the EWPT. Our study of TFT will primarily focus on: 1) The introduction and motivation of the imaginary time formalism; 2) The computation of a temperature dependent effective potential $V_{\text{eff}}(\varphi, T)$; and 3) Predictions of symmetry restoration at high temperature, critical temperature, and types of phase transitions.

3.1 Review of Statistical Mechanics

Let us consider a quantum system described by a Hamiltonian, H , and a set of conserved charges Q_A , (electric charge, lepton number, ...), at temperature T and chemical potentials μ_A . The equilibrium state is captured by the grand canonical density operator

$$\hat{\rho} = \frac{1}{\mathcal{Z}} e^{-\beta(H - \mu_A Q_A)}, \quad \beta = \frac{1}{T}, \quad (3.1)$$

where $\mathcal{Z}$ is the partition function

$$\mathcal{Z} = \text{Tr} \left\{ e^{-\beta(H - \mu_A Q_A)} \right\}, \quad (3.2)$$

and it is defined so as to ensure that $\hat{\rho}$ is normalized such that $\text{Tr}\{\hat{\rho}\} = 1$.

The thermal average of any observables $\mathcal{O}$ is given by

$$\langle \mathcal{O} \rangle = \text{Tr}\{\hat{\rho} \mathcal{O}\}. \quad (3.3)$$

Note: We can think of the terms $e^{-\beta H}$ and $e^{-\beta \mu_A Q_A}$ as Boltzmann weights that ensure energy and charge conservation, respectively. These terms ensure that the ground state of our theory is that of minimal grand canonical Hamiltonian

$$K = H - \mu_A Q_A. \quad (3.4)$$

This ensures that the lowest temperature state does not violate any conservation law, even if its energy is not minimized [6].

3.2 Partition Functions and Path Integrals

We start by considering the case of a single, changeless, quantum mechanical degree of freedom with Hamiltonian $H(p, q)$ [6]. The partition function is

$$\mathcal{Z} = \text{Tr} \left\{ e^{-\beta H} \right\} = \sum_n \langle n | e^{-\beta H} | n \rangle = \int dq \langle q | e^{-\beta H} | q \rangle, \quad (3.5)$$

where we wrote out the trace by using a complete set of energy and position eigen-states [40, 41].

We now recall how one can write the transition amplitude between an initial state q_i and final state q_f as

$$\langle q_f | e^{-iHt} | q_i \rangle = \int_{q(0)=q_i}^{q(t)=q_f} Dq Dp \exp \left[i \int_0^t dt' (p\dot{q} - H) \right], \quad (3.6)$$

where for Hamiltonians of the form $H = p^2/2m + V(q)$, the Dp results in a constant term absorbed into the measure [26]. Therefore the amplitude can be more compactly written as

$$\langle q_f | e^{-iHt} | q_i \rangle = \int_{q(0)}^{q(t)} Dq \exp \left[i \int_0^t dt \mathcal{L} \right] = \int_{q(0)}^{q(t)} Dq e^{iS[q]}. \quad (3.7)$$

Comparing the partition function (3.5) to the transition amplitude (3.7), we observe that:

- i) The trace sets $q_i = q_f$, i.e. it imposes cyclic boundary conditions [41]
- ii) The operators $e^{-\beta H}$ and e^{-iHt} are identical given that $t \rightarrow -i\beta$ [40].

This means that in TFT, t is periodic⁸ and its maximum value is given by β [29].

Making a wick rotation, $t \rightarrow -i\tau$, where τ is the Euclidean time, implies that

$$\tau \in [0, \beta] \quad \& \quad Z = \int_{q(0)=q(\tau)} Dq \exp \left(- \int_0^\beta d\tau \mathcal{L}_E \right), \quad (3.8)$$

where $\mathcal{L}_E$ is the Euclidean Lagrangian.

The extension to a quantum scalar field $\phi(\tau, \vec{x})$ is straightforward [26, 40]. The partition function of a scalar field theory with Hamiltonian H is given by

$$Z = \text{Tr} e^{-\beta H} = \int_{\phi(0, \vec{x})=\phi(\beta, \vec{x})} D\phi e^{-S_E[\phi]}, \quad (3.9)$$

where the Euclidean action over the compactified time is

$$S_E[\phi] = \int_0^\beta d\tau \int d^3x \mathcal{L}_E(\phi, \partial_\mu \phi), \quad \mathcal{L}_E = \frac{1}{2}(\partial_\tau \phi)^2 + \frac{1}{2}(\nabla \phi)^2 + V(\phi). \quad (3.10)$$

For fermion fields, the trace over the Fock space introduces an extra minus sign from the path integral which flips the boundary conditions [40, 41], leaving us with

$$\text{Bosons: } \phi(\tau, \vec{x}) = \phi(\tau + \beta, \vec{x}), \quad \text{Fermions: } \psi(\tau, \vec{x}) = -\psi(\tau + \beta, \vec{x}). \quad (3.11)$$

Note: At temperature T , the 4-dimensional Euclidean spacetime is not $\mathbb{R}^4$ but rather $S^1 \times \mathbb{R}^3$, where the circle S^1 has circumference $\beta = 1/T$. At $T \rightarrow 0$ the circle decompresses to $\mathbb{R}$, and we recover the standard Euclidean QFT. While at $T \rightarrow \infty$, the circle shrinks to a point and the system becomes effectively three-dimensional. This is the basis of dimensional reduction [44], although this topic is beyond the scope of this thesis.

3.3 Matsubara Formalism

Since the Euclidean time direction is compact with period β , any field configuration satisfying the periodic boundary conditions can be expanded in τ as a Fourier Series [6, 40, 41, 45].

For a bosonic field, the Fourier expansion is

$$\phi(\tau, \vec{x}) = \sum_{n=-\infty}^{\infty} \phi_n(\vec{x}) e^{-i\omega_n^B \tau}, \quad (3.12)$$

where the allowed frequencies ω_n^B are fixed by its periodicity (3.11)

$$e^{-i\omega_n^B(\tau+\beta)} = e^{-i\omega_n^B \tau} \Rightarrow \omega_n^B = 2\pi nT, \quad n \in \mathbb{Z}. \quad (3.13)$$

⁸The periodicity of the time direction can be also understood through the KMS condition [42, 43], which characterizes thermal equilibrium at the level of Green's functions and forces the periodicity [6, 28].

While for a fermionic field that satisfies (3.11), its Fourier expansion will be

$$\psi(\tau, \vec{x}) = \sum_{n=-\infty}^{\infty} \psi_n(\vec{x}) e^{-i\omega_n^F \tau}, \quad (3.14)$$

where the allowed frequencies ω_n^F will differ from those of (3.13) due to the anti-periodicity of fermions

$$e^{-i\omega_n^F(\tau+\beta)} = -e^{-i\omega_n^F \tau} \Rightarrow \omega_n^F = (2n+1)\pi T, \quad n \in \mathbb{Z}. \quad (3.15)$$

These frequencies $\omega_n^B(\omega_n^F)$ are called bosonic(fermionic) Matsubara Frequencies [40, 41, 45].

We can also Fourier transform in the spatial directions to obtain the full momentum space decomposition

$$\phi(\tau, \vec{x}) = \sum_n \int \frac{d^3 k}{(2\pi)^3} \phi(\omega_n, \vec{k}) e^{i\vec{k} \cdot \vec{x} - i\omega_n \tau}, \quad (3.16)$$

where we have left ω_n ambiguous as to represent both bosons and fermions.

From (3.16) we can see that while at $T = 0$ we carry out a continuous integration over the Euclidean 4-momentum, $k_E = (k_0, \vec{k})$, at finite T the k_0 direction has been discretized into the Matsubara Modes. The spacing between adjacent Matsubara frequencies is $\Delta\omega = 2\pi T$, therefore $T \sim \Delta\omega/2\pi$ plays the role of the integration measure $dk_0/2\pi$. This allows us to define the Matsubara sum rule as follows

$$\int \frac{d^4 k_E}{(2\pi)^4} \rightarrow T \sum_{n=-\infty}^{\infty} \int \frac{d^3 \vec{k}}{(2\pi)^3}, \quad (3.17)$$

thus replacing a continuous integration over k_0 by a discrete sum [28, 40].

3.3.1 Thermal Propagator

The time discretization also has implications for the propagator, at $T = 0$ we have

$$\Delta_E(k) = \frac{1}{k_0^2 + |\vec{k}|^2 + m^2}. \quad (3.18)$$

While at finite temperature the k_0 are replaced by the frequencies ω_n so the thermal propagator becomes

$$\Delta(\omega_n, \vec{k}) = \frac{1}{\omega_n^2 + |\vec{k}|^2 + m^2} = \frac{1}{\omega_n^2 + E^2}, \quad (3.19)$$

where $E^2 = |\vec{k}|^2 + m^2$ is the energy of a relativistic massive particle.

3.4 Evaluating Matsubara Sums

Any loop calculation at finite temperature will produce sums of the form $T \sum_n f(\omega_n)$, over Matsubara frequencies. In order to evaluate them analytically, we will trade the sums for contour integrals in the complex frequency plane, by exploiting meromorphic functions whose residues at $z = \pm i\omega_n$ are controlled [19, 40].

In order to do this, we use the distribution functions for bosons and fermions⁹

$$n_B(z) = \frac{1}{e^{\beta z} - 1}, \quad n_F(z) = \frac{1}{e^{\beta z} + 1}. \quad (3.20)$$

⁹Note that these functions emerge naturally as candidates from evaluating the Matsubara sums [6].

The functions of (3.20) have poles at $z = i\omega_n$ (for both bosonic and fermionic Matsubara frequencies) with residues T for bosons and $-T$ for fermions. One can therefore write

$$T \sum_n f(i\omega_n^B) = \sum_n \text{Res} [n_B(z)f(z), z = i\omega_n^B] = \oint_C \frac{dz}{2\pi i} n_B(z)f(z), \quad (3.21)$$

for bosons, while for fermions

$$T \sum_n f(i\omega_n^F) = - \sum_n \text{Res} [n_F(z)f(z), z = i\omega_n^F] = - \oint_C \frac{dz}{2\pi i} n_F(z)f(z). \quad (3.22)$$

For both (3.21) and (3.22), C is a contour that encircles all the periodic, bosonic and fermionic, Matsubara poles.

By deforming the contour C to instead pick up the poles of $f(z)$, one can relate the Matsubara sums to the residues of the original function as follows

$$T \sum_n f(i\omega_n^B) = - \sum_{\text{poles of } f} \text{Res}[n_B(z)f(z)], \quad (3.23)$$

for bosons, while for fermions

$$T \sum_n f(i\omega_n^F) = \sum_{\text{poles of } f} \text{Res}[n_F(z)f(z)]. \quad (3.24)$$

3.4.1 Example of a Matsubara Sum: Free Boson/Fermion Propagator

A very common factor that appears in most 1-loop integrals is the thermal propagator (3.19), i.e. $f(\omega_n) = (\omega_n^2 + E^2)^{-1}$. The poles of $f(z)$, where $z = i\omega_n$ are $z = \pm E$, and its residues are $\mp(2E)^{-1}$ [6].

For the bosonic case we use (3.23)

$$\begin{aligned} T \sum_n \frac{1}{\omega_n^2 + E^2} &= - \sum_{\text{poles of } f} \text{Res}[n_B(z)f(z)] = \frac{n_B(E)}{2E} - \frac{n_B(-E)}{2E} \\ &= \frac{1}{2E}(1 + 2n_B(E)) = \frac{1}{2E} \coth \frac{\beta E}{2} \quad (\text{Bosons}), \end{aligned} \quad (3.25)$$

while for the fermionic case we use (3.24)

$$\begin{aligned} T \sum_n \frac{1}{\omega_n^2 + E^2} &= \sum_{\text{poles of } f} \text{Res}[n_F(z)f(z)] = -\frac{n_F(E)}{2E} + \frac{n_F(-E)}{2E} \\ &= \frac{1}{2E}(1 - 2n_F(E)) = \frac{1}{2E} \tanh \frac{\beta E}{2} \quad (\text{Fermions}). \end{aligned} \quad (3.26)$$

These results have an interesting physical interpretation, the terms independent of the distribution functions are exactly what we expect from field theory at $T = 0$, while the remaining terms are purely thermal contributions from the heat bath [6].

3.5 Green's Functions at Finite Temperature

In TFT, the role of the VEV of an operator $\mathcal{O}$ is taken by the thermal average (3.3). We define the thermal green's function as the thermal average of the Euclidean ordered product of the n field operators

$$G^{(n)}(\tau_1, \vec{x}_1; \dots; \tau_n, \vec{x}_n) = \langle T_\tau \phi(\tau_1, \vec{x}_1) \cdots \phi(\tau_n, \vec{x}_n) \rangle, \quad (3.27)$$

where T_τ orders the field operators in increasing Euclidean time τ [6, 19, 40, 41].

The full machinery of generating functional directly carries over to finite temperature, the generating functional for thermal Green's functions is

$$Z^\beta[J] = \int_{\phi(0)=\phi(\beta)} D\phi \exp \left(-S_E[\phi] + \int_0^\beta d\tau \int d^3x J(\tau, \vec{x}) \phi(\tau, \vec{x}) \right). \quad (3.28)$$

The connected thermal Green's functions are generated by $W^\beta[J] = \ln Z^\beta[J]$, and the finite-temperature effective action is defined by the Legendre Transform of $W^\beta[J]$

$$\Gamma^\beta[\bar{\phi}] = W^\beta[J] - \int d^4x J(\tau, \vec{x}) \bar{\phi}(\tau, \vec{x}), \quad (3.29)$$

where $\bar{\phi}(x) = \langle \phi(x) \rangle = \delta W^\beta / \delta J(x)$.

For a translationally invariant theory, $\bar{\phi}$ is constant (φ_c). Ignoring the factor of spacetime volume, one defines the effective potential as

$$\Gamma^\beta[\varphi_c] = - \int d^4x V_{\text{eff}}^\beta(\varphi_c). \quad (3.30)$$

Finally, we know that SSB at temperature T occurs when the equation

$$\frac{\partial V_{\text{eff}}^\beta}{\partial \varphi_c} = 0, \quad (3.31)$$

has solutions for $\varphi_c \neq 0$ [6].

3.6 Finite Temperature Effective Potential

From the previous section, we know that the one-loop effective potential, at $T = 0$, can be computed as

$$V_{\text{eff}}(\varphi) = V_0(\varphi) + V_{\text{CW}}(\varphi) = V_0(\varphi) + \sum_j \frac{\bar{N}_j}{2} \int \frac{d^4k_E}{(2\pi)^4} \ln(k_E^2 + m_j^2(\varphi)), \quad (3.32)$$

where j runs over the particles that couple to the background field φ , $m_j^2(\varphi)$ is the field dependent mass, and $\bar{N}_j = (-1)^{2s_j} N_j$ where N_j counts the field's DoF and the s_j is the field's spin.

We now promote $V_{\text{eff}}(\varphi)$, (3.32), to finite temperature using the replacement rule (3.17)

$$V_{\text{eff}}(\varphi, T) = V_0(\varphi) + \sum_j \frac{\bar{N}_j T}{2} \sum_{n=-\infty}^{\infty} \int \frac{d^3k}{(2\pi)^3} \ln(\omega_n^2 + |\vec{k}|^2 + m_j^2(\varphi)). \quad (3.33)$$

In order to write V_{eff} as a sum of $T = 0$ and $T \neq 0$ terms we need to further develop the sums in (3.33). Let us write

$$\mathcal{S}_j \equiv T \sum_{n=-\infty}^{\infty} \ln(\omega_n^2 + E_j^2(\vec{k})), \quad (3.34)$$

where $E_j^2(\vec{k}) = |\vec{k}|^2 + m_j^2(\varphi)$.

Differentiating $\mathcal{S}_j$ with respect to E_j^2

$$\frac{\partial \mathcal{S}_j}{\partial E_j^2} = T \sum_{n=-\infty}^{\infty} \frac{1}{\omega_n^2 + E_j^2}. \quad (3.35)$$

This expression is exactly the Matsubara sum of the propagator we evaluated in (3.25) for bosons, and (3.26) for fermions. Therefore

$$\frac{\partial \mathcal{S}_j^B}{\partial E_j^2} = \frac{1}{2E_j} \left(1 + \frac{2}{e^{E_j/T} - 1} \right), \quad \frac{\partial \mathcal{S}_j^F}{\partial E_j^2} = \frac{1}{2E_j} \left(1 - \frac{2}{e^{E_j/T} + 1} \right), \quad (3.36)$$

where $\mathcal{S}_j^B$ refers to bosons, and $\mathcal{S}_j^F$ refers to fermions.

Integrating back over E_j^2 , we obtain the following exact result for the sum $\mathcal{S}_j$

$$\mathcal{S}_j = E_j + 2T \ln \left(1 \mp e^{-E_j/T} \right) + C, \quad (3.37)$$

where the $(-)$ is for bosons and $(+)$ is for fermions.

Substituting this result into (3.33)

$$V_{\text{eff}}(\varphi, T) = V_0(\varphi) + \sum_j \frac{\bar{N}_j}{2} \int \frac{d^3 k}{(2\pi)^3} \left[E_j(\vec{k}) + 2T \ln \left(1 \mp e^{-E_j(\vec{k})/T} \right) \right]. \quad (3.38)$$

Now, this can be easily separated into two separate pieces, the first of which can be identified as the $T = 0$ Coleman-Weinberg potential, while the second one will therefore be purely associated to thermal effects

$$V_{\text{eff}}(\varphi, T) = V_0(\varphi) + V_{T=0}(\varphi) + V_T(\varphi, T), \quad (3.39)$$

where

$$V_{T=0}(\varphi) = \sum_j \frac{\bar{N}_j}{2} \int \frac{d^3 k}{(2\pi)^3} E_j(\vec{k}) = \sum_j \frac{\bar{N}_j}{2} \int \frac{d^3 k}{(2\pi)^3} \sqrt{|\vec{k}|^2 + m_j^2(\varphi)}. \quad (3.40)$$

In order to properly identify this term as V_{CW} in (3.32), we need to show the following

$$\int \frac{d^3 k}{(2\pi)^3} \sqrt{|\vec{k}|^2 + m_j^2(\varphi)} = \int \frac{d^4 k_E}{(2\pi)^4} \ln(k_E^2 + m_j^2(\varphi)). \quad (3.41)$$

Differentiating both sides of (3.41) with respect to m_j^2

$$\text{LHS: } \int \frac{d^3 k}{(2\pi)^3} \frac{1}{2\sqrt{|\vec{k}|^2 + m_j^2(\varphi)}} = \int \frac{d^3 k}{(2\pi)^3} \frac{1}{2E}, \quad (3.42)$$

$$\text{RHS: } \int \frac{d^4 k_E}{(2\pi)^4} \frac{1}{k_E^2 + m_j^2(\varphi)} = \int \frac{d^3 k}{(2\pi)^3} \int_{-\infty}^{\infty} \frac{dk_4}{2\pi} \frac{1}{k_4^2 + E^2} = \int \frac{d^3 k}{(2\pi)^3} \frac{1}{2E}. \quad (3.43)$$

Since the derivatives match, the two expressions of (3.41) differ at most by a field-independent constant, which carries no physical information so we can say, without losing any sleep that

$$V_{T=0}(\varphi) = \sum_j \frac{\bar{N}_j}{2} \int \frac{d^3 k}{(2\pi)^3} E_j(\vec{k}) = \sum_j \frac{\bar{N}_j}{2} \int \frac{d^4 k_E}{(2\pi)^4} \ln(k_E^2 + m_j^2(\varphi)) = V_{CW}(\varphi). \quad (3.44)$$

The second term

$$V_T(\varphi, T) = \sum_j \bar{N}_j \int \frac{d^3 k}{(2\pi)^3} T \ln \left(1 \mp e^{-E_j(\vec{k})/T} \right), \quad (3.45)$$

is purely thermal as it vanishes when we set $T \rightarrow 0$. This expression can be rewritten

$$\begin{aligned} V_T(\varphi, T) &= \sum_j \bar{N}_j \frac{T}{2\pi^2} \int_0^\infty k^2 dk \ln \left[1 \mp \exp \left(-\frac{\sqrt{|\vec{k}|^2 + m_j^2(\varphi)}}{T} \right) \right] \\ &= \sum_j \frac{\bar{N}_j T^4}{2\pi^2} \int_0^\infty x^2 dx \ln \left(1 \mp e^{-\sqrt{x^2 + m_j^2(\varphi)/T^2}} \right) \\ &= \sum_j \frac{\bar{N}_j T^4}{2\pi^2} J_{B/F} \left(m_j^2(\varphi)/T^2 \right), \end{aligned} \quad (3.46)$$

where we have defined the thermal functions of bosons (J_B) and fermions (J_F) [6, 40, 41] as

$$J_B(y^2) = \int_0^\infty x^2 dx \ln \left(1 - e^{-\sqrt{x^2+y^2}} \right), \quad J_F(y^2) = \int_0^\infty x^2 dx \ln \left(1 + e^{-\sqrt{x^2+y^2}} \right). \quad (3.47)$$

These functions are UV-finite and they usually need to be integrated numerically¹⁰ [28, 46], although taking the high temperature limit allows us to compute analytical results and get an understanding of their behavior.

3.7 High Temperature Expansion

In order to obtain a qualitative understanding of the thermal potential, both in its dependence with temperature and with the fields, we expand the thermal functions (3.47) in the high temperature limit $T \gg m_j(\varphi)$.

Taking $y \ll 1$, one can expand the bosonic thermal function as

$$J_B(y^2) = -\frac{\pi^4}{45} + \frac{\pi^2}{12}y^2 - \frac{\pi}{6}(y^2)^{3/2} - \frac{1}{32}y^4 \ln \left(\frac{y^2}{a_B} \right) + \mathcal{O}(y^6), \quad (3.48)$$

where $a_B = 16\pi^2 \exp(3/2 - 2\gamma_E)$, and $\gamma_E \approx 0.5772$ is the Euler-Mascheroni constant[6].

The fermionic thermal function can be expanded as

$$J_F(y^2) = \frac{7\pi^4}{360} - \frac{\pi^2}{24}y^2 - \frac{1}{32}y^4 \ln \left(\frac{y^2}{a_F} \right) + \mathcal{O}(y^6), \quad (3.49)$$

where $a_F = \pi^2 \exp(3/2 - 2\gamma_E)$ [6].

Inserting these expansions in (3.46), and separating the particle sums into its bosonic and fermionic components, we can write the thermal part of the effective potential as

$$\begin{aligned} V_T(\varphi, T) = & \sum_{j=B} \frac{\bar{N}_j T^4}{2\pi^2} \left[-\frac{\pi^4}{45} + \frac{\pi^2}{12} \frac{m_j^2(\varphi)}{T^2} - \frac{\pi}{6} \left(\frac{m_j^2(\varphi)}{T^2} \right)^{3/2} - \frac{1}{32} \frac{m_j^4(\varphi)}{T^4} \ln \left(\frac{m_j^2(\varphi)}{a_B T^2} \right) \right] \\ & + \sum_{j=F} \frac{\bar{N}_j T^4}{2\pi^2} \left[\frac{7\pi^4}{360} - \frac{\pi^2}{24} \frac{m_j^2(\varphi)}{T^2} - \frac{1}{32} \frac{m_j^4(\varphi)}{T^4} \ln \left(\frac{m_j^2(\varphi)}{a_F T^2} \right) \right] + \dots \end{aligned} \quad (3.50)$$

Finally, inserting this into the full effective potential (3.39), and collecting the powers of T we write the full effective potential with both quantum and thermal corrections as

$$\begin{aligned} V_{\text{eff}}(\varphi, T) = & V_0(\varphi) - \frac{\pi^2}{90} \left(\sum_B \bar{N}_B - \frac{7}{8} \sum_F \bar{N}_F \right) T^4 \\ & + \frac{T^2}{24} \sum_B \bar{N}_B m_B^2(\varphi) - \frac{T^2}{48} \sum_F \bar{N}_F m_F^2(\varphi) \\ & - \frac{T}{12\pi} \sum_B \bar{N}_B m_B^3(\varphi) \\ & + \frac{1}{64\pi^2} \left[\sum_B \bar{N}_B m_B^4(\varphi) \left(\ln \left(\frac{a_B T^2}{\mu^2} \right) - c_B \right) \right. \\ & \left. + \sum_F \bar{N}_F m_F^4(\varphi) \left(\ln \left(\frac{a_F T^2}{\mu^2} \right) - c_F \right) \right]. \end{aligned} \quad (3.51)$$

¹⁰In this thesis we will use the Python code `CosmoTransitions` [46]

To study the physical interpretation of each of the terms it is useful to rearrange the previous expressions in powers of the background field φ . In order to do that we are going to make the following simplification, $m_j^2(\varphi) = k_j^2 \varphi^2$. Although there exist field-dependent masses with constant terms, note that this is general for fermions, bosons and some scalars [6, 25]. Assuming the usual tree-level potential $V_0(\varphi) = m^2 \varphi^2 + \lambda \varphi^4$

$$V_{\text{eff}}(\varphi, T) = -\frac{\pi^2}{90} g_* T^4 + M^2(T) \varphi^2 - B(T) \varphi^3 + \bar{\lambda}(T) \varphi^4, \quad (3.52)$$

where we have defined

$$\begin{aligned} g_* &= \sum_B \bar{N}_B - \frac{7}{8} \sum_F \bar{N}_F, \\ M^2(T) &= m^2 + \left(\sum_B \frac{\bar{N}_B k_B^2}{24} - \sum_F \frac{\bar{N}_F k_F^2}{48} \right) T^2, \\ B(T) &= \frac{T}{12\pi} \sum_B \bar{N}_B k_B^3, \\ \bar{\lambda}(T) &= \lambda + \frac{1}{64\pi^2} \left[\sum_B \bar{N}_B k_B^4 \left(\ln \left(\frac{a_B T^2}{\mu^2} \right) - c_B \right) + \sum_F \bar{N}_F k_F^4 \left(\ln \left(\frac{a_F T^2}{\mu^2} \right) - c_F \right) \right]. \end{aligned} \quad (3.53)$$

Let us now discuss the physical meaning of each term in (3.53) [6]:

1. **Pressure Term.** This is the free energy density of a massless boson (fermion) gas. It is field-independent and does not affect the shape of the potential; it only sets the overall vacuum energy at temperature T . It is nonetheless important for cosmology (it determines the energy density of the thermal bath and hence the Hubble rate).
2. **symmetry Restoration Term.** Since SSB is determined, at tree-level, by the presence of a tachyonic mass term. We see that this term adds a positive, symmetry restoring term, that increases with T , thus allowing for symmetry restoration in the early universe.
3. **Cubic/Barrier Term.** This term is unique to bosons and it is responsible for the creation or enhancement of a potential barrier due to thermal effects. This term is essential for models that require First-Order Phase Transition (FOPT)s, mainly those that transition due to a thermally induced barrier.
4. **Logarithmic Term.** This is a running-coupling effect. It supplements the $T = 0$ Coleman-Weinberg logarithm.

3.8 Phase Transitions

Having found the way to compute the thermal effective potential, we can now turn to the study of phase transitions. In the perturbative study we are performing we can differentiate between two types of phase transitions¹¹ by focusing on the position of the VEV at different temperatures $v(T)$. If the evolution $v(T)$ is continuous we have a second-order phase transition, but if $v(T)$ is not continuous we have a first-order phase transition [6].

Let us consider the following thermal effective potential

$$V_{\text{eff}} = \frac{1}{2} (cT^2 - \mu^2) \varphi^2 - BT \varphi^3 + \frac{\lambda(T)}{4} \varphi^4, \quad (3.54)$$

where μ^2 , c , B and $\lambda(T)$ are model dependent parameters, found by following the procedure of the previous section.

¹¹Crossovers differ from phase transitions since their thermal evolution is continuous and differentiable. They are highly relevant to our discussion, we know that in the SM the thermal evolution follows a crossover [10, 12, 19], but to prove this we require non perturbative calculations.

3.8.1 Second-Order Phase Transitions

In order to have a second order phase transition, we must have a model where $B \simeq 0$. In that case the thermal effective potential becomes

$$V_{\text{eff}} = \frac{1}{2} (cT^2 - \mu^2) \varphi^2 + \frac{\lambda(T)}{4} \varphi^4. \quad (3.55)$$

This potential has two critical points, one symmetrical one located at

$$v(T) = 0, \quad (3.56)$$

and an anti-symmetric one located at

$$v(T) = \frac{\sqrt{\mu^2 - cT^2}}{\sqrt{\lambda(T)}}. \quad (3.57)$$

We call the critical temperature T_c the temperature at which the multiple minima of the potential are degenerate [6]. For a second-order phase transition, it is clear that this can only occur at

$$T_c = \frac{\mu}{\sqrt{c}} \equiv T_0, \quad (3.58)$$

which coincides, by definition, with the temperature at which the second derivative at 0 becomes zero¹².

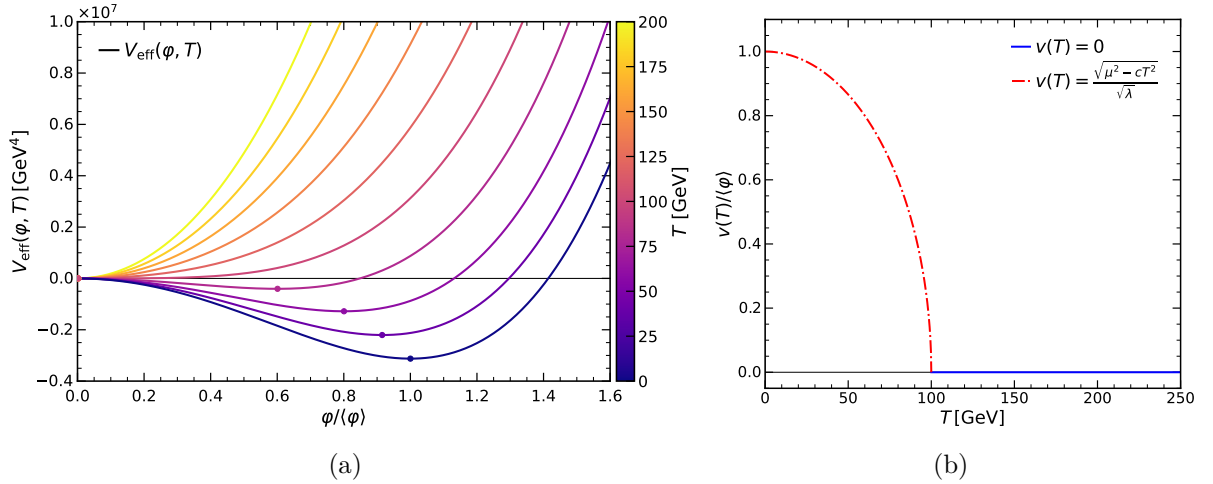


Figure 8: Thermal evolution of a second-order phase transition. (a) Thermal effective potential (3.55), at multiple temperatures. (b) Thermal evolution of the vacuum $v(T)$, the blue line corresponds to a symmetry preserving vacuum, while the red one is a symmetry breaking one.

3.8.2 First-Order Phase Transitions

FOPT are predicted by theories whose thermal effective potential can be written as (3.54), or most notably, theories with a barrier term between the two degenerate minima.

The critical points of our potential are found by the usual differentiation of (3.54), they are located at

$$\varphi = 0, \quad \varphi = \frac{3BT \pm \sqrt{9B^2T^2 - 4c\lambda T^2 + 4\lambda\mu^2}}{2\lambda}. \quad (3.59)$$

For a FOPT, we are interested in the case where, given all the parameters to be positive, the minima are located at 0 and at the solution with the + sign, while the - sign one is a maximum.

¹²For FOPT, in order to avoid confusion, we will label T_0 the temperature at which $V_{\text{eff}}''(0, T_0) = 0$.

In order for those conditions to be met, we first ensure the potential is a local minimum by imposing

$$T_c > T_0 = \frac{\mu}{\sqrt{c}}, \quad (3.60)$$

and we must also impose that the barrier forms before the T_c , this is done by imposing

$$T_1 = \frac{2\mu\sqrt{\lambda}}{\sqrt{4c\lambda - 9B^2}} > T_c. \quad (3.61)$$

The critical temperature can be found¹³ by substituting the non symmetric solution of (3.59) and imposing that the potential at that temperature be zero. The only solution that achieves that while keeping $T_1 > T_c > T_0$ [47], is

$$T_c = \frac{\mu}{\sqrt{c - \frac{2B^2}{\lambda}}}. \quad (3.62)$$

We define the strength of our FOPT as

$$\xi_c = \frac{v(T_c)}{T_c} = \frac{2B}{\lambda} \quad (3.63)$$

and we will require $\xi_c \geq 1$ in order for our transition to be considered a Strong First-Order Phase Transition (SFOPT)¹⁴ [25, 48, 49].

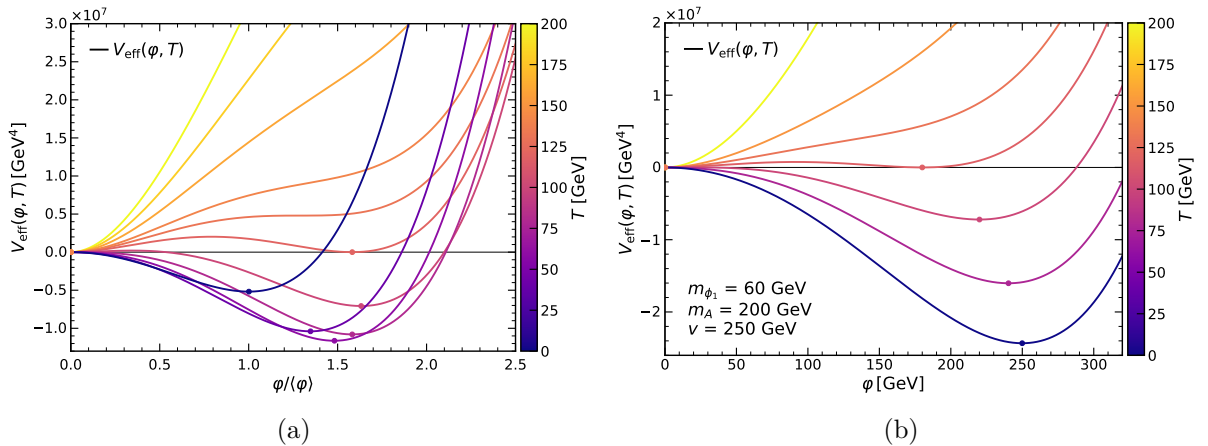


Figure 9: Examples of first-order phase transitions. (a) Thermal evolution of a effective potential of the form (3.54). (b) Full numerical thermal evolution of an abelian Higgs model like (2.47).

A particularly interesting results that comes from (3.59) is the fact that the barrier term shifts the VEV and makes it not monotonically increasing (see figure 9a). This is due to the nature of the high temperature expansion, since V_T vanishes at $T \rightarrow 0$, the high temperature limit does so too. But of course in the process of $T \rightarrow 0$ we are leaving the validity region of this limit and therefore those results are not valid. This is heuristically shown in figure 9b where we show the full numeric effective potential for the model (2.47), and that behavior is nowhere to be found.

¹³It can also be found by imposing that there are only two roots for (3.54), both approaches give (3.62).

¹⁴The definition of the strength of the phase transition is convention dependent [28, 29].

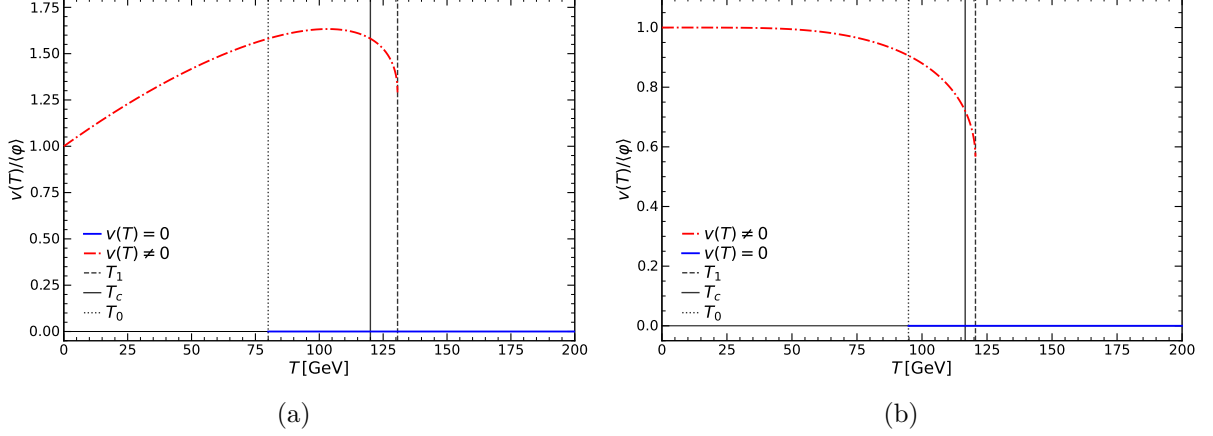


Figure 10: Tracking of the positions of the different vacua in theories with FOPTs. (a) Shows the minima of a high temperature potential (3.54), while (b) shows the minima as a function of T for the abelian Higgs model of figure 9b.

3.9 Thermal Tunneling and Bubble Nucleation

For a set of parameters of a certain model, finding that $\xi_c \geq 1$ does not guarantee that the SFOPT occurs in reality, it is possible for the universe to be trapped in that false vacuum and therefore impede the phase transition. This phenomenon is referred to as vacuum trapping [50], and it is highly relevant since it means that apart from finding the parameters that are compatible with a SFOPT, we will also have to ensure that nucleation occurs [29].

The probability of transition between the two minima across the potential barrier, also known as tunneling rate, per unit time and per unit volume is given by [51–54]

$$\Gamma(T) = A(T)e^{-S_3(T)/T}, \quad (3.64)$$

where $A(T) \sim T^4$ [50, 53], while S_3 is the three-dimensional Euclidean action for the “bounce” configuration ϕ_B that connects the false and true vacua¹⁵ for $T < T_c$ [6, 50],

$$S_3(T) = \int d^3x \left[\frac{1}{2} (\nabla \phi_B)^2 + V_{\text{eff}}(\phi_B, T) \right] = 4\pi \int_0^\infty r^2 dr \left[\frac{1}{2} \left(\frac{d\phi_B}{dr} \right)^2 + V_{\text{eff}}(\phi_B, T) \right]. \quad (3.65)$$

The bounce ϕ_B makes S_3 stationary, resulting in the equation of motion

$$\frac{d^2 \phi_B}{dr^2} + \frac{2}{r} \frac{d\phi_B}{dr} = V'_{\text{eff}}(\phi_B, T), \quad (3.66)$$

and boundary conditions

$$\frac{d\phi_B}{dr} = 0, \quad \lim_{r \rightarrow \infty} \phi_B = 0. \quad (3.67)$$

In order for the nucleation of the SFOPT to occur and outpace the Hubble expansion, we require that the transition rate (3.64), within a Hubble volume H become of order one [50, 53–55]. This allows us to define the nucleation temperature as [14, 29]

$$\int_{T_n}^{T_c} \frac{T^4}{H^4} \frac{A(T)}{T} e^{-S_3(T)/T} dT \sim 1 \Rightarrow \frac{S_3(T_n)}{T_n} \sim 140, \quad (3.68)$$

where the Hubble parameter H is given by [56]

$$H^2(T) = \frac{8\pi^3 g_*(T) T^4}{90 M_{Pl}^2}. \quad (3.69)$$

If (3.68) is not satisfied for any $T_n < T_c$, we will consider that the vacuum becomes trapped.

¹⁵It is understood that the bounce solution starts close to the global minimum (the broken phase) [55].

3.10 On the Validity of Thermal Perturbation Theory

In this chapter we have seen how to compute the thermal component of the effective potential $V_T(\varphi, T)$, (3.46), but the calculation is still incomplete as we need to take care of a well know issue in perturbative TFT.

The existence of symmetry restoration has a very profound implication on the perturbative validity of a theory. The fact that thermal-loop effects overpower a temperature independent tree-level effect serves as a warning sign for our theory, which now has two different scales μ (mass term) and T [29, 50, 57].

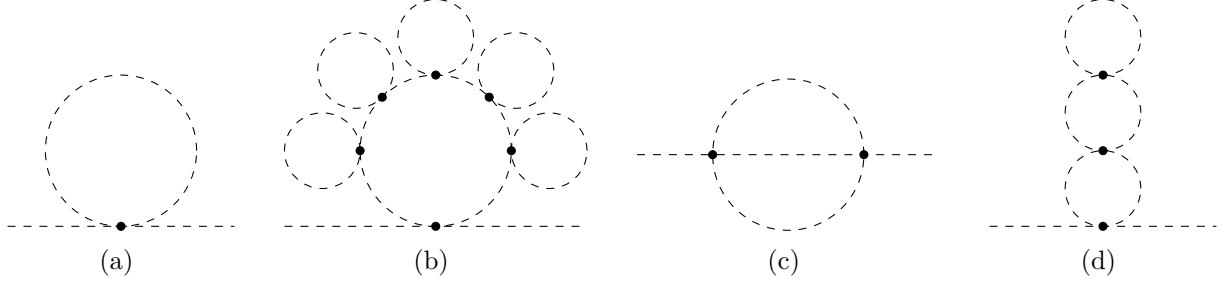


Figure 11: Contributions to the scalar mass for a scalar field theory (3.70). Their leading order temperature scaling are [57]: (a) one-loop diagram, $\sim \lambda NT^2$. (b) Six-loop daisy, $\sim \lambda^6 N^6 \frac{T^{11}}{\mu^9}$. (c) Two-loop sunset, $\sim \lambda^2 NT^2$. (d) Three-loop super-daisy $\sim \lambda^3 N^3 \frac{T^4}{\mu^2}$.

In order to show this breakdown of perturbation theory let us consider the high temperature behavior of a $\lambda\phi^4$ theory with N fields

$$V = -\frac{1}{2}\mu^2\phi_i\phi^i + \frac{1}{4}\lambda\left(\phi_i\phi^i\right)^2. \quad (3.70)$$

The phase transition of this theory occurs when the thermal mass and the tree-level μ^2 cancel, resulting in the following

$$N\lambda T^2 - \mu^2 \approx 0 \Rightarrow \rho \equiv \lambda N \frac{T^2}{\mu^2} \approx 1. \quad (3.71)$$

From figure 11b we can extract that the thermal contribution of each n -loop daisy diagram, at leading order in T , they are [57]

$$\delta m_{D_n}^2 \sim \lambda^n N^n \frac{T^{2n-1}}{\mu^{2n-3}}, \quad n \geq 2. \quad (3.72)$$

The ratio of the n to the $n+1$ loop daisy contribution scales as

$$\frac{\delta m_{D_n}^2}{\delta m_{D_{n-1}}^2} \sim \rho, \quad (3.73)$$

which, during the phase transition, we have it is of order 1, showcasing the breakdown of perturbation theory. This is solved via daisy resummation [29, 57].

There are multiple resummation methods¹⁶, but in this thesis we will focus on the one introduced by Parwani [59]. it consists in shifting the masses of all the Matsubara modes for the bosonic fields (for gauge bosons only the longitudinal component is shifted) thus giving them finite mass and avoiding the IR-divergence

$$m_j^2(\phi_i) \rightarrow m_j^2(\phi_i) + \Pi_j(T). \quad (3.74)$$

¹⁶One might also use the Arnold-Espinoza method [58] which only resums the non-analytic Matsubara zero mode, via the introduction of a new term in the effective potential [29].

3.11 Electroweak Phase Transitions and Baryogenesis

In the previous sections we developed the machinery needed to properly study the thermal effective potential and find a model's phase transitions. In this section we will apply that machinery to the SM noting that this is a perturbative approach so we will not be able to predict a crossover like in [10, 12, 19].

Instead of carrying out a high temperature expansion, we will use `CosmoTransitions` [46] to numerically evaluate the thermal functions $J_{B/F}$ (3.47). We will build the `generic_potential` class using the information of the table 1, including the particles up to the mass of the b quark. The thermal corrections to the field-dependent masses for the Higgs and the longitudinal components of the gauge bosons are computed using the procedure described in [28], and they are

$$\begin{aligned} \Pi_{h,G^{\pm,0}}(T) &= \frac{T^2}{16} (4y_b^2 + 3g^2 + g_w^2 + 8\lambda + 4g_t^2), \quad \Pi_W(T) = \frac{11g^2T^2}{6}, \\ M_{Z/\gamma}^2(\varphi, T) &= M_{W_3/B}^2(\varphi) + \Pi_{W_3/B}(T) = \begin{pmatrix} \frac{g^2\varphi^2}{4} + \frac{11g^2T^2}{6} & -\frac{gg'\varphi^2}{4} \\ -\frac{gg'\varphi^2}{4} & \frac{g'^2\varphi^2}{4} + \frac{11g'^2T^2}{6} \end{pmatrix}. \end{aligned} \quad (3.75)$$

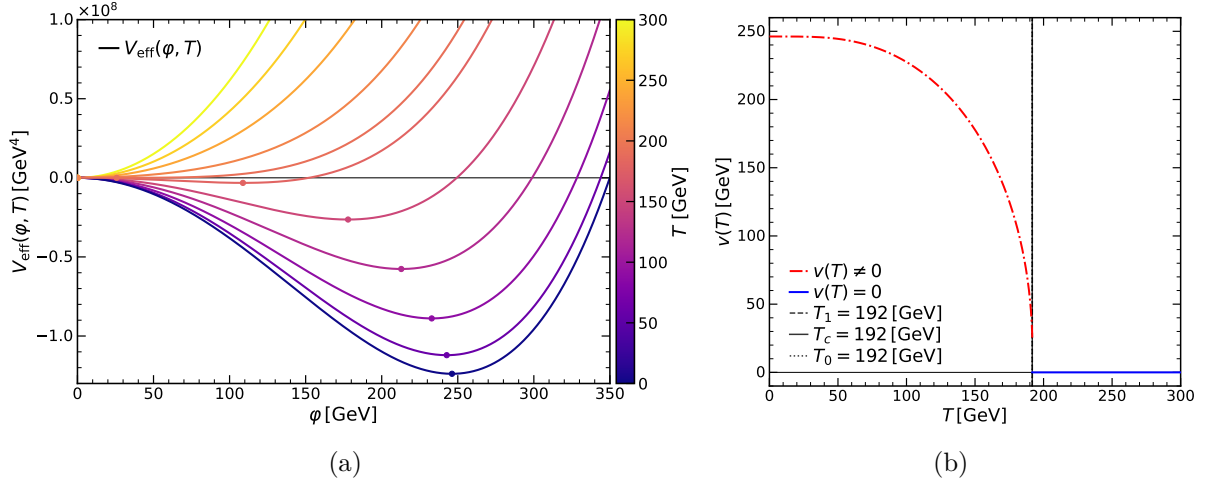


Figure 12: Thermal evolution of the SM effective potential computed using perturbative thermal field theory. (a) One-loop thermal effective potential at different temperatures. (b) Thermal evolution of the minima predicted by our numerical model.

As we can see from figure 12, our numerical model predicts that there is a FOPT in the SM, albeit a very weak one with $\xi_c = 0.13$. This is actually to be expected since as we have said above, perturbative TFT predicts that any QFT with bosons, will have a barrier term that will induce FOPTs. In order to check whether they are not crossovers, which is the case of the SM, one must turn to lattice calculation [12, 19]. Still we can perform these perturbative calculation since in order to allow for baryogenesis, one must have a SFOPT, and therefore ignore results with $\xi_c < 1$.

3.11.1 Electroweak Baryogenesis

As we saw in the introduction, one of the most compelling reasons to study the EWPT is that it offers a promising scenario to explain the BAU. Baryogenesis is the hypothetical physical process that allows us to explain the observed BAU. Assuming initial conditions in which baryon number was a symmetry of the Universe, three conditions, known as the Sakharov conditions [4], are necessary to generate an excess of baryons over antibaryons:

1. **Baryon number violation.** This is trivially required to reach different quantities of baryons and antibaryons, given that the initial state is symmetric. In the Standard Model, baryon number is violated through sphaleron processes whose rate is unsuppressed at temperatures around the electroweak scale [5, 6, 25].
2. **C and CP violation.** These are necessary to ensure that the processes generating an excess of particles and those generating an excess of antiparticles do not mirror each other and cancel out. In the Standard Model, these symmetries are violated in electroweak processes, mainly through quark mixing due to the complex phase of CKM matrix [25].
3. **Departure from thermal equilibrium.** If all particles in the universe remained in thermal equilibrium, then no direction for time would be defined and the baryon violating processes could be reversed, rendering the previous two conditions irrelevant [6].

Electroweak Baryogenesis provides a minimal scenario where these conditions are met in the SM. Baryon number and CP violation are already present in the SM at the energy scales of the EWPT, and the latter serves as our out of equilibrium process. However, there are two main issues in the SM that prevent this: the first is that the CP violation is not sufficient [60–63], and the second is that we require an First-Order Electroweak Phase Transition (FOEWPT), something that we previously saw is excluded in the SM. Therefore new BSM physics is required to account for electroweak baryogenesis.

In the following sections, we will study a minimal extension of the SM consisting of a real singlet coupled to the Higgs doublet. We will see what parameter regions allow for a FOEWPT, and finally check the validity of our model against current experimental measurements.

4 Real Singlet Extension of the Standard Model

In the previous chapters we have learned about the effective potential at both zero and finite temperature. One of the main applications we have focused on is the study of phase transitions and their application in EW baryogenesis. We saw that one of the necessary conditions to induce EW baryogenesis was to have an FOEWPT. We saw that this was excluded by the SM, and therefore motivated the need to consider new BSM theories.

In this chapter we will introduce a minimal extension of the SM that consists of coupling a new singlet scalar field to the Higgs/scalar sector of the SM. The phenomenology of the Real Singlet Extension of the Standard Model (RxSM) is dependent on the assumptions used to build the model, although a common factor is that the Higgs boson measured at the LHC will couple to a new scalar particle.

4.1 Scalar and Tree-Level Potential

The most generic, renormalizable, scalar potential for a RxSM is

$$\begin{aligned} V = & -\mu_H^2 |H|^2 + \lambda_H |H|^4 \\ & + \mu_1^3 S - \frac{1}{2}\mu_S^2 S^2 - \frac{1}{3}\mu_3 S^3 + \frac{1}{4}\lambda_S S^4 \\ & + \mu_{HS} |H|^2 S + \frac{1}{2}\lambda_{HS} |H|^2 S^2, \end{aligned} \quad (4.1)$$

where H is the $SU(2)_L$ Higgs Doublet of the SM (2.74) and S is a real scalar singlet under all symmetry groups.

The scalar fields can be parametrized, in the unitary gauge, as

$$H(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \varphi_h + h(x) \end{pmatrix}, \quad S(x) = (\varphi_s + s(x)), \quad (4.2)$$

where $\varphi_{h(s)}$ is the background field $h(s)$, and whose VEVs are defined as

$$\langle \varphi_h \rangle = v = 246 \text{ GeV}, \quad \langle \varphi_s \rangle = u. \quad (4.3)$$

Using the definitions of H and S , we can write the tree-level effective potential as

$$\begin{aligned} V_0(\varphi_h, \varphi_s) = & -\frac{1}{2}\mu_H^2 \varphi_h^2 + \frac{1}{4}\lambda_H \varphi_h^4 \\ & + \mu_1^3 \varphi_s - \frac{1}{2}\mu_S^2 \varphi_s^2 - \frac{1}{3}\mu_3 \varphi_s^3 + \frac{1}{4}\lambda_S \varphi_s^4 \\ & + \frac{1}{2}\mu_{HS} \varphi_h^2 \varphi_s + \frac{1}{4}\lambda_{HS} \varphi_h^2 \varphi_s^2. \end{aligned} \quad (4.4)$$

4.2 Tadpole Equations and Field Dependent Masses

The singlet, S , described by the potential (4.1), possesses a shift freedom that allows one to either get rid of the tadpole term μ_1 [64] or the singlet's vev u [31, 48]. Through the remainder of this thesis we will adopt the latter, thus setting the EW vacuum to be located at $(\varphi_h, \varphi_s) = (v, 0)$.

The tadpole equations are found by imposing that said EW vacuum be a minimum of our potential

$$\begin{aligned} \left. \frac{\partial V_0}{\partial \varphi_h} \right|_{(v,0)} &= 0 \Rightarrow \mu_H^2 = v^2 \lambda_H, \\ \left. \frac{\partial V_0}{\partial \varphi_s} \right|_{(v,0)} &= 0 \Rightarrow \mu_1^3 = -\frac{v^2 \mu_{HS}}{2}. \end{aligned} \quad (4.5)$$

The field dependent mass matrix of the states h and s is given by

$$M^2(\varphi_h, \varphi_s) = \begin{pmatrix} 3\lambda_H \varphi_h^2 - \mu_H^2 + \frac{1}{2}\lambda_{HS}\varphi_s^2 + \mu_{HS}\varphi_s & \varphi_h(\mu_{HS} + \lambda_{HS}\varphi_s) \\ \varphi_h(\mu_{HS} + \lambda_{HS}\varphi_s) & \frac{1}{2}\varphi_h^2\lambda_{HS} + \varphi_s(3\lambda_S\varphi_s - 2\mu_3) - \mu_S^2 \end{pmatrix}, \quad (4.6)$$

while the field dependent mass of the would-be goldstone bosons is

$$m_{G^{\pm,0}}(\varphi_h, \varphi_s) = \lambda_H \varphi_h^2 - \mu_H^2 + \frac{1}{2}\lambda_{HS}\varphi_s^2 + \mu_{HS}\varphi_s. \quad (4.7)$$

The physical masses are obtained by computing them in the EW vacuum. We recover that the physical masses of the would-be goldstone bosons is zero, but for the states h and s we get a non diagonal mass matrix

$$M_{h/s}^2 = \begin{pmatrix} M_{hh}^2 & M_{hs}^2 \\ M_{hs}^2 & M_{ss}^2 \end{pmatrix} = \begin{pmatrix} 2v^2\lambda_H & v\mu_{HS} \\ v\mu_{HS} & \frac{v^2\lambda_{HS}}{2} - \mu_S^2 \end{pmatrix}, \quad (4.8)$$

meaning that these states are mixed, and therefore are not the physical eigen-states.

We define the mass eigen-states of our theory as h_1 and h_2 with their respective masses m_1, m_2 . This allows us to write a rotation matrix $R(\theta)$ between the interaction and mass eigen-states such that,

$$\begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = R(\theta) \begin{pmatrix} h \\ s \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} h \\ s \end{pmatrix}, \quad (4.9)$$

which must satisfy

$$\begin{aligned} \begin{pmatrix} h_1 & h_2 \end{pmatrix} \begin{pmatrix} m_1^2 & 0 \\ 0 & m_2^2 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} &= \begin{pmatrix} h & s \end{pmatrix} \begin{pmatrix} M_{hh}^2 & M_{hs}^2 \\ M_{hs}^2 & M_{ss}^2 \end{pmatrix} \begin{pmatrix} h \\ s \end{pmatrix} \\ \Rightarrow \begin{pmatrix} m_1^2 & 0 \\ 0 & m_2^2 \end{pmatrix} &= R(\theta) M^2(\varphi_h, \varphi_s) R^T(\theta). \end{aligned} \quad (4.10)$$

Comparing element-wise the previous matrix equality we arrive at

$$\begin{aligned} m_1^2 &= M_{hh}^2 \cos^2 \theta + M_{ss}^2 \sin^2 \theta + M_{hs}^2 \sin(2\theta), \\ m_2^2 &= M_{hh}^2 \sin^2 \theta + M_{ss}^2 \cos^2 \theta - M_{hs}^2 \sin(2\theta), \\ 0 &= M_{hs}^2 \cos(2\theta) + (M_{ss}^2 - M_{hh}^2) \frac{\sin(2\theta)}{2} \Rightarrow \tan(2\theta) = \frac{2M_{hs}^2}{M_{hh}^2 - M_{ss}^2}. \end{aligned} \quad (4.11)$$

This allows us to invert those relations and write the field dependent mass matrix using the physical masses and mixing angle

$$\begin{aligned} M_{hh}^2 &= m_1^2 \cos^2 \theta + m_2^2 \sin^2 \theta, \\ M_{ss}^2 &= m_1^2 \sin^2 \theta + m_2^2 \cos^2 \theta, \\ M_{hs}^2 &= (m_1^2 - m_2^2) \cos \theta \sin \theta. \end{aligned} \quad (4.12)$$

Substituting in the values of the mass matrix elements in the interaction basis (4.8) we find the following relations between parameters

$$\begin{aligned} \lambda_H &= \frac{1}{2v^2} (m_1^2 \cos^2 \theta + m_2^2 \sin^2 \theta), \\ \mu_{HS} &= \frac{\sin \theta \cos \theta}{v} (m_1^2 - m_2^2), \\ \mu_S^2 &= \frac{\lambda_{HS} v^2}{2} - (m_1^2 \sin^2 \theta + m_2^2 \cos^2 \theta). \end{aligned} \quad (4.13)$$

From (4.11) we can see that due to our definition of the mixing angle θ , for a $\theta > 0$, $m_1 > m_2$. This means, if we want exotic Higgs bosons decays, that we must identify h_1 with the Higgs boson discovered at the LHC [8, 9], with mass $m_1 = 125 \text{ GeV}$ [3, 39].

4.3 Parameter Prescription

By rotating our fields into a physical basis, we have seen that we can relate the original (Lagrangian) parameters of the model

$$\{\mu_H, \lambda_H, \mu_1, \mu_S, \mu_3, \lambda_S, \mu_{HS}, \lambda_{HS}\},$$

to their physical counterparts

$$\{m_1, m_2, \theta, v, \lambda_S, \mu_3, \lambda_{HS}\}.$$

Since our objective is to relate our theoretical predictions to experimental observables, it is important to create a mapping between these two prescriptions.

In the previous section we saw how imposing that the EW vacuum sits at $(v, 0)$ automatically fixes μ_H and μ_1 (4.5), the latter of which is actually fixed by the shift freedom in order to ensure that there is no VEV for the singlet field. The parameters λ_H , μ_{HS} and μ_S , are determined by rotating to the physical basis, and are fixed by the masses of our eigen-states (4.13).

Combining all those results together, we arrive at the following parameter prescription

$$\begin{aligned} \mu_H^2 &= \frac{1}{2} (m_1^2 \cos^2 \theta + m_2^2 \sin^2 \theta), \\ \lambda_H &= \frac{1}{2v^2} (m_1^2 \cos^2 \theta + m_2^2 \sin^2 \theta), \\ \mu_1^3 &= \frac{v \sin \theta \cos \theta}{2} (m_2^2 - m_1^2), \\ \mu_S^2 &= \frac{\lambda_{HS} v^2}{2} - (m_1^2 \sin^2 \theta + m_2^2 \cos^2 \theta), \\ \mu_{HS} &= \frac{\sin \theta \cos \theta}{v} (m_1^2 - m_2^2). \end{aligned} \quad (4.14)$$

4.4 Feynman Rules and Higgs-Singlet Coupling

After the rotation to the physical basis (4.9), we changed the scalar sector of the SM and it is therefore important to find how these particles interact with the rest of the SM, and also how they interact with each other.

Directly from the definition of (4.9) we know that the coupling of $h_1(h_2)$ to the particles of the SM are proportional to those between a Higgs boson of mass $m_1(m_2)$ multiplied by $\cos \theta(\sin \theta)$. By identifying h_1 with the Higgs boson measured at the LHC, we can see that since its couplings are closely aligned to those predicted by the SM [65, 66] we have that θ must be very small. Therefore we have an alignment limit where h_1 is aligned with the Higgs doublet H , while h_2 is aligned with the singlet S .

Looking now into the interactions between h_1 and h_2 , since we are interested in the case where $m_2 < m_1/2$ the only new relevant process that we need consider is the decay of $h_1 \rightarrow h_2 h_2$.

In order to find the coupling of this process, $\lambda_{h_1 h_2 h_2}$, we need to collect all the terms contributing to $\mathcal{O}(3)$ in both h or s , since those are the ones that will contribute to the $h_1 h_2 h_2$ vertex. After SSB, the cubic potential becomes

$$V_{\text{cubic}} = \lambda_H v h^3 + \frac{1}{2} \lambda_{HS} v h s^2 + \frac{1}{2} \mu_{HS} h^2 s - \frac{1}{3} \mu_3 s^3. \quad (4.15)$$

Making the rotation (4.9), and collecting all the terms that contain $h_1 h_2 h_2$ we arrive at the following vertex

$$V_{\text{cubic}} \supset \frac{1}{2} \lambda_{h_1 h_2 h_2} v h_1 h_2^2, \quad (4.16)$$

where

$$\lambda_{h_1 h_2 h_2} = \frac{\cos \theta}{2v^2} \left[2m_1^2 \sin^2 \theta + 4m_2^2 \sin^2 \theta + (3 \cos(2\theta) - 1) \lambda_{HS} v^2 - 2v \sin(2\theta) \mu_3 \right]. \quad (4.17)$$

Once we have this vertex, we can compute the width for h_1 to decay into a pair of h_2 scalars. After some elementary QFT computations we arrive at

$$\Gamma(h_1 \rightarrow h_2 h_2) = \frac{v^2 \lambda_{h_1 h_2 h_2}^2}{32\pi m_1} \sqrt{1 - 4 \frac{m_2^2}{m_1^2}}. \quad (4.18)$$

4.5 Tree-Level vacua

In this section, we will carry out a tree-level analysis of the vacuum structure of this model. We use the procedure described in [31] to find the minima at tree level, a strategy that will be useful for the following analyses.

All the minima will be located along two paths on the direction of φ_h , the symmetric path $\varphi_h = 0$, and the broken one $\varphi_h \neq 0$, where our EW vacuum will be located.

First, let us look for all the minima along $\varphi_h = 0$, the tree-level potential along this path is

$$V_0(\varphi_h = 0, \varphi_s) = \mu_1^3 \varphi_s - \frac{1}{2} \mu_S^2 \varphi_s^2 - \frac{1}{3} \mu_3 \varphi_s^3 + \frac{1}{4} \lambda_S \varphi_s^4. \quad (4.19)$$

The minima can be found by solving

$$\mu_1^3 - \mu_S^2 \varphi_s - \mu_3 \varphi_s^2 + \lambda_S \varphi_s^3 = 0, \quad (4.20)$$

and to ensure that they are minimum, they must satisfy

$$3\lambda_S \varphi_s^2 - \mu_S^2 - 2\mu_3 \varphi_s > 0. \quad (4.21)$$

Having located those possible symmetric vacua, let us now look at the extremal points with $\varphi_h \neq 0$. First, we derive w.r.t. φ_h , and equate to 0 to find the path, $D_h(\varphi_s)$, along which these minima will lie.

$$\begin{aligned} \frac{\partial V_0}{\partial \varphi_h} = 0 &\Rightarrow -\varphi_h \mu_H^2 + \lambda_H \varphi_h^3 + \mu_{HS} \varphi_h \varphi_s + \frac{1}{2} \lambda_{HS} \varphi_h \varphi_s^2 = 0 \\ &\Rightarrow \varphi_h^2 \equiv D_h^2(\varphi_s) = \frac{2\mu_H^2 - 2\mu_{HS} \varphi_s - \lambda_{HS} \varphi_s^2}{2\lambda_H}. \end{aligned} \quad (4.22)$$

By doing this, the tree-level potential reduces from two dimensions to one,

$$\begin{aligned} V_0(D_h(\varphi_s), \varphi_s) &= -\frac{\mu_H^4}{4\lambda_H} + \left(\frac{\mu_H^2 \mu_{HS}}{2\lambda_H} + \mu_1^3 \right) \varphi_s \\ &\quad - \frac{2\lambda_H \mu_S^2 + \mu_{HS}^2 - \mu_H^2 \lambda_{HS}}{4\lambda_H} \varphi_s^2 \\ &\quad - \left(\frac{\lambda_{HS} \mu_{HS}}{4\lambda_H} + \frac{\mu_3}{3} \right) \varphi_s^3 + \left(\frac{\lambda_S}{4} - \frac{\lambda_{HS}^2}{16\lambda_H} \right) \varphi_s^4. \end{aligned} \quad (4.23)$$

The minima will come from the solutions of the following equation

$$\frac{\mu_H^2 \mu_{HS}}{2\lambda_H} + \mu_1^3 + \frac{\mu_H^2 \lambda_{HS} - 2\lambda_H \mu_S^2 - \mu_{HS}^2}{2\lambda_H} \varphi_s - \left(\frac{3\lambda_{HS} \mu_{HS}}{4\lambda_H} + \mu_3 \right) \varphi_s^2 + \left(\lambda_S - \frac{\lambda_{HS}^2}{4\lambda_H} \right) \varphi_s^3 = 0 \quad (4.24)$$

while the minima will satisfy

$$\frac{\mu_H^2 \lambda_{HS} - 2\lambda_H \mu_S^2 - \mu_{HS}^2}{2\lambda_H} > \left(\frac{3\lambda_{HS} \mu_{HS}}{2\lambda_H} + 2\mu_3 \right) \varphi_s + \left(\frac{3\lambda_{HS}^2}{4\lambda_H} - 3\lambda_S \right) \varphi_s^2. \quad (4.25)$$

Although equation (4.24) is hard to analytically solve, in our analysis we will find the minima numerically. But substituting the values of the EW vacuum $(v, 0)$, one can show that it is a solution for that equation.

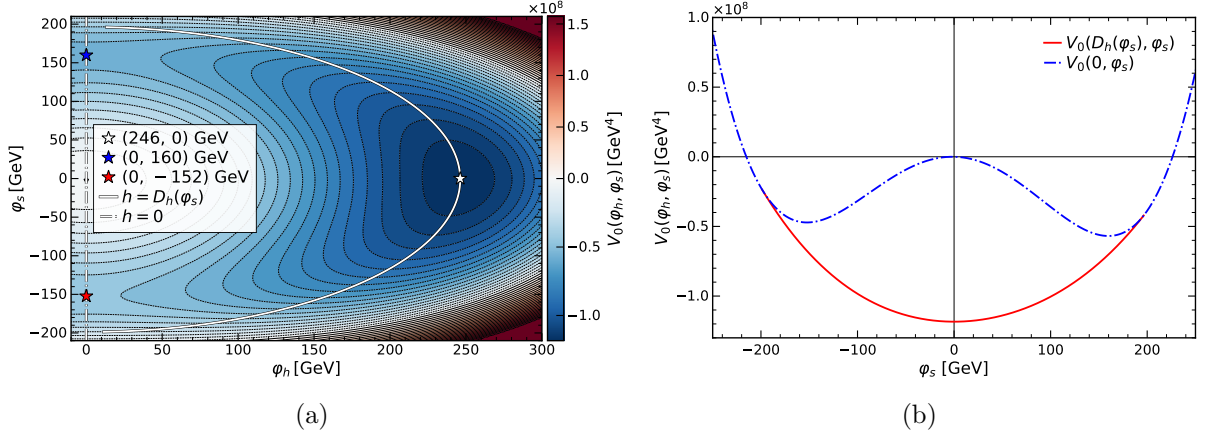


Figure 13: (a) Contour plot of the tree-level effective potential (4.4). The white solid line is the path $D_h(\varphi_s)$ where all the non-symmetric minima lie (4.22), while the dash-dotted line corresponds to $\varphi_h = 0$. The red and blue stars are solutions for the symmetric minima (4.20), while the white star is the EW vacuum $(v, 0)$, also a solution of (4.24). (b) Values of $V_0(\varphi_h, \varphi_s)$ along the paths φ_s (blue), and $D_h(\varphi_s)$ (red). Same parameters as in figure 14

4.6 Coleman-Weinberg and Counter Term Potentials

Having studied the tree-level structure of the effective potential, we can now use those result to write the full one-loop effective potential

$$V_{\text{eff}}(\varphi_h, \varphi_s, T) = V_0(\varphi_h, \varphi_s) + V_{\text{CW}}(\varphi_h, \varphi_s) + V_{\text{CT}}(\varphi_h, \varphi_s) + V_T(\varphi_h, \varphi_s, T), \quad (4.26)$$

where V_0 is the tree-level effective potential of (4.4), V_{CW} the CW potential (4.27), V_{CT} the counter term potential (4.28), and V_T is the thermal effective potential, that we will define in the following sections.

4.6.1 Coleman-Weinberg Potential

From our review of the effective potential in chapter 2, we know that the CW effective potential, in the $\overline{MS}$ renormalization scheme can be computed for any model using (2.37). For our RxSM model we will have

$$V_{\text{CW}}(\varphi_h, \varphi_s) = \sum_i (-1)^{2s_i} N_i \frac{m_i^4(\varphi_h, \varphi_s)}{64\pi^2} \left[\ln \left(\frac{m_i^2(\varphi_h, \varphi_s)}{Q^2} \right) - C_i \right], \quad (4.27)$$

where i runs over all the particles of our model, s_i is the particle's spin and N_i its degrees of freedom, Q is the renormalization scale and C_i is 3/2 for all scalars and the longitudinal polarizations of the gauge bosons, and 1/2 for transverse gauge bosons [46].

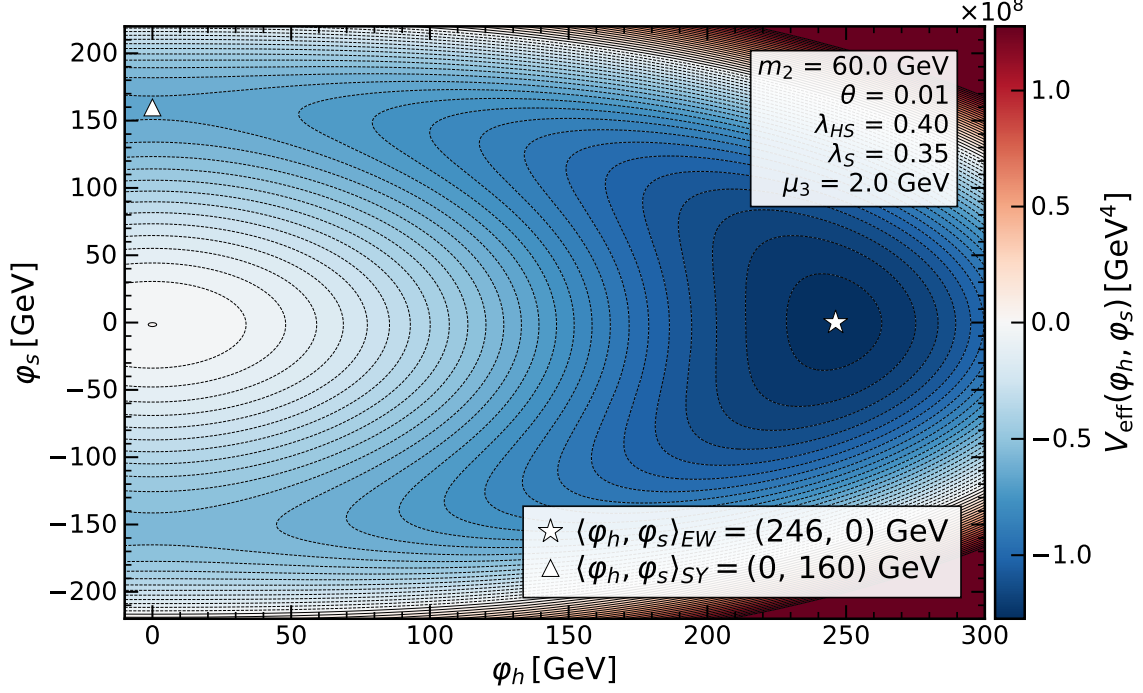


Figure 14: Contour plot of the full one-loop effective potential, at $T = 0$, for the general RxSM (4.26). The star corresponds to the EW vacuum located at $(v, 0)$. The triangle corresponds to the (false) symmetric vacuum located at $(0, w)$.

4.6.2 Counter Term Potential

We will build the counter term potential V_{CT} to preserve the tree-level values and relations found in the previous sections

$$\begin{aligned}
 V_{CT}(\varphi_h, \varphi_s) = & -\delta\mu_H \frac{\varphi_h^2}{2} + \delta\lambda_H \frac{\varphi_h^4}{4} - \delta\mu_S \frac{\varphi_s^2}{2} + \delta\lambda_S \frac{\varphi_s^4}{4} \\
 & + \delta\mu_1^3 \varphi_s - \delta\mu_3 \frac{\varphi_s^3}{3} + \delta\mu_{HS} \frac{\varphi_h^2 \varphi_s}{2} + \delta\lambda_{HS} \frac{\varphi_h^2 \varphi_s^2}{4} + \delta V_0,
 \end{aligned} \quad (4.28)$$

allowing us to directly perform phenomenological studies using the observables as inputs.

In section 4.5 we found that there are at least two vacua in our theory, a vacuum that preserves the EW symmetry located at $(0, w)$ obtained from (4.20), and the true vacuum of the model at $T = 0$ $(v, 0)$ [31].

Our choice of renormalization conditions will be chosen in order to preserve the properties of those vacua. Explicitly they are:

1. Preserve the tree-level positions of the symmetric and EW vacua

$$\left. \frac{\partial (V_{CW} + V_{CT})}{\partial \varphi_h} \right|_{(v,0)} = 0, \quad \left. \frac{\partial (V_{CW} + V_{CT})}{\partial \varphi_s} \right|_{(v,0)} = 0, \quad \left. \frac{\partial (V_{CW} + V_{CT})}{\partial \varphi_s} \right|_{(0,w)} = 0. \quad (4.29)$$

2. Preserve the tree-level values of the mass matrix in the EW vacuum

$$\left. \frac{\partial^2 (V_{CW} + V_{CT})}{\partial \varphi_h^2} \right|_{(v,0)} = 0, \quad \left. \frac{\partial^2 (V_{CW} + V_{CT})}{\partial \varphi_h \partial \varphi_s} \right|_{(v,0)} = 0, \quad \left. \frac{\partial^2 (V_{CW} + V_{CT})}{\partial \varphi_s^2} \right|_{(v,0)} = 0. \quad (4.30)$$

3. Preserve the same singlet cubic coupling

$$\left. \frac{\partial^3 (V_{\text{CW}} + V_{\text{CT}})}{\partial \varphi_s^3} \right|_{(v,0)} = 0. \quad (4.31)$$

4. Preserve the value of the potential at the minima

$$(V_{\text{CW}} + V_{\text{CT}})|_{(v,0)} = 0, \quad (V_{\text{CW}} + V_{\text{CT}})|_{(0,w)} = 0. \quad (4.32)$$

We can now solve the system of equations to arrive at the following counter terms:

$$\begin{aligned} \delta\mu_H &= \frac{3V_{\text{CW}}^h(v,0) - v V_{\text{CW}}^{hh}(v,0)}{2v}, \\ \delta\lambda_H &= \frac{V_{\text{CW}}^h(v,0) - v V_{\text{CW}}^{hh}(v,0)}{2v^3}, \\ \delta\mu_S &= -\frac{1}{6w^2} \left(24V_{\text{CW}}(v,0) - 15v V_{\text{CW}}^h(v,0) + 3v^2 V_{\text{CW}}^{hh}(v,0) - 24V_{\text{CW}}(0,w) \right. \\ &\quad \left. - 9vw V_{\text{CW}}^{hs}(v,0) + 18w V_{\text{CW}}^s(v,0) + 6w V_{\text{CW}}^s(0,w) + w^3 V_{\text{CW}}^{sss}(v,0) \right), \\ \delta\lambda_S &= -\frac{1}{6w^4} \left(24V_{\text{CW}}(v,0) - 15v V_{\text{CW}}^h(v,0) + 3v^2 V_{\text{CW}}^{hh}(v,0) - 24V_{\text{CW}}(0,w) \right. \\ &\quad \left. - 6vw V_{\text{CW}}^{hs}(v,0) + 12w V_{\text{CW}}^s(v,0) + 12w V_{\text{CW}}^s(0,w) - 2w^3 V_{\text{CW}}^{sss}(v,0) \right), \\ \delta\mu_{HS} &= -\frac{V_{\text{CW}}^{hs}(v,0)}{v}, \\ \delta\lambda_{HS} &= -\frac{1}{3v^2 w^2} \left(24V_{\text{CW}}(v,0) - 15v V_{\text{CW}}^h(v,0) + 3v^2 V_{\text{CW}}^{hh}(v,0) \right. \\ &\quad \left. - 24V_{\text{CW}}(0,w) - 9vw V_{\text{CW}}^{hs}(v,0) + 18w V_{\text{CW}}^s(v,0) + 6w V_{\text{CW}}^s(0,w) \right. \\ &\quad \left. + 6w^2 V_{\text{CW}}^{ss}(v,0) + w^3 V_{\text{CW}}^{sss}(v,0) \right), \\ \delta\mu_1^3 &= \frac{1}{2} \left(v V_{\text{CW}}^{hs}(v,0) - 2V_{\text{CW}}^s(v,0) \right), \\ \delta\mu_3 &= \frac{V_{\text{CW}}^{sss}(v,0)}{2}, \\ \delta V_0 &= \frac{1}{8} \left(-8V_{\text{CW}}(v,0) + 5v V_{\text{CW}}^h(v,0) - v^2 V_{\text{CW}}^{hh}(v,0) \right), \end{aligned} \quad (4.33)$$

where we have used the notation $V_{\text{CW}}^{ij}(a,b)$ to denote the derivative of V_{CW} w.r.t i and j , evaluated at the point (a,b) .

4.7 Thermal Potential and Thermal Masses

To compute the thermal potential V_T , we will numerically evaluate the thermal functions (3.47) using **CosmoTransitions**. We will use the Parwani method (3.74) to perform the daisy resummation.

Using the computation method defined in [28], we find that the thermal masses for the bosons (scalars and longitudinal components of gauge bosons) of our theory are

$$\begin{aligned} \Pi_{h,G^{\pm,0}}(T) &= \frac{1}{48} T^2 \left(12y_b^2 + 9g^2 + 3g_w^2 + 24\lambda_H + 2\lambda_{HS} + 12y_t^2 \right), \\ \Pi_s(T) &= \frac{1}{12} T^2 (2\lambda_{HS} + 3\lambda_S), \quad \Pi_W(T) = \frac{11g^2 T^2}{6}, \\ M_{Z/\gamma}^2(\varphi, T) &= \begin{pmatrix} \frac{g^2 \varphi^2}{4} + \frac{11g^2 T^2}{6} & -\frac{gg' \varphi^2}{4} \\ -\frac{gg' \varphi^2}{4} & \frac{g'^2 \varphi^2}{4} + \frac{11g'^2 T^2}{6} \end{pmatrix}. \end{aligned} \quad (4.34)$$

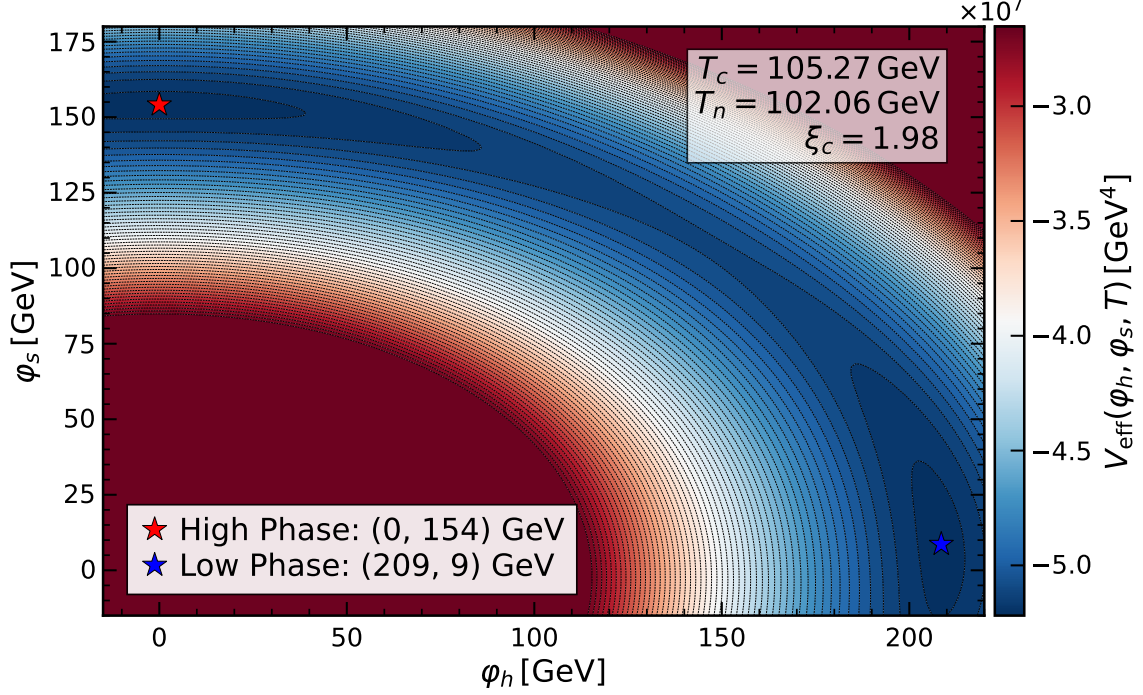


Figure 15: Contour plot of the thermal effective potential at $T = T_c$, where T_c is the critical temperature. The red star indicates the phase the universe is currently in, which for this model is the false symmetric vacuum. The blue star indicates the phase the universe will transition to, which for this model is the true EW vacuum. Same parameters as in figure 14.

The main advantage of using `CosmoTransitions` for the computation of our thermal effective potential is that we can use the built-in functions to compute the critical temperature T_c and the nucleation temperature T_n , as shown in figure 15. Therefore allowing us to compute the strength of the phase transition, and check that the universe does not remain trapped in the false vacuum. An observation we can make from figure 16 is that $T_n > T_0$ if T_0 exists, since at $T < T_0$, the false vacuum becomes a saddle point, and therefore unstable.

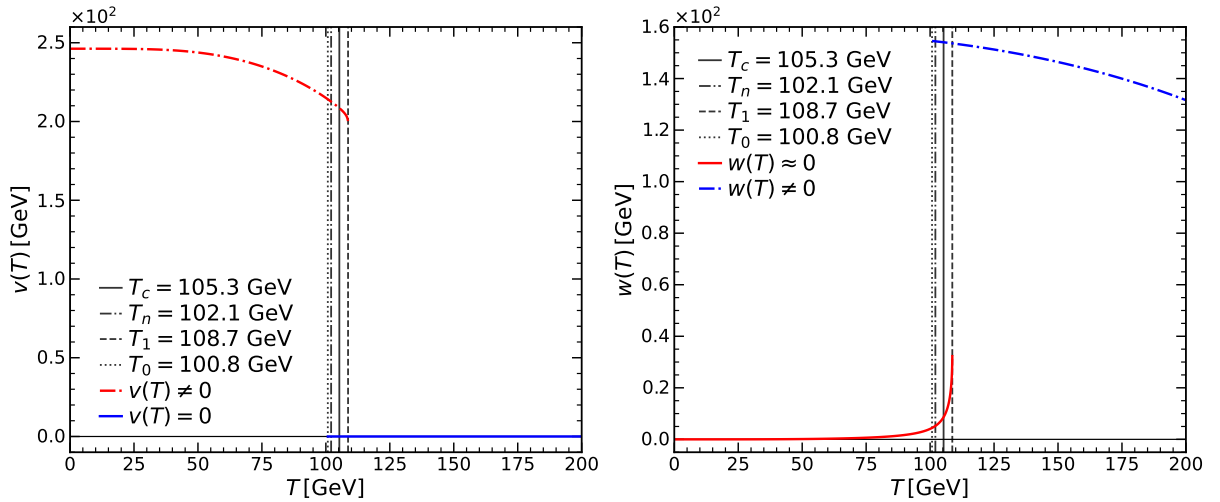


Figure 16: Thermal evolution of the position of the symmetric (blue) and EW (red) minima. On the left(right) plot we present the evolution along the $\varphi_h(\varphi_s)$ axis as a function of temperature. We also show as vertical lines the critical and nucleation temperatures, along with T_0 (3.58) and T_1 (3.61). Same parameters as in figure 14.

4.8 On the Z_2 Symmetric Models

Throughout the previous section we explored the behavior of the most general version of the RxSM model. Setting m_1 and v to their SM experimental values, the model has 5 remaining free parameters: the mass of the singlet m_2 , the mixing angle θ , the portal and singlet quartic couplings λ_{HS} and λ_S , and the singlet cubic coupling μ_3 . An important subset of RxSM impose a Z_2 symmetry on the singlet field, i.e. $S \rightarrow -S$ [16, 48].

These Z_2 symmetric models offer the main advantage of reducing the parameter space from a five-dimensional one to a three-dimensional one. The choice of parameters is highly dependent on whether or not the Z_2 is spontaneously broken at the EW vacuum. In the totally generic model we have a shift symmetry that allowed us to tune the tadpole term μ_1 and the singlet VEV u in order to set one of the two to zero, imposing a Z_2 symmetry automatically sets μ_1 , alongside μ_3 and μ_{HS} , to zero, hence losing the shift symmetry that allowed us to set $u = 0$ [31].

We can therefore identify two different families of Z_2 symmetric models, those with $u = 0$ and those with $u \neq 0$. The phenomenology of those models is vastly different mainly due to the fact that in the unbroken model ($u = 0$): the fields H and S are no longer mixed¹⁷, therefore the model becomes one where the 125 GeV Higgs boson couples to the singlet s , and since this one does not couple to any other SM particles, it becomes a stable particle. The next chapter will be devoted to this specific model, and we will try to find the regions in the parameter space compatible with a FOEWPT, and compare those points against current LHC constraints.

The spontaneously broken Z_2 RxSM is studied in [64]. The main phenomenological difference between this and the Z_2 preserving model is that the H and S fields remain mixed due to the singlet's VEV. This can not be seen directly in (4.8) since we have derived all of our relations using the shift freedom and setting $u = 0$, but in appendix A we re-derive all of the relations but now allowing for $u \neq 0$ and setting $\mu_1 = 0$. This new prescription of the general RxSM is identical to the one of this chapter due to the shift freedom, but by imposing a Z_2 symmetry we can recover the spontaneously broken Z_2 RxSM, something impossible in our current prescription. The studies of this model, alongside the fully generic RxSM are left as possible future works.

¹⁷See equation (4.8) and set $\mu_{HS} = 0$.

5 First Order Phase Transitions in a Z_2 preserving RxSM

In this chapter we will study the phase transitions of a RxSM with an unbroken Z_2 symmetry. In order to achieve this we need to set the Lagrangian parameters μ_1 , μ_3 and μ_{HS} to zero, translating this into our physical parametrization (4.14) we see that it becomes setting $\theta = 0$ and $\mu_3 = 0$. The fact that the mixing angle θ is zero is an interesting consequence of the Z_2 preserving model, this means that the mass matrix is diagonal and therefore h and s are physical particles.

5.1 Model and Parameters

Taking the Z_2 limit, as previously defined, we get that the tree-level potential becomes

$$V_0(\varphi_h, \varphi_s) = -\frac{1}{2}\mu_H^2\varphi_h^2 + \frac{1}{4}\lambda_H\varphi_h^4 - \frac{1}{2}\mu_S^2\varphi_s^2 + \frac{1}{4}\lambda_S\varphi_s^4 + \frac{1}{4}\lambda_{HS}\varphi_h^2\varphi_s^2. \quad (5.1)$$

By looking at the mass matrix (4.8) we see that it is diagonal, so the particle spectrum can be easily found since there is no mixing between the Higgs and Singlet field. The particle information of this model is presented in table 2.

Particle(s)	$m_j^2(\varphi_h, \varphi_s)$	$\bar{N}_j$
χ	$\lambda_H\varphi_h^2 + \frac{1}{2}\lambda_{HS}\varphi_s^2 - \mu_H^2$	3
h	$3\lambda_H\varphi_h^2 + \frac{1}{2}\lambda_{HS}\varphi_s^2 - \mu_H^2$	1
s	$\frac{1}{2}\lambda_{HS}\varphi_h^2 + 3\lambda_S\varphi_s^2 - \mu_S^2$	1
$t(b)$	$\frac{1}{2}y_{t(b)}^2\varphi_h^2$	$-12(-12)$
$W^\pm$	$\frac{1}{4}g^2\varphi_h^2$	6
Z	$\frac{1}{4}(g^2 + g'^2)\varphi_h^2$	3

Table 2: Particle content of the RxSM model with a Z_2 , along with their respective field dependent masses and degrees of freedom.

For our phenomenological parameter scan it is convenient to use the physical parametrization defined in (4.14). In this limit, the parametrization becomes

$$\mu_H^2 = \frac{1}{2}m_h^2, \quad \lambda_H = \frac{1}{2}\frac{m_h^2}{v^2}, \quad \mu_S^2 = \frac{\lambda_{HS}v^2}{2} - m_s^2, \quad (5.2)$$

where due to our diagonal basis it is convenient to use $m_{1(2)} \equiv m_{h(s)}$.

5.2 Tree-Level Vacua and Stability Constraints

Due to the reduction of parameters, the equations that determine the positions of the possible tree-level vacua now have analytical solutions. Solving them, we find that there are 4 types of possible vacua:

1. The totally symmetric vacuum

$$(\varphi_h, \varphi_s) = (0, 0). \quad (5.3)$$

2. The EW vacuum

$$(\varphi_h, \varphi_s) = \left(\pm \frac{\mu_H}{\sqrt{\lambda_H}}, 0\right) = (\pm v, 0). \quad (5.4)$$

3. The (Z_2) symmetry breaking vacuum

$$(\varphi_h, \varphi_s) = \left(0, \pm \frac{\mu_S}{\sqrt{\lambda_S}}\right) = (0, \pm w). \quad (5.5)$$

4. And, the totally asymmetric vacuum

$$(\varphi_h, \varphi_s) = \left(\pm \sqrt{\frac{2\lambda_{HS}\mu_S^2 - 4\lambda_S\mu_H^2}{\lambda_{HS}^2 - 4\lambda_H\lambda_S}}, \pm \sqrt{\frac{2\lambda_{HS}\mu_H^2 - 4\lambda_H\mu_S^2}{\lambda_{HS}^2 - 4\lambda_H\lambda_S}}\right). \quad (5.6)$$

Imposing that the EW vacuum is the true one with respect to the Z_2 symmetric one i.e. $V_0(v, 0) < V_0(0, w)$, we arrive at the following constraint on λ_S

$$\lambda_S > \frac{(\lambda_{HS}v^2 - 2m_s^2)^2}{2v^2m_h^2} \quad (5.7)$$

There is no need to impose that the EW vacuum be deeper than the totally asymmetric vacua since if the EW exists, they will be saddle points¹⁸. This can be verified by actually carrying out the computation, where we arrive at

$$-\frac{\mu_H^4}{4\lambda_H} < \frac{\lambda_S\mu_H^4 - \lambda_{HS}\mu_H^2\mu_S^2 + \lambda_H\mu_S^4}{\lambda_{HS}^2 - 4\lambda_H\lambda_S}. \quad (5.8)$$

for $\lambda_{HS}^2 - 4\lambda_H\lambda_S > 0$ it follows that

$$0 < \lambda_{HS}^2\mu_H^4 + 4\lambda_H^2\mu_S^4 - 4\lambda_{HS}\lambda_H\mu_H^2\mu_S^2 \Rightarrow 0 < \left(2\lambda_H\mu_S^2 - \lambda_{HS}\mu_H^2\right)^2, \quad (5.9)$$

which is always true. While for $\lambda_{HS}^2 - 4\lambda_H\lambda_S < 0$ we obtain the same relation (5.9) but with the sign flipped $0 > \left(2\lambda_H\mu_S^2 - \lambda_{HS}\mu_H^2\right)^2$, which is unphysical. Therefore this proves that when those totally asymmetric points exist, they can not be the true vacuum of V_0 .

5.3 High Temperature Limit

Let us now perform the high temperature expansion of the thermal functions $J_{B/F}(m_i^2/T^2)$ (3.46). The main reason for doing this is that through this limit we can obtain an analytic expression for the thermal effective potential. This will allow us to get analytical estimates for both the critical temperature T_c and the strength of the phase transition ξ_c .

From our review of Thermal Field Theory we know that our thermal effective potential (3.52) can be expanded, in a high temperature regime, as

$$\begin{aligned} V_{\text{eff}}(\varphi_h, \varphi_s, T) \approx & V_0(\varphi_h, \varphi_s) - \frac{\pi^2}{90} \left(\sum_B \bar{N}_B - \frac{7}{8} \sum_F \bar{N}_F \right) T^4 \\ & + \frac{T^2}{24} \sum_B \bar{N}_B m_B^2(\varphi_h, \varphi_s) - \frac{T^2}{48} \sum_B \bar{N}_F m_F^2(\varphi_h, \varphi_s) \\ & - \frac{T}{12\pi} \sum_B \bar{N}_B m_B^3(\varphi_h, \varphi_s), \end{aligned} \quad (5.10)$$

where we have truncated our expansion to $\mathcal{O}(m_j^3/T^3)$, resulting in no CW-like contributions, and omitted the thermal Debye masses (3.74).

¹⁸This is only true for this model since the Z_2 imposes that both the $+$ and $-$ solutions have the same curvature. Therefore if the EW vacuum is a minima along φ_s , this points must be maxima.

Writing out the sum of (5.10) using the information of table 2, and collecting the terms in powers of φ_h and φ_s , we arrive at [64]

$$V_{\text{eff}}(\varphi_h, \varphi_s, T) \approx \frac{1}{2} (c_H T^2 - \mu_H^2) \varphi_h^2 + \frac{1}{2} (c_S T^2 - \mu_S^2) \varphi_s^2 - B^{SM} T \varphi_h^3 - B^{BSM}(\varphi_h, \varphi_s) T + \frac{1}{4} \lambda_H \varphi_h^4 + \frac{1}{4} \lambda_S \varphi_s^4 + \frac{1}{4} \lambda_{HS} \varphi_h^2 \varphi_s^2, \quad (5.11)$$

where we have eliminated the constant terms and we have defined the thermal coefficients $c_{H(S)}$, and the barrier terms $B^{SM(BSM)}$ as

$$\begin{aligned} c_H &= \frac{1}{48} (9g^2 + 3g'^2 + 12y_b^2 + 12y_t^2 + 24\lambda_H + 2\lambda_{HS}), \\ c_S &= \frac{1}{4} \lambda_S + \frac{1}{6} \lambda_{HS}, \\ B^{SM} &= \frac{1}{32\pi} [2g^3 + (g^2 + g'^2)^{3/2}], \\ B^{BSM}(\varphi_h, \varphi_s) &= \frac{1}{2\pi} \left[(m_h^2(\varphi_h, \varphi_s))^{3/2} + (m_s^2(\varphi_h, \varphi_s))^{3/2} + 3 (m_G^2(\varphi_h, \varphi_s))^{3/2} \right]. \end{aligned} \quad (5.12)$$

For the remainder of this analysis we are going to ignore both of the barrier terms [64]. This will allow us to find exact solutions for the estimates of T_c and ξ_c , but also it is a justified choice since $B^{SM} \approx 0.01$ is small¹⁹, and as we will see the strength of our FOPT will be driven by tree-level renormalizable effects [57].

5.4 Types of Phase Transitions

This model has 3 different types of transitions, all of them triggered in different regions of the parameter space. Those transitions, shown in figure 17, are: a) A direct SM-Like phase transition, b) a Two-Step phase transition from a (EW) symmetric vacuum to the to-be EW vacuum, and c) a three-step phase transition similar to b but now involving an intermediate vacuum.

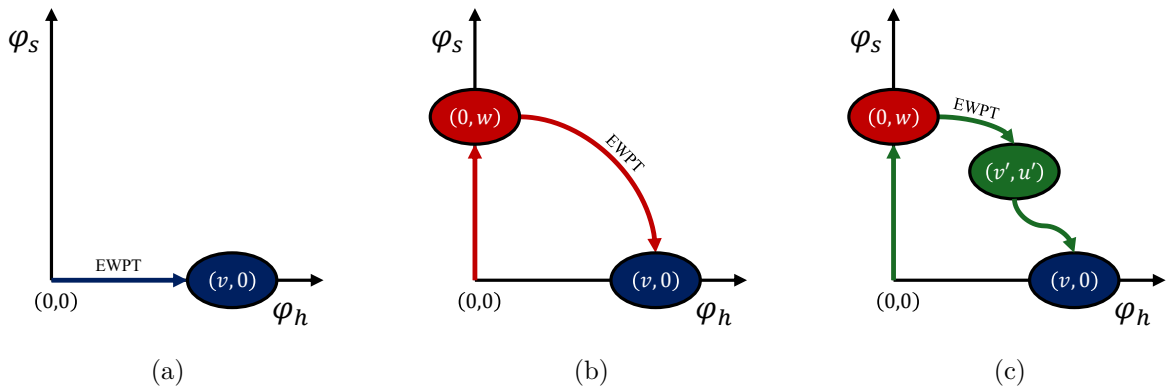


Figure 17: Types of Phase Transitions in a Z_2 symmetric RxSM: a) SM-like phase transition from the totally symmetric vacuum and the EW vacuum. b) Two-step phase transition, where the EWPT occurs between the symmetric vacuum $(0, w)$ and the EW vacuum $(v, 0)$. c) Three-step phase transition, where the EWPT occurs between the symmetric vacuum and an intermediate vacuum (v', u') , where this one as the temperature decreases thermally evolves into the EW vacuum.

¹⁹Note that this term is entirely due to SM effects. As we saw when we studied the thermal effective potential of the SM, this radiative term is not large enough to cause a SFOEWPT.

In the following sections we will use the previously established high temperature thermal effective potential (5.11) in order to study the parameter regions corresponding to each type of phase transition.

The SM-like transitions, like those of figure 17a only deviate from those of the SM due to the effects of B^{BSM} . That is to say, if B^{BSM} is small, like what we are expect to be [64], the barrier separating the totally symmetric vacuum and the EW vacuum does not receive any relevant correction and we therefore expect all of these transitions to be weak and therefore not worth studying if we are interested in SFOEWPTs.

5.4.1 Two-Step Phase Transitions

The two-step phase transitions like those in figure 17b, occur between the symmetric and EW vacua. Differentiating the high temperature potential (5.11), we obtain that those vacua are located, at finite temperature, at

$$v(T) = \sqrt{\frac{\mu_H^2 - T^2 c_H}{\lambda_H}}, \quad w(T) = \sqrt{\frac{\mu_S^2 - T^2 c_S}{\lambda_S}}. \quad (5.13)$$

A necessary (but not sufficient) condition for this kind of phase transition is the existence of $w(T)$ [48]. Imposing this results in the following constraint on λ_{HS}

$$\lambda_{HS} > \frac{3(4m_s^2 + T^2 \lambda_S)}{2(3v^2 - T^2)}, \quad (5.14)$$

but if we set $T = 0$, i.e. imposing that the symmetric vacuum exists at zero temperature, we arrive at the temperature independent condition

$$\lambda_{HS} > \frac{2m_s^2}{v^2}. \quad (5.15)$$

The critical temperature, T_c , can be found by imposing that the potential (5.11) be degenerate for both minima, at the same temperature, i.e. $V_{\text{eff}}(v(T_c), 0, T_c) = V_{\text{eff}}(0, w(T_c), T_c)$. Substituting the results of (5.13), we can find the value of the potential in those points as a function of T , allowing us to find T_C as

$$\begin{aligned} V_{\text{eff}}(v(T_c), 0, T_c) = V_{\text{eff}}(0, w(T_c), T_c) &\Rightarrow \frac{(\mu_H^2 - c_H T_c^2)^2}{4\lambda_H} = \frac{(\mu_S^2 - c_S T_c^2)^2}{4\lambda_S} \\ &\Rightarrow T_c^2 = \frac{\mu_S^2 \sqrt{\lambda_H} - \mu_H^2 \sqrt{\lambda_S}}{c_S \sqrt{\lambda_H} - c_H \sqrt{\lambda_S}}. \end{aligned} \quad (5.16)$$

With the critical temperature (5.16) and the position of the vacua (5.13), we can compute the strength of the phase transition via

$$\xi_c = \frac{v(T_c)}{T_c} = \sqrt{\frac{c_S \mu_H^2 - c_H \mu_S^2}{\lambda_H \mu_S^2 - \mu_H^2 \sqrt{\lambda_H \lambda_S}}}. \quad (5.17)$$

Since we are interested in SFOEWPTs we will need $\xi_c \geq 1$, and to ensure that the universe is not trapped in the false symmetric vacuum we will require T_n to exists, where T_n will be computed using `CosmoTransitions`.

Using the expression (5.17) we find that one set of parameters that produce a two-step phase transition is $m_s = 54.29 \text{ GeV}$, $\lambda_{HS} = 0.184$, $\lambda_S = 0.059$, with $T_c = 114.8 \text{ GeV}$ and $\xi_c = 1.21$. From figure 18 we see the values of T_c and ξ_c from `CosmoTransitions` for the same parameters.

Comparing both values we see discrepancies of 7 % for T_c and 17 % for ξ_c . This is to be expected since this procedure was done to find estimates for those observables, although due to the small

enough difference, we will use (5.17) to perform a high temperature parameter scan, to compare with the one performed using CosmoTransitions.

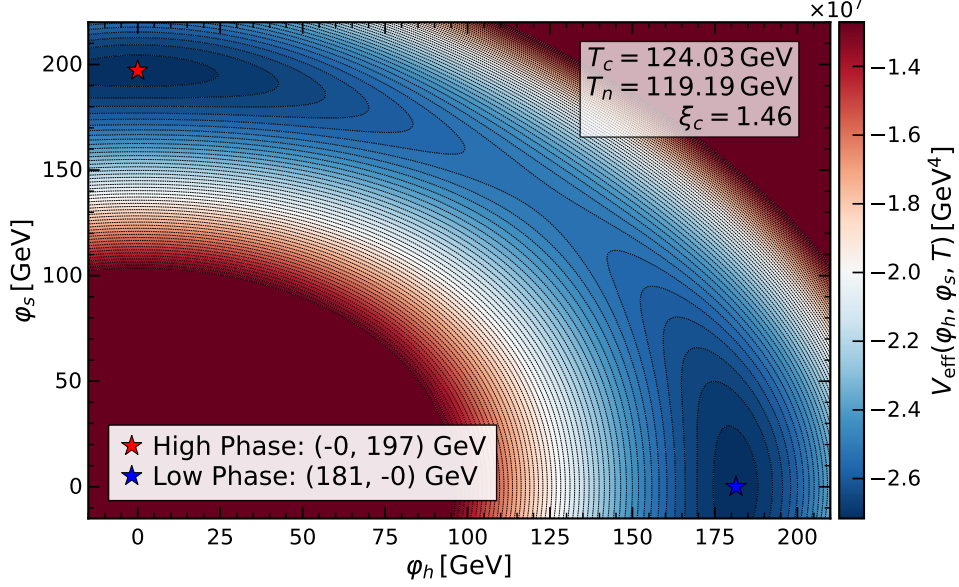


Figure 18: Thermal effective potential, at the critical temperature T_c , for a Z_2 symmetric RxSM with a two-step strong first-order phase transition.

5.4.2 Three-Step Phase Transitions

The three-step phase transitions in our model are interesting to characterize for two main reasons: the first is that they occur due to radiative effects, and the second is that they reduce significantly the strength of the transition so we will want to be able to exclude that region from our scans.

In order to study these types of transitions we need to find the location of the vacua involved, see figure (17c). In order to do that we minimize the high temperature potential (5.11), and differentiating w.r.t. φ_s , we find that the minima are located along the following paths

$$\varphi_s = 0, \quad \& \quad \varphi_s^2 = D_s^2(\varphi_h, T) \equiv -\frac{T^2 c_S - \mu_S^2}{\lambda_S} - \frac{\lambda_{HS}}{2\lambda_S} \varphi_h^2. \quad (5.18)$$

The EW vacuum will be located along the $\varphi_s = 0$ path, while the intermediate and symmetric vacua will be located along $D_s(\varphi_h, T)$. The intermediate vacuum thermally evolves along this path until it coincides with the EW vacuum.

The potential along the path $D_s(\varphi_h)$ is

$$\begin{aligned} V_{\text{eff}}(\varphi_h, D_s(\varphi_h), T) = & -\frac{(\mu_S^2 - T^2 c_S)^2}{4\lambda_S} + \frac{1}{2} \left((T^2 c_H - \mu_H^2) - \frac{\lambda_{HS}(T^2 c_S - \mu_S^2)}{2\lambda_S} \right) \varphi_h^2 \\ & - B^{SM} T \varphi_h^3 + \frac{1}{4} \left(\lambda_H - \frac{\lambda_{HS}^2}{4\lambda_S} \right) \varphi_h^4, \end{aligned} \quad (5.19)$$

where we have kept the SM barrier term in order to remarks its relevance in creating FOPTs.

The potential of (5.19) allows us to simplify the two dimensional effective potential by only considering the 1D potential along that path. For this potential we can make two observations:

1. As we have mentioned before, a three-step FOPT is only possible due to radiative effects. This is directly shown by removing the barrier term in (5.19) and seeing that the transition becomes second order.

2. In order for an intermediate point to form that intermediate point must be a stable minima, if not it would be a tree-level barrier and we would recover a two-step PT. This translates into imposing the potential to be bounded from below, i.e.

$$\left(\lambda_H - \frac{\lambda_{HS}^2}{4\lambda_S} \right) > 0 \Rightarrow \lambda_{HS} < 2\sqrt{\lambda_H\lambda_S}. \quad (5.20)$$

If $\lambda_{HS} < 2\sqrt{\lambda_H\lambda_S}$, at $T = T_c$ we will have that both the symmetric minima and a false intermediate vacuum located at $v'(T), u'(T)$, with $u'(T) \neq 0$, are both degenerate minima as shown in figure 19. As the temperature drops, the universe transitions from that symmetric vacuum to the intermediate one, and this one as the temperature drops to $T \rightarrow 0$ thermally evolves into the true EW vacuum $(v, 0)$ (see figure 20).

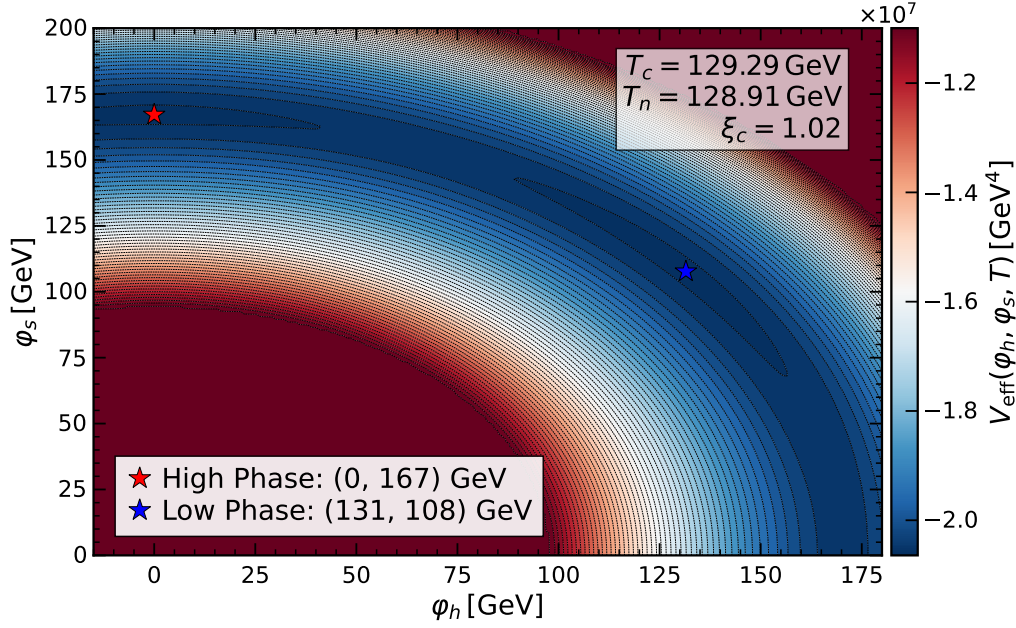


Figure 19: Thermal effective potential, at the critical temperature T_c , for a Z_2 symmetric RxSM with a three-step strong first-order phase transition.

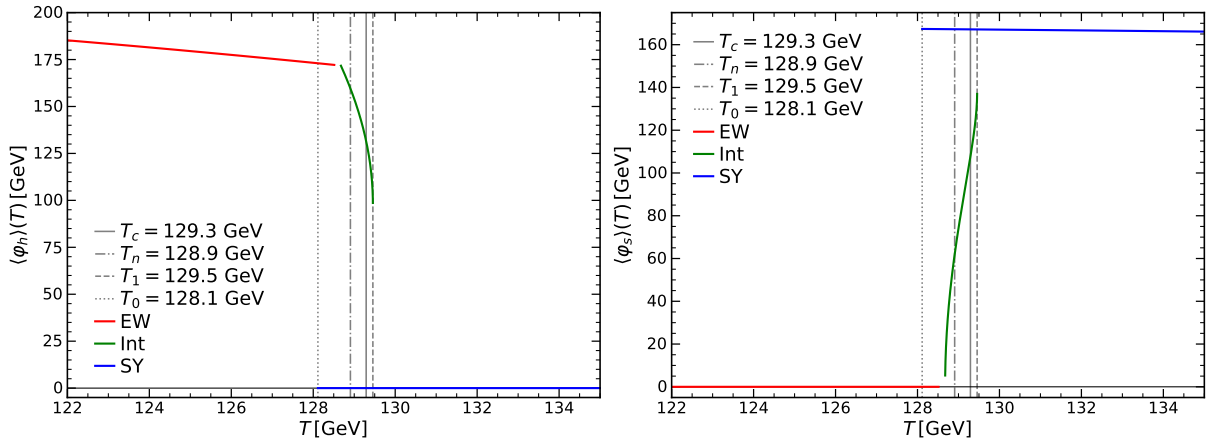


Figure 20: Thermal evolution of the position of the symmetric (blue), intermediate (green) and EW (red) minima. The left(right) shows the evolution along the $\phi_h(\phi_s)$ axis as a function of temperature. Same parameters as in figure 19.

If $\lambda_{HS} > 2\sqrt{\lambda_H \lambda_S}$, we have a two-step phase transition like those we studied in the previous section. It is important to note, that due to the radiative nature of the three-step phase transitions we can not correctly predict them with the high temperature effective potential (5.10). It is for that reason that when we perform the high temperature parameter scan we will have to cut all regions compatible with (5.20).

5.5 Analytical Estimate for Vacuum Trapping

On chapter three we saw the concept of vacuum trapping, and how one can be sure it does not happen to a specific set of parameters by computing the nucleation temperature T_n [29]. Through this chapter we have seen the usefulness of being able to analytically compute the parameters of a phase transition in the high temperature limit, it is then natural to want to do the same to the computation of T_n in order to check whether our phase transitions are free from vacuum trapping.

When we first studied first- and second-order phase transitions, we defined T_0 as the temperature at which the symmetric vacuum becomes unstable, as clearly seen in the examples of figures 16 and 20. In both cases see that $T_n > T_0$, which is expected: trapping requires the universe to sit in a stable minimum, and this is only possible for $T > T_0$. Below T_0 the symmetric vacuum is no longer a stable minimum, so the universe cannot remain trapped there.

Therefore, for our high temperature study we will substitute the vacuum trapping condition from imposing the existence of T_n to imposing the existence of T_0 . Of course, this is only a crude approximation, but as we will see when we compare both the high temperature and numeric scans, we will see that it is very efficient at getting rid of the phase transition candidates that become trapped.

Applying this condition to our Z_2 symmetric RxSM, imposing that T_0 exists is the same as imposing that the symmetric vacuum $(0, w)$ be a saddle-point at $T = 0$. Imposing that the φ_h curvature at $(0, w(T = 0))$ be negative we arrive at the following condition

$$\left. \frac{\partial^2 V_0(\varphi_h, \varphi_s)}{\partial \varphi_h^2} \right|_{(0, w)} < 0 \Rightarrow \frac{\lambda_{HS} \mu_S^2}{2\lambda_S} - \mu_H^2 < 0 \Rightarrow \lambda_S > \frac{\lambda_{HS} (v^2 \lambda_{HS} - 2m_s^2)}{2m_h^2}. \quad (5.21)$$

To understand the physical meaning of this condition we need to remember that when we compute the tunneling temperature we are saying that the field will tunnel through the barrier term in order to reach the true vacuum. This condition is imposing that the barrier term vanishes at $T = 0$ enforcing that there must always be a point where the barrier is small enough to allow tunneling. Of course this is only an approximation, as we will see once we carry out the full numerical parameter scan, the barrier can be small enough and still exist as to allow nucleation at $T = 0$.

5.6 Parameter Scans

For this model we will perform two parameter scans, an analytical high temperature one and a numerical one enabled by `CosmoTransitions`, over the same regions of the parameter space in order to be able to compare both of their predictions.

Since we are interested in the possibility of exotic Higgs decays we scan $m_h \in [5 \text{ GeV}, m_h/2]$, and we scan logarithmically $\lambda_{HS}, \lambda_S \in [10^{-5}, 1]$. While we also need impose that the EW vacuum is the global minima of the potential, which algebraically corresponds to the constraint (5.7).

5.6.1 High Temperature Limit

The high temperature parameter scans will be performed using the analytical expressions found in section 5.4. The high temperature expansion was performed up to $\mathcal{O}(m_j^3/T^3)$, resulting in the thermal effective potential (5.11). Since we are neglecting the thermal barrier terms B^{SM} and B^{BSM} , we will be only able to reliably scan over the parameter region that results in a two-step phase transition, this means that we will have to keep only the parameters that satisfy $\lambda_{HS} > 2\sqrt{\lambda_H\lambda_S}$ (see section 5.4.2). Finally, the last thing we will need to satisfy is our vacuum trapping estimate (5.21).

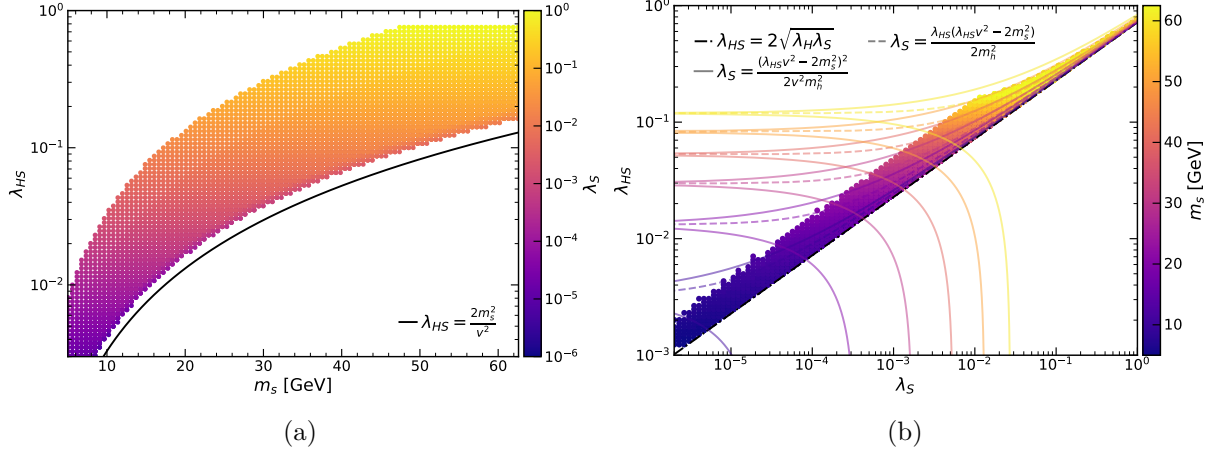


Figure 21: High temperature scan of the values of λ_{HS} consistent with a two-step SFOEWPT as a function of m_s , and λ_S in plots (a) and (b) respectively, the remaining variable is represented in a color map. The black curve in plot (a) corresponds to imposing that the symmetric vacuum $w(T)$ exists at $T = 0$ (5.15). While the dot-dashed, solid, and dashed lines in plot (b) correspond to imposing a two-step phase transition (5.20), imposing that the EW vacuum is the true vacuum (5.7), and imposing the symmetric vacuum to be a saddle-point (5.21) as to induce nucleation at tree-level, respectively.

In figure 21 we show the results of the high temperature parameter scan. Plot 21a(21b) shows the points compatible with a SFOEWPT in the subspace formed by the mass(quartic coupling) of the singlet $m_s(\lambda_S)$ and the portal quartic coupling λ_{HS} . This subspace is a two-dimensional projection of the full three-dimensional parameter space $\{m_s, \lambda_{HS}, \lambda_S\}$, the remaining parameter $\lambda_S(m_s)$ for plot 21a(21b) is plotted over, and to represent this, we color-code the points with the values of that third variable²⁰.

5.6.2 Numeric Scan

Using the `CosmoTransitions` built-in functions to compute the critical and nucleation temperatures, as well as the positions of the vacua at different temperatures, we can repeat the analytic parameter scan of figure 21 but now with the full numeric model (4.26).

The results for the numeric scan are shown in figure 22. Comparing the numerical scan to the analytic high temperature results of figure 21 we can see a clear overlap between the regions that are compatible with SFOEWPTs. The overlap serves as a validity check since although both approaches are not equivalent, one being an approximation of the other, we would expect an agreement between both in the points where the transition admits the high temperature hypothesis $T \gg m_j$. We see that this is in fact the case by identifying that the regions where

²⁰Since all the points with different values of λ_S or m_s are degenerate, only the color of the maximum value of those variables is shown.

there is more discrepancy are the regions where the critical temperature is lowest, this can be seen in figures 24 and 25 where we have color coded our points according to the values of the critical temperature.

This agreement between both strategies can therefore be used to our advantage in trying to minimize the computational time of the parameter scans. In order to compute all the points of figure 22 it took a personal laptop around 4 hours of computational time, this is mainly due to the reliance on the built-in functions of `CosmoTransitions` to compute the phase transition observables per each point. On the other hand, on the previous section we saw that by taking the high temperature limit of the thermal potential we can find analytic closed form relations for many of the phase transition observables, using those relations we can complete a scan like that of figure 21 in fractions of a second. By first finding analytic relations in the high temperature limit, we can perform a fast preliminary scan using them, after having found the high temperature candidate points, we can then use them to tune our parameter space so the numeric scans cover that same area without having to waste computational time on areas incompatible with SFOEWPTs.

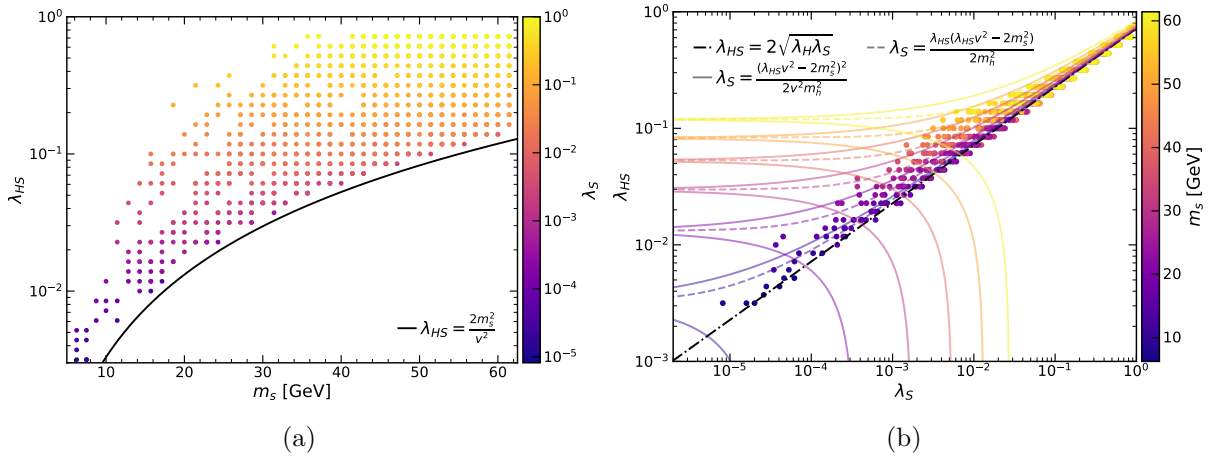


Figure 22: Parameter scan of the values of $m_s, \lambda_{HS}, \lambda_S$ consistent with a two-step SFOEWPT. The plots are structure in the same way as in figure 21. In contrast to the high temperature results of figure 21, this scan is done using a fully numeric treatment of the one-loop thermal effective potential (4.26).

While the overlap areas of both analyses are interesting from a validation point of view, it is also important to look at the points not accessible by each approach. Looking at 22b there are two families of points that are not present in the high temperature scan of figure 21b: points where we have a three-step phase transition i.e. points that satisfy (5.20), and points where the symmetric vacuum is a stable minima at $T = 0$, i.e. points that violate (5.21).

Focusing on the three-step phase transitions, they can be clearly seen in figure 22b since they are the points below the boundary line $\lambda_{HS} = 2\sqrt{\lambda_H \lambda_S}$ (i.e. $\lambda_{HS} < 2\sqrt{\lambda_H \lambda_S}$). The high temperature is not able to find these points, by construction, since we chose to neglect the barrier terms, and we saw in section 5.4 that they were essential for predicting three-step phase transitions. The dependence on radiative barrier terms weakens these kinds of transitions and this is translated into the parameter space, the region compatible with three-step phase transitions is subdominant to the region compatible with two-step phase transitions, and even those points live very near to the boundary line. This subdominant nature often gives rise to their omission like in [48], but still they need to be accounted for since as we can see in figure 19 they can indeed provide SFOEWPTs.

Another subdominant class of points are those where the symmetric vacuum $(0, w)$ is a stable minima at $T = 0$ and they have a nucleation temperature. Those points are interesting since they reinforce the condition (5.21) as only an approximation, a good one since 90 % of the points in figure 22 still satisfy it. The computation of the bounce action is a quantum mechanical one since we are asking our field to nucleate via tunneling through the potential barrier [14, 29]. As we mentioned section 5.5 the condition (5.21) is enforcing that the barrier disappears at $T = 0$ as to ensure the tunneling is possible. Still, one can have a small enough barrier at zero-temperature like it is shown in figure 26 and still allow for nucleation to occur.

Finally, the overlap between both approaches is interesting from the point of view of gauge invariance. The CW is a gauge dependent object [49, 67], that we compute in the Landau gauge for convenience, but if we were to compute it in a general R_ξ gauge the ξ dependence would not vanish [20]. Since we use the CW to build the numeric effective potential the results of figure 22 are gauge dependent as well. This gauge dependence is a well know issue and there are ways to quantify the dependencies of our observables and transform them into theoretical uncertainties of our predictions [20]. However, a consequence of comparing the numerical results to those of the high temperature limit is that the points where they both coincide must be free from gauge dependence. This is due to the fact that the high temperature thermal effective potential (3.51) is gauge independent since it consists only of the tree-level effective potential and the thermal corrections to induce symmetry restoration.

5.7 Experimental Constraints on Exotic Higgs Decays at the LHC

Having studied the regions of the parameter space compatible with SFOEWPTs, we now need to apply the experimental bounds on these kinds of BSM extensions. Our Z_2 preserving model predicts a new colorless and chargeless particle that the Higgs can decay into, the width for this decay can be computed from (4.18) and for the Z_2 limit becomes

$$\Gamma(h \rightarrow ss) = \frac{v^2 \lambda_{HS}^2}{32\pi m_h} \sqrt{1 - 4 \frac{m_s^2}{m_h^2}}. \quad (5.22)$$

Since our scalar is undetectable by the LHC experiments due to it not interacting with any other SM particles (in the Z_2 limit, see the discussion on section 5.7.1 for non Z_2 exact models), the experimental constraints will come from the LHC upper bound on the invisible branching ratio of the Higgs boson $\mathcal{B}(h \rightarrow \text{inv})$ [68, 69]. This branching fractions is defined as the width of Higgs to invisible decay, which for our model is (5.22), divided by the Higgs total width, which is the width for all the SM decays of the Higgs boson $\Gamma^{SM}(h_{125}) = 4.1 \text{ MeV}$ [70] plus the new singlet decay

$$\mathcal{B}(h \rightarrow \text{inv}) \equiv \mathcal{B}(h \rightarrow ss) = \frac{\Gamma(h \rightarrow ss)}{\Gamma^{SM}(h_{125}) + \Gamma(h \rightarrow ss)}. \quad (5.23)$$

In order for our parameters to be allowed by the experimental bounds our branching ratio must satisfy $\mathcal{B}(h \rightarrow ss) \leq 0.107$ [68]. Inserting this bound in (5.23) and substituting in expression of the Higgs to two singlets decay width (5.22), we arrive at the following bounds on our parameter space

$$\lambda_{HS} \leq \left[\frac{32\pi m_h}{v^2 \sqrt{1 - 4m_s^2/m_h^2}} \frac{\mathcal{B}(h \rightarrow \text{inv})}{1 - \mathcal{B}(h \rightarrow \text{inv})} \Gamma^{SM}(h_{125}) \right]^{1/2}. \quad (5.24)$$

Introducing the upper limit on the Higgs to invisible branching fraction (0.107) [68], and the value for the SM predicted width for the $m_h = 125 \text{ GeV}$ ($\Gamma^{SM}(h_{125}) = 4.1 \text{ MeV}$) [70] we arrive at the bounds shown in figure 23. As we can see from the experimental bound on invisible decays, a large region of the parameter space has already been excluded, mainly those regions associated to the heaviest singlet masses. These bounds are expected to become more restrictive

as the LHC operations continue and the HL-LHC becomes operational [48], considering future colliders like the FCC-ee as well, most of the parameter space for this Z_2 preserving RxSM could be accessible for experimental observation [71].

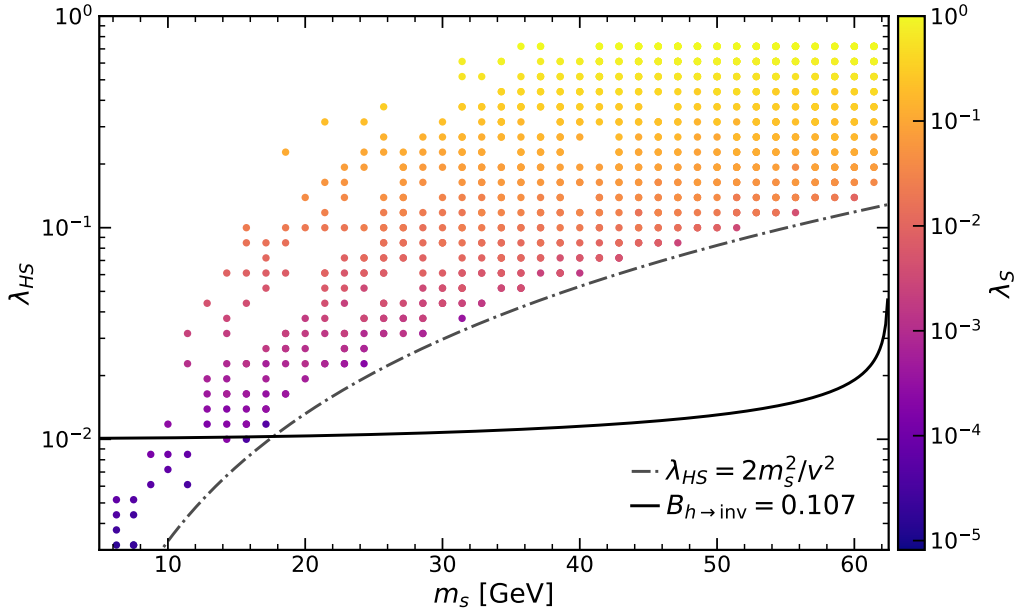


Figure 23: Current experimental bounds on the numerical results for parameters compatible with a SFOEWPT. The solid line corresponds to the current upper limit on the $h \rightarrow \text{inv}$ branching fraction [68], assuming that the only invisible decay is $h \rightarrow ss$.

5.7.1 Generalization to Mixed Scenarios

Through this chapter we chose to analyze the Z_2 preserving RxSM since it provided the simplest and most minimal extension of the SM compatible with SFOEWPT. We saw that by imposing an exact Z_2 symmetry the singlet S and the Higgs H remain unmixed providing then a very simple experimental signal. Let us now consider the case where the Z_2 symmetry is not exact by either having it spontaneously broken or by considering the most general model, and discuss briefly what would be the experimental bounds we would have to consider.

We restrict this discussion, as we have throughout this work, to the case relevant for exotic Higgs decays, $m_2 < m_1/2$, where h_1 denotes the 125 GeV state and h_2 the lighter, mostly-singlet eigen-state. Once we relax the Z_2 symmetry, mixing is introduced and characterized by the angle θ (4.11). Every coupling of h_1 to SM particles is universally rescaled by $\cos \theta$ relative to its SM value, while h_2 inherits the complementary $\sin \theta$ suppressed couplings, with the same values corresponding to a SM Higgs. In addition, the trilinear coupling $\lambda_{h_1 h_2 h_2}$ is no longer just λ_{HS} and we need to consider the full expression (4.17). From this introduction of mixing two consequences follow immediately at the level of collider observables.

Universal suppression of h_1 . Since every h_1 coupling is rescaled by $\cos \theta$, all visible production and decay rates of the 125 GeV Higgs boson are suppressed by $\cos^2 \theta$ relative to their SM prediction. This is precisely the scenario probed by global fits to the measured Higgs signal strengths and couplings that allow for a universal coupling modifier $\kappa_i \equiv \cos \theta$ [72, 73], those results and their allowed deviations from the SM provide the bounds on our mixing angle θ .

Nature of h_2 at the LHC. In our Z_2 preserving model the singlet S had no way to decay back into the SM, now due to the mixing it has the same $\sin \theta$ suppressed couplings the SM

Higgs boson would have. Because h_2 's partial widths are suppressed by $\sin^2 \theta$, its total width, and hence its decay length, can vary over many orders of magnitude depending on θ and m_2 . Due to this, three qualitatively distinct experimental signal types arise for the h_2 decay:

- **Prompt decays.** For values of $\sin \theta$ not too small²¹, h_2 decays promptly within the detector. The predominant decay channel (when it is kinematically possible) is $b\bar{b}$ with subdominant $\tau^+\tau^-$, $c\bar{c}$, gg and $\mu^+\mu^-$, these therefore give rise to decay topologies of the form $h_1 \rightarrow h_2 h_2 \rightarrow 4b, 2b2\tau, \dots$. These are signature targeted by dedicated exotic Higgs decay searches [74].
- **Displaced decays.** As $\sin \theta$ decreases, the h_2 total width $\Gamma(h_2)$ shrinks and the decay length $c\tau_{h_2}$ grows due to it being $\propto \sin^{-2} \theta$. For sufficiently small mixing and/or low m_2 , this length becomes macroscopic on detector scales, and h_2 manifests as a long-lived particle, producing displaced vertices in the tracker, calorimeter, or muon tracker. This regime is constrained by dedicated long-lived-particle searches [75] providing bounds on the branching ratio of $h_1 \rightarrow h_2 h_2$ for different values of $\sin^2 \theta$ and different regions of m_2 .
- **Invisible decays.** In the limit of very small $\sin \theta$, $c\tau_{h_2}$ will exceed the detector size entirely, and h_2 escapes undetected before it decays. The collider signature for this parameter region, is an invisible particle like those of the Z_2 preserving model. Therefore, this regime feeds back into the same $\mathcal{B}(h_1 \rightarrow \text{inv})$ bounds, with the main differences that now there will not be an analytic expression for the bounds as in (5.24), and more importantly, we will need to weigh the widths by their corresponding proportionalities w.r.t. the mixing angle [76].

These three regimes are highly dependent on the regions of the parameter space we are probing and therefore the proper experimental bounds should be applied according to the type of signal they entail. This is considerably more involved than the single, one dimensional bound on $\lambda_{HS}(m_s)$ we had for the Z_2 preserving case, and it is left outside the scope of this thesis.

²¹The exact values depend on both the decay length but also on m_2 so an order of magnitude is hard to give but when we say not too small we mean values of $\sin \theta$ close to the experimental limit that sits around 10^{-2} .

6 Conclusions and Outlook

In this thesis we have set out to build, from first principles, the quantum field theory machinery needed to describe the electroweak phase transition, and to apply it to the real singlet extension of the Standard Model. Through chapters 2 and 3 we developed the zero- and finite-temperature effective potential: the Coleman-Weinberg effective potential, renormalization, the Matsubara formalism, the high-temperature expansion, the study of phase transitions, and finally, introduced nucleation and daisy resummation. Through these chapters we used example models to help settle these ideas, starting from toy models and then moving to the full SM. This approach was carried through in chapters 4 and 5, where we derived the one-loop thermal effective potential of the Z_2 preserving RxSM in full, allowing us to both evaluate it analytically in the high-temperature limit, and also implement it numerically in *CosmoTransitions*.

This double approach was used in the latter sections of Chapter 5 in order to map out the regions of our model's parameter space compatible with a strong first-order EWPT. We further confronted this parameter space with the current experimental status of exotic Higgs boson decay searches at the LHC [68], to assess which regions of the SFOEWPT-compatible parameter space remain viable and which have already been probed by searches for invisible Higgs decays.

A result we consider worth highlighting on its own is the comparison between the parameter scan results of the analytic high-temperature expansion and the numerical *CosmoTransitions* analysis. Although the two methods are not equivalent, the high-temperature expansion neglects thermal corrections beyond the leading order and sidesteps the bounce-action nucleation computation in favor of a classical limit, we found substantial overlap between the regions of parameter space that both methods identify as compatible with a SFOEWPT.

Comparing the enormous difference in computational cost between the numerical scan (multiple hours) and the high-temperature scan (fractions of a second), we can highlight a possible use for the high-temperature limit as a preselection tool for the numerical parameter scan. The analytic approximation should not become a substitute for the full numerical analysis of *CosmoTransitions* or any other specialized software, but since we find, from the results of this thesis, a substantial overlap in the candidate regions for SFOEWPTs, we can use the high-temperature results to focus the parameter scan on regions of interest, and therefore avoid wasting computational time on parameter points that will not yield a SFOEWPT.

The analysis presented in this thesis has been restricted to the Z_2 preserving RxSM, in which the singlet does not mix with the Higgs doublet and is stable. This symmetric imposition made it possible to obtain closed formed relations using the high temperature limit, but it also limits the phenomenological implications at the collider level. A stable singlet can only be probed through invisible Higgs decays and missing-energy searches [68, 76]. Relaxing the Z_2 symmetry, either softly as in the spontaneously broken RxSM discussed briefly at the end of chapter 4 and in appendix A, or fully, as discussed extensively in chapter 4, is the natural next step of our analysis since in those models we can not set the mixing angle θ to be zero.

Singlet-Higgs mixing drastically changes the phenomenology of our model. Depending on θ and the singlet mass m_2 , the new scalar can decay visibly, allowing for direct detection at the LHC, including even the possibility of macroscopic decay lengths, opening the door to long-lived-particles signatures. This means that the constraints for these kinds of new signals will have to be included in our studies [74, 75], alongside our invisible and missing-energy constraints when they apply, in order to be able to properly probe the validity of SFOEWPTs in these models at colliders. At the same time, the mixing unavoidably rescales all SM Higgs couplings by $\cos \theta$ and modifies double-Higgs production, so the mixing angle in these extended scenarios is no longer constrained only by exotic-decay searches but also by precision Higgs-coupling measurements and by di-Higgs searches [72, 73]. Incorporating these bounds, together with existing single-

Higgs coupling fits [65, 66], into a chapter 5-like scan would be an essential ingredient of a full study of the spontaneously broken and/or generic RxSM.

Beyond extending the model, the gauge-dependence cross-check presented in Chapter 5 points to a natural improvement of the formalism itself. Rather than relying on the agreement between the leading-order high-temperature expansion and a fixed-gauge numerical result as an indirect argument for the smallness of gauge artifacts, we could carry out a detailed analysis like that of [20], where the authors compare commonly used perturbative frameworks (including the formalism used in this thesis) and keep track of the resulting theoretical uncertainty in the predicted gravitational-wave signal arising from the renormalization scale and gauge dependence of the phase transition parameters. Being able to repeat the analysis at those levels of precision, and quantifying how much of the SFOEWPT-compatible parameter space survives, would result in more robust predictions for both the gravitational wave and collider observables.

Finally, we can extend the study of phase transition parameters by computing the transition strength α , the inverse duration β/H and the bubble wall velocity. Those parameters, alongside T_n , can be used to predict the resulting stochastic gravitational wave spectrum, opening a complementary avenue for probing the electroweak phase transition through observatories such as LISA [13, 14]. Extending our scan in this direction would make it possible to overlay current and projected LISA sensitivity curves directly onto the SFOEWPT-compatible regions identified in Chapter 5, translating our exotic-decay collider constraints into gravitational-wave discovery prospects and vice versa. Future work along these lines could therefore combine collider probes, from ongoing LHC and future HL-LHC exotic Higgs decay searches, with gravitational-wave observations, providing genuinely multi-messenger constraints on the electroweak phase transition.

A RxSM with a Spontaneously Broken Z_2 Symmetry

In this section we will perform the same type of analysis of the RxSM as in chapter 4, but now instead of using the shift symmetry to impose that the VEV of the field S is 0, we will use it to impose that the tadpole μ_1 is.

Through this section we will still be considering the generic RxSM model so this prescription is equivalent to that of chapter 4. The main difference, as discussed in section 4.8 is that when we impose a Z_2 via the procedure $\mu_3 \rightarrow 0$ and $\mu_{HS} \rightarrow 0$, we will be able to obtain a spontaneously broken Z_2 model, that preserves the mixing between the H and S fields [64].

A.1 Scalar and Tree-Level Potential

The most generic, renormalizable, scalar potential for a RxSM is

$$\begin{aligned} V = & -\mu_h^2 |H|^2 + \lambda_h |H|^4 \\ & -\frac{1}{2}\mu_s^2 S^2 + \frac{1}{4}\lambda_s S^4 - \frac{1}{3}\mu_3 S^3 \\ & + \frac{1}{2}\lambda_{hs} |H|^2 S^2 + \mu_{hs} |H|^2 S, \end{aligned} \quad (\text{A.1})$$

where H is the $SU(2)_L$ Higgs Doublet of the SM (2.74) and S is a real scalar singlet under all symmetry groups, and we have already used the shift symmetry [31] to set the tadpole term μ_1 to zero.

The scalar fields can be parametrized, in the unitary gauge, as

$$H(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \varphi_h + h(x) \end{pmatrix}, \quad S(x) = (\varphi_s + s(x)), \quad (\text{A.2})$$

where $\varphi_{h(s)}$ is the background field $h(s)$, and whose VEVs are defined as

$$\langle \varphi_h \rangle = v = 246 \text{ GeV}, \quad \langle \varphi_s \rangle = u. \quad (\text{A.3})$$

Using the definitions of H and S , we can write the tree-level effective potential as

$$\begin{aligned} V_0(\varphi_h, \varphi_s) = & -\frac{1}{2}\mu_H^2 \varphi_h^2 + \frac{1}{4}\lambda_H \varphi_h^4 \\ & -\frac{1}{2}\mu_S^2 \varphi_s^2 - \frac{1}{3}\mu_3 \varphi_s^3 + \frac{1}{4}\lambda_S \varphi_s^4 \\ & + \frac{1}{2}\mu_{HS} \varphi_h^2 \varphi_s + \frac{1}{4}\lambda_{HS} \varphi_h^2 \varphi_s^2. \end{aligned} \quad (\text{A.4})$$

A.2 Tadpole Equations and Field Dependent Masses

Since we have used the shift freedom to already set $\mu_1 = 0$, the EW vacuum will therefore have a non-zero value for the VEV of the S field, i.e. the EW vacuum will be located at (v, u) .

Imposing that the EW vacuum is a minima of the potential we arrive at the following tadpole equations

$$\begin{aligned} \left. \frac{\partial V_0}{\partial \varphi_h} \right|_{(v,u)} = 0 \Rightarrow \mu_H^2 &= v^2 \lambda_H + \frac{u^2 \lambda_{HS}}{2} + u \mu_{HS}, \\ \left. \frac{\partial V_0}{\partial \varphi_s} \right|_{(v,u)} = 0 \Rightarrow \mu_S^2 &= \frac{1}{2} \left(\frac{v^2 \mu_{HS}}{u} + v^2 \lambda_{HS} + 2u^2 \lambda_S - 2\mu_3 u \right). \end{aligned} \quad (\text{A.5})$$

The field dependent mass matrix of the scalar particles are scheme independent, therefore field dependent masses for the would-be goldstone bosons is given by (4.7), while for the radial fields h and s have a non-diagonal matrix (4.6).

The physical masses are obtain those field dependent masses at the EW vacuum (v, u) . Doing that for (4.7) we get that it is zero, as expected, while for the scalar fields we get

$$M_{h/s}^2 = \begin{pmatrix} M_{hh}^2 & M_{hs}^2 \\ M_{hs}^2 & M_{ss}^2 \end{pmatrix} = \begin{pmatrix} 2v^2\lambda_H & v\mu_{HS} + vu\lambda_{HS} \\ v\mu_{HS} + vu\lambda_{HS} & 2u^2\lambda_S - \mu_3u - \frac{v^2\mu_{HS}}{2u} \end{pmatrix}. \quad (\text{A.6})$$

The mass matrix is non-diagonal meaning that the interaction eigen-states are mixed and we will need to perform a rotation to a physical basis to get the mass eigen-states of our theory. Following the same procedure as in chapter 4, and substituting the values of (A.6) in (4.12), we arrive at the following relations

$$\begin{aligned} \lambda_H &= \frac{1}{2v^2} (m_1^2 \cos^2 \theta + m_2^2 \sin^2 \theta), \\ \lambda_{HS} &= \frac{1}{uv} (m_1^2 \sin \theta \cos \theta - m_2^2 \sin \theta \cos \theta - v\mu_{HS}), \\ \lambda_S &= \frac{1}{4u^3} (v^2\mu_{HS} + 2m_1^2u \sin^2 \theta + 2m_2^2u \cos^2 \theta + 2\mu_3u^2). \end{aligned} \quad (\text{A.7})$$

A.3 Parameter Prescription

We define the parameters the same way as in chapter 4, θ is the mixing angle, and by its choice of sign we arrive at $m_1 > m_2$, where $m_{1(2)}$ is the mass of the physical particle $h_{1(2)}$. Setting $m_1 = 125 \text{ GeV}$, we identify the h_1 as the Higgs measured at the LHC.

By rotating our field into a physical basis, we have seen that we can relate the original model's (Lagrangian) parameters

$$\{\mu_H, \lambda_H, \mu_S, \mu_3, \lambda_S, \mu_{HS}, \lambda_{HS}\},$$

to their physical counterparts

$$\{m_1, m_2, \theta, v, u, \mu_3, \mu_{HS}\}.$$

The relations between these parameter prescriptions are given by the tadpole equations (A.5), and from the rotation to a physical basis (A.7).

If we want to impose a Z_2 , we need only set $\mu_3 = \mu_{HS} = 0$. The main difference with the Z_2 preserving case is that now θ is non-zero due to the fact that SSB still induces mixing between the H and S fields.

A.4 Higgs-Singlet Coupling

As we saw in chapter 4, the interactions between the state $h_1(h_2)$ and the SM particles will be the same as the standard model's H couplings but multiplied by $\cos\theta(\sin\theta)$.

The only new relevant interaction that we need to compute is the $h_1h_2h_2$ vertex since it will be the responsible for the decay of $h_1 \rightarrow h_2h_2$. The terms that contribute to this vertex are those cubic in the fields h and s , after SSB they are

$$V_{\text{cubic}} = h^3v\lambda_H + \frac{1}{2}h^2s\mu_{HS} + \frac{1}{2}h^2su\lambda_{HS} + \frac{1}{2}hs^2v\lambda_{HS} - \frac{\mu_3s^3}{3} + s^3u\lambda_S. \quad (\text{A.8})$$

Making the rotation (4.9), and collecting all the terms that contain $h_1h_2h_2$ we arrive at the following vertex

$$V_{\text{cubic}} \supset \frac{1}{2}\lambda_{h_1h_2h_2}vh_1h_2^2, \quad (\text{A.9})$$

where $\lambda_{h_1 h_2 h_2}$ is given by

$$\lambda_{h_1 h_2 h_2} = \frac{\cos \theta}{4u^2 v^2} \left[2u \left(2 \sin \theta \left(m_H^2 + 2m_S^2 \right) (u \sin \theta + v \cos \theta) + \mu_3 u v \sin(2\theta) \right) + v^2 \mu_{HS} (-6u \cos(2\theta) + 2u + 3v \sin(2\theta)) \right]. \quad (\text{A.10})$$

we can compute the width for h_1 to decay into a pair of h_2 scalars as

$$\Gamma(h_1 \rightarrow h_2 h_2) = \frac{v^2 \lambda_{h_1 h_2 h_2}^2}{32\pi m_1} \sqrt{1 - 4 \frac{m_2^2}{m_1^2}}. \quad (\text{A.11})$$

A.5 Tree-Level Vacuums

We know that all the minima of our tree-level potential (2.50) will be located along two paths: $\varphi_h = 0$, and $\varphi_h = D_h(\varphi_s)$ where

$$\begin{aligned} \frac{\partial V_0}{\partial \varphi_h} = 0 &\Rightarrow -\varphi_h \mu_H^2 + \lambda_H \varphi_h^3 + \mu_{HS} \varphi_h \varphi_s + \frac{1}{2} \lambda_{HS} \varphi_h \varphi_s^2 = 0 \\ &\Rightarrow \varphi_h^2 \equiv D_h^2(\varphi_s) = \frac{2\mu_H^2 - 2\mu_{HS} \varphi_s - \lambda_{HS} \varphi_s^2}{2\lambda_H}. \end{aligned} \quad (\text{A.12})$$

The potential along the path $\varphi_h = 0$ is given by

$$V_0(0, \varphi_s) = \frac{1}{4} \lambda_S \varphi_s^4 - \frac{1}{3} \mu_3 \varphi_s^3 - \frac{1}{2} \mu_S^2 \varphi_s^2. \quad (\text{A.13})$$

Thanks to there being no tadpole term, we know that the minima will therefore be located on

$$\varphi_s = 0 \quad \text{or} \quad \varphi_s = \frac{\mu_3 \pm \sqrt{\mu_3^2 + 4\lambda_S \mu_S^2}}{2\lambda_S} \equiv w, \quad (\text{A.14})$$

where the symmetric minima is defined as $(0, w)$.

The potential along the path $\varphi_h = D_h(\varphi_s)$ is given by

$$\begin{aligned} V_0(D_h(\varphi_s), \varphi_s) &= -\frac{\mu_H^4}{4\lambda_H} + \frac{\mu_H^2 \mu_{HS}}{2\lambda_H} \varphi_s \\ &\quad - \frac{2\lambda_H \mu_S^2 + \mu_{HS}^2 - \mu_H^2 \lambda_{HS}}{4\lambda_H} \varphi_s^2 \\ &\quad - \left(\frac{\lambda_{HS} \mu_{HS}}{4\lambda_H} + \frac{\mu_3}{3} \right) \varphi_s^3 + \left(\frac{\lambda_S}{4} - \frac{\lambda_{HS}^2}{16\lambda_H} \right) \varphi_s^4, \end{aligned} \quad (\text{A.15})$$

and the minima will come from the solutions of the following equation

$$\frac{\mu_H^2 \mu_{HS}}{2\lambda_H} + \frac{\mu_H^2 \lambda_{HS} - 2\lambda_H \mu_S^2 - \mu_{HS}^2}{2\lambda_H} \varphi_s - \left(\frac{3\lambda_{HS} \mu_{HS}}{4\lambda_H} + \mu_3 \right) \varphi_s^2 + \left(\lambda_S - \frac{\lambda_{HS}^2}{4\lambda_H} \right) \varphi_s^3 = 0. \quad (\text{A.16})$$

This equation has three solutions as well but in this case it is more convenient to find them numerically. Although we can not solve it analytically, it can be checked that after substituting the tadpole relations (A.5) $s = u$, i.e. the EW vacuum, is a solution.

B Auxiliary Figures

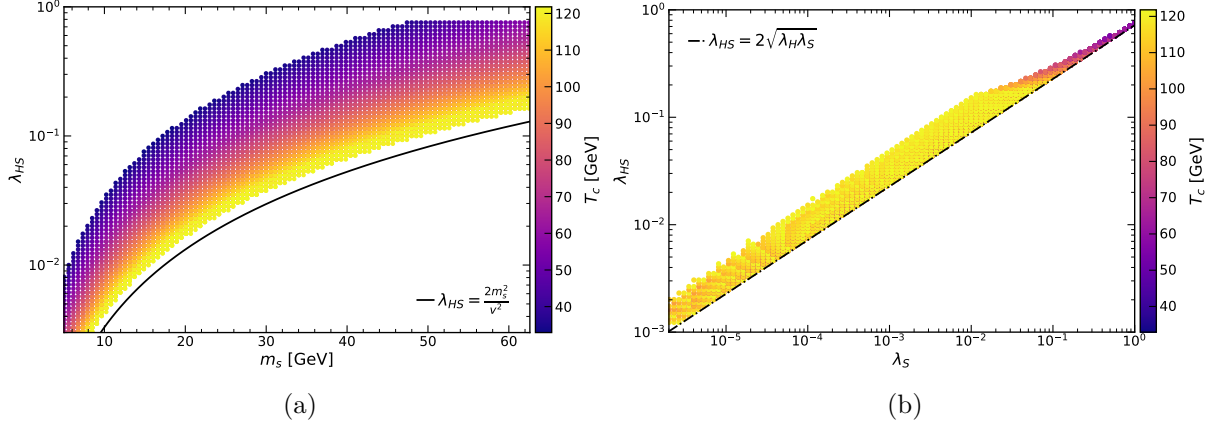


Figure 24: High temperature parameter scan of the Z_2 preserving RxSM, same points as in figure 21. (a)[(b)] Values of $m_s[\lambda_S]$ and λ_{HS} compatible with a SFOEWPT. Each point has been color coded according to its value of the critical temperature T_c .

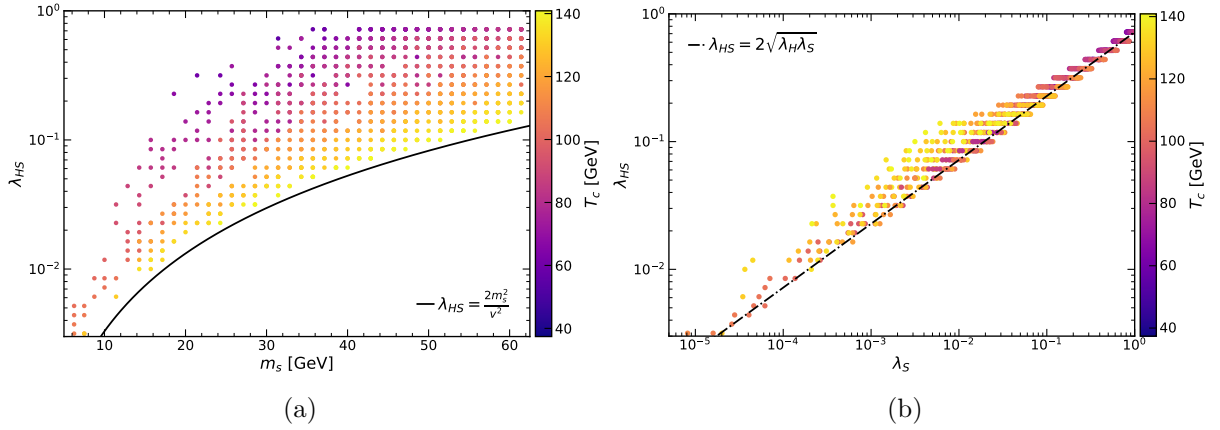


Figure 25: Same as figure 24 but for the numerical parameter scan shown in figure 22.

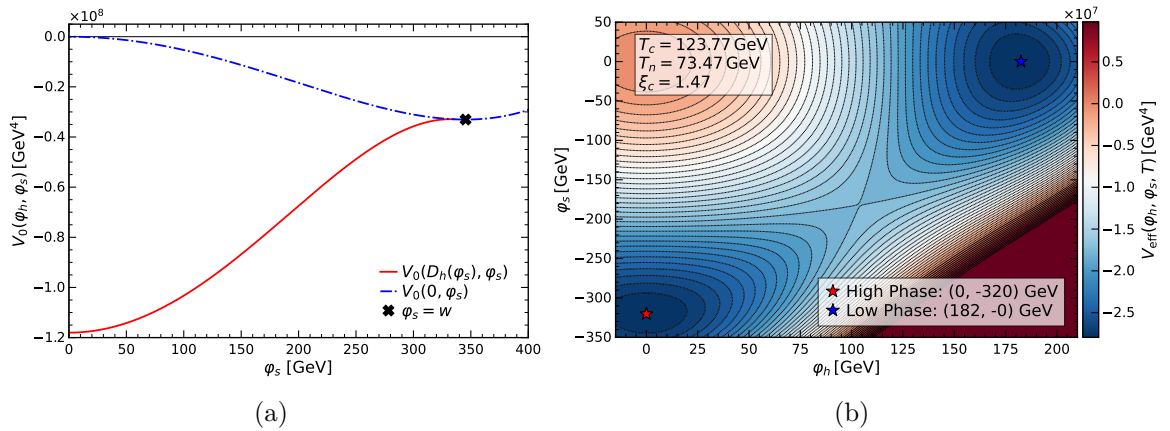


Figure 26: (a) Shape of the tree-level effective potential along the paths $\varphi_h = 0$ and $\varphi_h = D_h(\varphi_h)$ (4.22) and (b) thermal Potential at T_c for a set of parameters with a (tiny) tree-level barrier at zero-temperature, violating (5.21).

Bibliography

- [1] T.-H. Yeh, J. Shelton, K. A. Olive, and B. D. Fields, “Probing physics beyond the standard model: limits from bbn and the cmb independently and combined,” *Journal of Cosmology and Astroparticle Physics*, vol. 2022, no. 10, p. 046, Oct. 2022. [Online]. Available: <http://dx.doi.org/10.1088/1475-7516/2022/10/046>
- [2] N. Aghanim *et al.*, “Planck 2018 results: Vi. cosmological parameters,” *Astronomy & Astrophysics*, vol. 641, p. A6, Sep. 2020. [Online]. Available: <http://dx.doi.org/10.1051/0004-6361/201833910>
- [3] F. Takahashi *et al.*, “Review of Particle Physics,” *Int. J. Mod. Phys. A*, vol. 41, p. 2630011, 2026.
- [4] A. D. Sakharov, “Violation of CP Invariance, C asymmetry, and baryon asymmetry of the universe,” *Pisma Zh. Eksp. Teor. Fiz.*, vol. 5, pp. 32–35, 1967.
- [5] V. A. Kuzmin, V. A. Rubakov, and M. E. Shaposhnikov, “On the Anomalous Electroweak Baryon Number Nonconservation in the Early Universe,” *Phys. Lett. B*, vol. 155, p. 36, 1985.
- [6] M. Quiros, “Finite temperature field theory and phase transitions,” 1999. [Online]. Available: <https://arxiv.org/abs/hep-ph/9901312>
- [7] D. E. Morrissey and M. J. Ramsey-Musolf, “Electroweak baryogenesis,” *New Journal of Physics*, vol. 14, no. 12, p. 125003, Dec. 2012. [Online]. Available: <http://dx.doi.org/10.1088/1367-2630/14/12/125003>
- [8] G. Aad *et al.*, “Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC,” *Physics Letters B*, vol. 716, no. 1, pp. 1–29, 2012. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S037026931200857X>
- [9] S. Chatrchyan *et al.*, “Observation of a new boson at a mass of 125 GeV with the CMS experiment at the LHC,” *Physics Letters B*, vol. 716, no. 1, pp. 30–61, 2012. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0370269312008581>
- [10] K. Kajantie, M. Laine, K. Rummukainen, and M. Shaposhnikov, “Is there a hot electroweak phase transition at $m_H \gtrsim m_W$?” *Phys. Rev. Lett.*, vol. 77, pp. 2887–2890, Sep 1996. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.77.2887>
- [11] M. Laine and M. Meyer, “Standard model thermodynamics across the electroweak crossover,” *Journal of Cosmology and Astroparticle Physics*, vol. 2015, no. 07, p. 035–035, 7 2015. [Online]. Available: <http://dx.doi.org/10.1088/1475-7516/2015/07/035>
- [12] M. D’Onofrio and K. Rummukainen, “Standard model cross-over on the lattice,” *Physical Review D*, vol. 93, no. 2, 1 2016. [Online]. Available: <http://dx.doi.org/10.1103/PhysRevD.93.025003>
- [13] C. Caprini *et al.*, “Detecting gravitational waves from cosmological phase transitions with lisa: an update,” *Journal of Cosmology and Astroparticle Physics*, vol. 2020, no. 03, p. 024–024, Mar. 2020. [Online]. Available: <http://dx.doi.org/10.1088/1475-7516/2020/03/024>
- [14] P. Auclair *et al.*, “Cosmology with the laser interferometer space antenna,” *Living Reviews in Relativity*, vol. 26, no. 1, Aug. 2023. [Online]. Available: <http://dx.doi.org/10.1007/s41114-023-00045-2>

-
- [15] M. Carena, J. Kozaczuk, Z. Liu, T. Ou, M. J. Ramsey-Musolf, J. Shelton, Y. Wang, and K.-P. Xie, “Probing the Electroweak Phase Transition with Exotic Higgs Decays,” *LHEP*, vol. 2023, p. 432, 2023.
 - [16] D. Curtin, P. Meade, and C.-T. Yu, “Testing electroweak baryogenesis with future colliders,” *Journal of High Energy Physics*, vol. 2014, no. 11, Nov. 2014. [Online]. Available: [http://dx.doi.org/10.1007/JHEP11\(2014\)127](http://dx.doi.org/10.1007/JHEP11(2014)127)
 - [17] S. Coleman and E. Weinberg, “Radiative corrections as the origin of spontaneous symmetry breaking,” *Phys. Rev. D*, vol. 7, pp. 1888–1910, Mar 1973. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevD.7.1888>
 - [18] E. J. Weinberg, “Radiative corrections as the origin of spontaneous symmetry breaking,” PhD thesis, Harvard University, Cambridge, Massachusetts, 1973.
 - [19] M. Laine and A. Vuorinen, *Basics of Thermal Field Theory: A Tutorial on Perturbative Computations*. Springer International Publishing, 2016. [Online]. Available: <http://dx.doi.org/10.1007/978-3-319-31933-9>
 - [20] T. Biekötter, A. Dashko, M. Löschner, and G. Weiglein, “Perturbative aspects of the electroweak phase transition with a complex singlet and implications for gravitational wave predictions,” 2025. [Online]. Available: <https://arxiv.org/abs/2511.14831>
 - [21] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*. CRC Press, 2018.
 - [22] W. Heisenberg and H. Euler, “Consequences of dirac theory of the positron,” 2006. [Online]. Available: <https://arxiv.org/abs/physics/0605038>
 - [23] J. Schwinger, “On gauge invariance and vacuum polarization,” *Phys. Rev.*, vol. 82, pp. 664–679, Jun 1951. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRev.82.664>
 - [24] J. Goldstone, A. Salam, and S. Weinberg, “Broken Symmetries,” *Phys. Rev.*, vol. 127, pp. 965–970, 1962.
 - [25] D. Alonso Gonzalez, “The higgs boson and the early universe,” MSc thesis, Universidad Autónoma de Madrid, Cantoblanco, Madrid, 2020.
 - [26] A. Zee, *Quantum Field Theory in a Nutshell*, 2nd ed. Princeton University Press, 2010.
 - [27] M. D. Schwartz, *Quantum Field Theory and the Standard Model*. Cambridge: Cambridge University Press, 2014.
 - [28] P. Basler, “Vacuum Structure of Models beyond the Standard Model,” Ph.D. dissertation, KIT, Karlsruhe, 2019.
 - [29] M. O. Olea Romacho, “Higgs physics as a window to the electroweak epoch,” Ph.D. dissertation, U. Hamburg (main), Hamburg U., 2022.
 - [30] P. Basler, M. Mühlleitner, and J. Müller, “Electroweak phase transition in non-minimal higgs sectors,” *Journal of High Energy Physics*, vol. 2020, no. 5, May 2020. [Online]. Available: [http://dx.doi.org/10.1007/JHEP05\(2020\)016](http://dx.doi.org/10.1007/JHEP05(2020)016)
 - [31] J. Espinosa, T. Konstandin, and F. Riva, “Strong electroweak phase transitions in the standard model with a singlet,” *Nuclear Physics B*, vol. 854, no. 3, p. 592–630, Jan. 2012. [Online]. Available: <http://dx.doi.org/10.1016/j.nuclphysb.2011.09.010>
 - [32] F. Englert and R. Brout, “Broken symmetry and the mass of gauge vector mesons,” *Phys. Rev. Lett.*, vol. 13, pp. 321–323, Aug 1964. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.13.321>

- [33] P. W. Higgs, “Broken symmetries, massless particles and gauge fields,” *Phys. Lett.*, vol. 12, pp. 132–133, 1964.
- [34] G. S. Guralnik, C. R. Hagen, and T. W. B. Kibble, “Global conservation laws and massless particles,” *Phys. Rev. Lett.*, vol. 13, pp. 585–587, Nov 1964. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.13.585>
- [35] P. W. Higgs, “Broken symmetries and the masses of gauge bosons,” *Phys. Rev. Lett.*, vol. 13, pp. 508–509, Oct 1964. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.13.508>
- [36] T.-P. Cheng and L.-F. Li, *Gauge Theory of Elementary Particle Physics*. Oxford, UK: Oxford University Press, 1984.
- [37] D. M. Webber *et al.*, “Publisher’s Note: Measurement of the Positive Muon Lifetime and Determination of the Fermi Constant to Part-per-Million Precision [Phys. Rev. Lett. 106, 041803 (2011)],” *Phys. Rev. Lett.*, vol. 106, p. 079901, Feb 2011. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.106.079901>
- [38] V. Tishchenko *et al.*, “Detailed report of the MuLan measurement of the positive muon lifetime and determination of the Fermi constant,” *Phys. Rev. D*, vol. 87, p. 052003, Mar 2013. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevD.87.052003>
- [39] CMS Collaboration, “A measurement of the higgs boson mass in the diphoton decay channel in proton-proton collisions at $\sqrt{s} = 13$ tev,” 2026. [Online]. Available: <https://arxiv.org/abs/2607.28396>
- [40] J. I. Kapusta and C. Gale, *Finite-temperature field theory: Principles and applications*, ser. Cambridge Monographs on Mathematical Physics. Cambridge University Press, 2011.
- [41] M. L. Bellac, *Thermal Field Theory*, ser. Cambridge Monographs on Mathematical Physics. Cambridge University Press, 3 2011.
- [42] R. Kubo, “Statistical-mechanical theory of irreversible processes. i. general theory and simple applications to magnetic and conduction problems,” *Journal of the Physical Society of Japan*, vol. 12, no. 6, pp. 570–586, 1957. [Online]. Available: <https://doi.org/10.1143/JPSJ.12.570>
- [43] P. C. Martin and J. Schwinger, “Theory of Many-Particle Systems. I,” *Phys. Rev.*, vol. 115, pp. 1342–1373, Sep 1959. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRev.115.1342>
- [44] K. Kajantie, M. Laine, K. Rummukainen, and M. Shaposhnikov, “Generic rules for high temperature dimensional reduction and their application to the standard model,” *Nuclear Physics B*, vol. 458, no. 1, pp. 90–136, 1996. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0550321395005498>
- [45] T. Matsubara, “A New approach to quantum statistical mechanics,” *Prog. Theor. Phys.*, vol. 14, pp. 351–378, 1955.
- [46] C. L. Wainwright, “Cosmotransitions: Computing cosmological phase transition temperatures and bubble profiles with multiple fields,” *Computer Physics Communications*, vol. 183, no. 9, p. 2006–2013, 9 2012. [Online]. Available: <http://dx.doi.org/10.1016/j.cpc.2012.04.004>
- [47] G. W. Anderson and L. J. Hall, “Electroweak phase transition and baryogenesis,” *Phys. Rev. D*, vol. 45, pp. 2685–2698, Apr 1992. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevD.45.2685>

-
- [48] J. Kozaczuk, M. J. Ramsey-Musolf, and J. Shelton, “Exotic higgs boson decays and the electroweak phase transition,” *Physical Review D*, vol. 101, no. 11, p. 6 2020. [Online]. Available: <http://dx.doi.org/10.1103/PhysRevD.101.115035>
 - [49] H. H. Patel and M. J. Ramsey-Musolf, “Baryon washout, electroweak phase transition, and perturbation theory,” *Journal of High Energy Physics*, vol. 2011, no. 7, July 2011. [Online]. Available: [http://dx.doi.org/10.1007/JHEP07\(2011\)029](http://dx.doi.org/10.1007/JHEP07(2011)029)
 - [50] T. Biekötter, S. Heinemeyer, J. M. No, M. O. Olea-Romacho, and G. Weiglein, “The trap in the early universe: impact on the interplay between gravitational waves and lhc physics in the 2hdm,” *Journal of Cosmology and Astroparticle Physics*, vol. 2023, no. 03, p. 031, Mar. 2023. [Online]. Available: <http://dx.doi.org/10.1088/1475-7516/2023/03/031>
 - [51] S. Coleman, “Fate of the false vacuum: Semiclassical theory,” *Phys. Rev. D*, vol. 15, pp. 2929–2936, May 1977. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevD.15.2929>
 - [52] C. G. Callan and S. Coleman, “Fate of the false vacuum. ii. first quantum corrections,” *Phys. Rev. D*, vol. 16, pp. 1762–1768, Sep 1977. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevD.16.1762>
 - [53] A. D. Linde, “Fate of the False Vacuum at Finite Temperature: Theory and Applications,” *Phys. Lett. B*, vol. 100, pp. 37–40, 1981.
 - [54] A. D. Linde, “Decay of the False Vacuum at Finite Temperature,” *Nucl. Phys. B*, vol. 216, p. 421, 1983, [Erratum: *Nucl.Phys.B* 223, 544 (1983)].
 - [55] J. R. Espinosa, T. Konstandin, J. M. No, and M. Quirós, “Some cosmological implications of hidden sectors,” *Physical Review D*, vol. 78, no. 12, Dec. 2008. [Online]. Available: <http://dx.doi.org/10.1103/PhysRevD.78.123528>
 - [56] E. W. Kolb and M. S. Turner, *The Early Universe*. Taylor and Francis, 5 2019, vol. 69.
 - [57] D. Curtin, P. Meade, and H. Ramani, “Thermal resummation and phase transitions,” 2016. [Online]. Available: <https://arxiv.org/abs/1612.00466>
 - [58] P. Arnold and O. Espinosa, “Effective potential and first-order phase transitions: Beyond leading order,” *Phys. Rev. D*, vol. 47, pp. 3546–3579, Apr 1993. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevD.47.3546>
 - [59] R. R. Parwani, “Resummation in a hot scalar field theory,” *Phys. Rev. D*, vol. 45, pp. 4695–4705, Jun 1992. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevD.45.4695>
 - [60] M. Gavela, P. Hernandez, J. Orloff, and O. Pène, “Standard model cp-violation and baryon asymmetry,” *Modern Physics Letters A*, vol. 09, no. 09, p. 795–809, Mar. 1994. [Online]. Available: <http://dx.doi.org/10.1142/S0217732394000629>
 - [61] M. Gavela, M. Lozano, J. Orloff, and O. Pène, “Standard model cp-violation and baryon asymmetry (i). zero temperature,” *Nuclear Physics B*, vol. 430, no. 2, p. 345–381, Nov. 1994. [Online]. Available: [http://dx.doi.org/10.1016/0550-3213\(94\)00409-9](http://dx.doi.org/10.1016/0550-3213(94)00409-9)
 - [62] M. Gavela, P. Hernandez, J. Orloff, O. Pène, and C. Quimbay, “Standard model cp-violation and baryon asymmetry (ii). finite temperature,” *Nuclear Physics B*, vol. 430, no. 2, p. 382–426, Nov. 1994. [Online]. Available: [http://dx.doi.org/10.1016/0550-3213\(94\)00410-2](http://dx.doi.org/10.1016/0550-3213(94)00410-2)
 - [63] P. Huet and E. Sather, “Electroweak baryogenesis and standard model $\langle i \rangle_{\text{cp}} \langle /i \rangle$ violation,” *Physical Review D*, vol. 51, no. 2, p. 379–394, Jan. 1995. [Online]. Available: <http://dx.doi.org/10.1103/PhysRevD.51.379>

- [64] M. Carena, Z. Liu, and Y. Wang, “Electroweak phase transition with spontaneous z_2 -breaking,” *Journal of High Energy Physics*, vol. 2020, no. 8, 8 2020. [Online]. Available: [http://dx.doi.org/10.1007/JHEP08\(2020\)107](http://dx.doi.org/10.1007/JHEP08(2020)107)
- [65] ATLAS Collaboration, “A detailed map of higgs boson interactions by the atlas experiment ten years after the discovery,” *Nature*, vol. 607, no. 7917, p. 52–59, July 2022. [Online]. Available: <http://dx.doi.org/10.1038/s41586-022-04893-w>
- [66] CMS Collaboration, “A portrait of the higgs boson by the cms experiment ten years after the discovery,” *Nature*, vol. 607, no. 7917, p. 60–68, July 2022. [Online]. Available: <http://dx.doi.org/10.1038/s41586-022-04892-x>
- [67] N. K. Nielsen, “On the Gauge Dependence of Spontaneous Symmetry Breaking in Gauge Theories,” *Nucl. Phys. B*, vol. 101, pp. 173–188, 1975.
- [68] G. Aad *et al.*, “Combination of searches for invisible decays of the Higgs boson using 139 fb⁻¹ of proton-proton collision data at $\sqrt{s}=13$ TeV collected with the ATLAS experiment,” *Phys. Lett. B*, vol. 842, p. 137963, 2023.
- [69] H. Bahl, T. Biekötter, S. Heinemeyer, C. Li, S. Paasch, G. Weiglein, and J. Wittbrodt, “Higgstools: Bsm scalar phenomenology with new versions of higgsbounds and higgssignals,” *Computer Physics Communications*, vol. 291, p. 108803, Oct. 2023. [Online]. Available: <http://dx.doi.org/10.1016/j.cpc.2023.108803>
- [70] D. de Florian *et al.*, “Handbook of LHC Higgs Cross Sections: 4. Deciphering the Nature of the Higgs Sector,” *CERN Yellow Rep. Monogr.*, vol. 2, pp. 1–869, 2017.
- [71] M. Carena, J. Kozaczuk, Z. Liu, T. Ou, M. J. Ramsey-Musolf, J. Shelton, Y. Wang, and K.-P. Xie, “Probing the electroweak phase transition with exotic higgs decays,” 2022. [Online]. Available: <https://arxiv.org/abs/2203.08206>
- [72] ATLAS Collaboration, “Characterising the higgs boson with atlas data from the lh run-2,” *Physics Reports*, vol. 1116, pp. 4–56, 2025, breaking boundaries — ATLAS physics highlights and milestones from the LHC Run 2. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S037015732400382X>
- [73] The ATLAS and CMS Collaborations, “Combination of atlas and cms searches for higgs boson pair production at $\sqrt{s} = 13$ tev,” 2026. [Online]. Available: <https://arxiv.org/abs/2602.23991>
- [74] M. Cepeda, S. Gori, V. M. Outchoorn, and J. Shelton, “Exotic Higgs Decays,” *Annual Review of Nuclear and Particle Science*, vol. 72, no. Volume 72, 2022, pp. 119–149, 2022. [Online]. Available: <https://www.annualreviews.org/content/journals/10.1146/annurev-nucl-102319-024147>
- [75] A. E. del Valle, “Dark scalar reinterpretation of searches for higgs boson decays into long-lived particles at the lh,” *Journal of High Energy Physics*, vol. 2025, no. 12, p. 129, Dec 2025. [Online]. Available: [https://doi.org/10.1007/JHEP12\(2025\)129](https://doi.org/10.1007/JHEP12(2025)129)
- [76] T. Biekötter and M. Pierre, “Higgs-boson visible and invisible constraints on hidden sectors,” *The European Physical Journal C*, vol. 82, no. 11, Nov. 2022. [Online]. Available: <http://dx.doi.org/10.1140/epjc/s10052-022-10990-x>